

Mathematical inference

science and logic

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*“I got bored one day and I put everything on a bagel...
everything: All my hopes and dreams, my old report
cards, every breed of dog, every last personal ad on
Craigslist, sesame, poppy seed, salt... and it collapsed
in on itself. ’Cause you see, when you really put
everything on a bagel, it becomes this: The truth.”*

Jobu Tupaki in Everything Everywhere All At Once

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CHAPTER 1

Prelude: Andante sostenuto

“Science, including philosophy, is intrinsically polemical, and the polemical cannot be separated from the agonistic. In epochs in which great new things arise, the agonistic factor usually comes to the fore”

Johan Huizinga in *Homo Ludens*

1.1 Inspiration

When I was an undergraduate studying physics, probability theory often felt oddly under-defined. In statistical physics, and even more sharply in quantum physics, probability seemed to appear sometimes as ignorance, sometimes as frequency, sometimes as a physical tendency, and sometimes merely as a useful rule for calculation. The mathematical formalism worked; that was not the problem. The problem was that the interpretations surrounding it seemed to pull in different directions, while the underlying logical story remained incoherent.

After my studies I discovered E. T. Jaynes' *Probability Theory: The Logic of Science* [Jay03]. That book was, for me, the book I would have liked to have read much earlier. Jaynes, also trained as a physicist, refused to treat probability as a bag of unrelated conventions. He presented it, instead, as a disciplined extension of logic: a way of saying what one should believe, given what one already knows. The attraction of this view was not only technical: it also restored a kind of intellectual honesty to the subject, since hidden assumptions and convenient interpretations could no longer be smuggled in without being noticed.

Yet, Jaynes' book stops before the story is finished. Part of this is literal: the book was posthumously released on the basis of an unfinished manuscript. Part of it is conceptual: Jaynes gave a powerful solution for probability, following and extending Cox's programme [Cox46; Cox61], but several of his broader complaints about mathematics, subjectivity, causality and scientific practice remain only partially resolved. This thesis is therefore both a tribute and an attempt at continuation. It asks what Jaynes' programme might look like if it were done in an epistemological, constructive, type-theoretic and causal setting.

1.2 Introduction

The central concern of this work is the process by which new knowledge is added to existing information. I will call this process "mathematical inference". It is not a marginal technicality; it is, or at the very least should be, the core of scientific reasoning. Jaynes was therefore not simply concerned about the correctness of a few, specific statistical formulae: he thought that the core of science itself was at stake, because large parts of mathematical and statistical practice were built on foundations that were too permissive, too tolerant. His polemical tone makes sense in this context: if the problem is pervasive and foundational, then a mere technical correction would miss part of the point.

The urgency that is felt in Jaynes' text also perfectly fits the opening quotation from Huizinga on the polemical and agonistic nature of defending new scientific paradigms [Hui49]. This urgency becomes even more justified if one looks at two interrelated phenomena: independent analysts reaching divergent conclusions from the same data [Bot+20; Bre+22; Sch+21; Sil+18]; and large replication projects that reveal failures to replicate important and accepted scientific findings [Cam+16; Kle+14; Ope15]. Based on these phenomena, it becomes abundantly clear that a well-defined theory and formalism for mathematical inference is not a mere theoretical exercise, but an urgent matter.

Three concerns run through this thesis:

1. Mathematics should have foundations that make assumptions explicit and prevent uncontrolled use of ideal objects like infinity;
2. Mathematical inference should be treated as a principled extension of logic, while also being honest about the limits of that derivation;
3. Scientific practice should be organised in such a way that good inference is encouraged rather than punished.

Jaynes addresses all three, but solves only part of the second. The present work takes these concerns together, since separating them would miss part of the problem.

The formal answer developed here starts from Martin-Löf-style type theory [DP24; Mar84; NPS90]. This is not chosen for foundational purity: it is chosen because constructive type theory forces one to be explicit about what is being constructed, in which context, and by which rules. It also connects mathematics to computation, so that the reconstruction is not just a philosophical exercise, but computationally checkable. For this reason, it is also a natural environment for rebuilding finite probability theory, especially in light of finitary refinements of Cox-style arguments [CW17].

Moreover, if probability is called subjective, then subjectivity should not merely mean that one may freely choose a prior. A subjective theory should also be able to represent which questions are in scope, which facts are remembered, which possibilities are being imagined, and how relations between concepts change. Also for this reason, Martin-Löf-style type theory proves to be an excellent basis, as its intuitionistic and epistemological foundations naturally support these operations on contexts [DP24; SN25]. However, to explicate that specific operations on the context change someone's knowledge, someone's epistemic state, modes and modalities are introduced. This introduction is made possible by modal dependent type theory, which extends Martin-Löf-style type theory so that these modes can be described natively [Gra+21; Gra23; GSB19].

The probabilistic part is not the whole story. Jaynes presents probability theory as a universal logic of science, but the mature formal theory of causal models was developed largely outside his work. To extend Jaynes' probabilistic framework beyond prediction, Pearl's structural causal models and do-calculus therefore have to be incorporated into the framework [Pea09; PM18].

Finally, inference is not performed in a vacuum. McElreath's presentation of Bayesian modelling is valuable not only because it teaches calculations, but also because it keeps statistical inference close to modelling choices, scientific questions and workflow [McE20]. The reproducibility crisis and the diverging conclusions from competent researchers mentioned earlier illustrate that even technically competent inference can become unreliable inside a bad scientific pipeline [GL13; Ioa05]. Good science needs good inference, but also a good environment in which inference takes place.

Scope and assumptions

This is a thesis in artificial intelligence, where artificial intelligence is understood broadly as the formalisation and implementation of inference. In this sense probability theory, logic, causality and proof checking all fall naturally within its

1. Prelude: Andante sostenuto

scope. The aim is not to build a large applied AI system, but to investigate what kind of mathematical inference such systems should ideally perform when they reason under uncertainty, act under causal assumptions, or revise an epistemic state.

The thesis therefore looks at inference on two levels. Internally, it asks which formal rules should ideally govern uncertain, subjective and causal inference. Externally, it asks in what scientific environment mathematical and statistical inferences are generally made. This distinction matters, because the proposed formal machinery may not fit comfortably inside every existing scientific habit; and if that turns out to be the case, the problem may lie – at least partly – with those habits.

The mathematical scope is deliberately restricted: finite probability spaces; finite causal models; explicitly representable contexts and epistemic states; and central objects that are constructible and implementable. Thus the technical development is discrete throughout, but more narrowly so: the results below are finite results, not a general theory of arbitrary discrete spaces. This restriction is not merely a convenience: it follows Jaynes’ suspicion of uncontrolled infinities; it follows the constructive spirit of Martin-Löf-style type theory; and it fits the computational aspect of artificial intelligence.

Selected formal parts are implemented in Lean 4 [MU21] where this is useful and feasible. Lean is used here as a proof checker for representative definitions, proofs and examples, not as a promise that every philosophical claim in the thesis will be mechanised: the point is more modest and more important. The central mathematical constructions should be expressible in a form precise enough for independent mechanical checking. Recent discussions of digitised proofs and AI-assisted mathematics make clear that this is becoming an increasingly relevant part of mathematical practice [Cas26; Kak26; Orn26; Slo26].

Several limitations follow directly from this scope: this work does not reconstruct all of classical probability theory; it does not provide a general theory of continuous probability spaces; it does not attempt to solve measure theory; it does not claim that Martin-Löf-style type theory is the only acceptable foundation for mathematics; it does not claim that Pearl’s framework captures everything that can meaningfully be called causality; and it does not pretend to give a complete empirical sociology of science. The aim is narrower: to construct a coherent finite framework in which probability, subjectivity and causality can be represented with as little hand-waving as possible.

Research questions

The thesis is guided by the following research questions:

1. Can Jaynes’ probabilistic framework be cast in the language of Martin-Löf-style type theory?
2. Can Jaynes’ probabilistic framework be extended so as to include subjectivity as part of the formal machinery, rather than as an interpretation added afterwards?

3. Can Jaynes' probabilistic framework be extended so as to include Pearl-style causal reasoning?
4. Can Jaynes' commentary on bad scientific practice be placed in a wider institutional context?

Expected contributions

My first hope is to clarify Jaynes' programme. Jaynes' probability theory will be presented as an attempt to treat probability as extended logic, while the thesis carefully separates what he actually derives from what he claims. The distinction matters, because some of Jaynes' strongest claims — around universality, subjectivity and completeness — are exactly where later objections become sharpest [Col04; CW17; Hal99a; Hal99b; Sha04; Van03].

Second, the thesis offers a constructive reconstruction of the relevant slice of Jaynes' programme inside a Martin-Löf-style type-theoretic setting. In the finite case, probability assignments should be representable as explicit objects, and the familiar rules of probability should either be derived or obeyed by construction. Probability types are the natural vehicle for this, and they connect naturally to finitary versions of Cox-style reasoning [CW17].

A constructive account of causality follows. Pearl-style structural causal models are not replaced; they are embedded in a broader type-theoretic and epistemic setting. Interventions, a representative model-level counterfactual calculation and changes of causal structure are then treated as changes between explicitly represented epistemic states.

Alongside causality comes a more explicit account of subjectivity. Subjectivity is not reduced to the choice of a prior – instead, epistemic position is encoded through modes and transitions between them. This makes it possible to speak formally about learning, forgetting, interest and hypothesising, which matters especially for artificial intelligence, where agents may have bounded resources, limited memory and different contexts [HP07; NR19; Sim55].

A limited Lean 4 formalisation accompanies the formal chapters. This is not solely decorative; it is a guard against hidden assumptions. When a definition or proof cannot be made precise, even in the finite setting, this indicates not an implementation nuisance, but mathematical information.

The thesis closes with a diagnosis of scientific practice that connects back to the formal work. Hidden assumptions in mathematics and hidden incentives in science are different phenomena, but they are intertwined: both make it possible for conclusions to look cleaner than the process that produced them. The thesis therefore ends with practical suggestions concerning policies, incentives and research culture.

Compactly put: the formal part succeeds if finite Jaynes-style probabilistic updating and finite Pearl-style causal reasoning can be represented in the chosen Martin-Löf-style framework, with representative constructions checked in Lean 4. The wider thesis succeeds if this formal work can be convincingly connected to the claim that good science needs both good inference and a good environment in which this inference takes place.

Relevance

The relevance for artificial intelligence is direct. If artificial intelligence is, at least in part, the formalisation and implementation of inference, then its reliability depends on the quality of the reasoning methods on which it is built. A finite, constructive and computational account of probability and causality would provide a clearer route from abstract reasoning to executable systems.

The causal extension is especially important. Modern intelligent systems are increasingly expected not only to predict, but to answer questions about intervention, responsibility and alternative outcomes. Prediction alone is too weak for that task. Pearl’s causal framework shows why prediction has to be supplemented by intervention and counterfactual reasoning [Pea09; PM18]. This thesis asks how such reasoning can be embedded in a more explicitly constructive and subjective framework.

The broader scientific-practice component is also relevant to artificial intelligence. AI systems trained on, used within, or used to accelerate science inherit the weaknesses of the scientific pipeline. If the surrounding environment rewards fragile results, opaque reasoning or publication gamesmanship, then faster inference may simply produce bad science more efficiently. The formal and institutional questions therefore have to be considered together.

1.3 Outline

The thesis is structured as follows:

The First movement expounds Jaynes’ central claims: it discusses his demand for solid foundations in mathematics, solid foundations in statistics, and a good and honest attitude for doing these sciences; it also introduces the mathematical manifesto that will guide the rest of the work.

The Second movement discusses Jaynes’ solution for statistics and the problems surrounding it: it reconstructs the Jaynes–Cox view; it separates derivation from interpretation; and it discusses objections concerning consistency, reasonableness, universality, subjectivity and causality.

The Third movement proposes a more suitable mathematical setting: it discusses alternative logics and type theory; it then presents the Martin–Löf-style type theory used in the constructive parts of the thesis.

The Fourth movement uses this mathematical setting to address the problems raised earlier: it develops context manipulation, modalities, probability types and structural causal models inside the chosen type-theoretic framework; it also collects the formal results.

The Fifth movement returns to scientific practice: it expands Jaynes’ complaints about bad attitudes into a wider analysis of perverse incentives in modern science; it also proposes practical changes.

The appendices contain definitions, proofs, arguments and code-oriented material that would interrupt the main line of the text. They are not less important, but they are placed where they can be checked without interrupting the narrative.

1.4 Idiosyncrasies

Before continuing, a few habits of this text deserve some attention:

The structure is intentionally unconventional. The thesis is arranged as a symphony, with movements rather than ordinary parts. This is inspired by Huizinga’s insistence on the playful character of culture and science, and by the suspicion that some of this playfulness has been replaced by a more anxious and institutionalised competition. The musical structure is therefore not only decorative: it is a reminder that scientific thought is not just a sequence of deliverables, but also a playful activity. The tempo markings are mood cues rather than a strict musical programme: the Prelude is deliberate, the First movement polemical, the Second reflective, the Third slow and foundational, the Fourth broad and constructive, and the Fifth quick and heated.

Acronyms will be avoided where they do not genuinely help. Science already contains enough abbreviations that sound useful mainly because they sound fundable. Likewise, ordinary terms will not be visually inflated with unnecessary capital letters. This means there will be no grand acronym for “Mathematical Inference as a Logical Extension” (MILE); there will simply be “mathematical inference”, in the sense developed in this text.

The thesis is written with active AI collaboration: ChatGPT has been used for source discovery and general framing; Lumo has been used for stylistic assistance; Claude and ChatGPT are used where appropriate for coding and proof-checking work, and Claude is also co-responsible for the figures and diagrams; this belongs to the wider movement towards AI-assisted formal mathematics [Cas26; Kak26; Orn26; Slo26]. This thesis in AI is therefore, rather appropriately, 100% AI-made, 100% human-reviewed. Every paragraph was (re-)written, weighed and revised by me; the intellectual responsibility, including responsibility for errors, remains completely mine.

The intended audience is broad, and this is risky. The thesis lies at the crossing of logic, mathematics, artificial intelligence, philosophy of probability, causality and the sociology of science. Readers from any one of these domains may sometimes want more detail exactly where another reader needs less. The compromise adopted here is narrative-first: the main text preserves the flow of ideas, while technical definitions, proofs and code are moved to the appendices whenever they would otherwise break the story.

These idiosyncrasies are not meant as a decorative ornament: they are part of the method. A thesis about explicit reasoning, constructive mathematics and scientific honesty should itself try to be explicit about the way it reasons and the choices it makes. The reader is invited to follow this movement.

CHAPTER 2

First movement: Allegro con brio

“Indeed, an insistence that all of mathematics be stated in that language all of the time can place unnecessary burdens on a theory, particularly one intended for application in the real world. It can also degenerate into an affectation, used only linguistically rather than functionally. To give every old, familiar notion a new, impressive name and symbol unknown to Gauss and Cauchy has nothing to do with rigor. It is, more often than not, a form of gamesmanship whose real purpose is to conceal the Mickey Mouse triviality of what is being done. One would blush to state it in plain English”

E. T. Jaynes in *Probability Theory: The Logic of Science*

2.1 Solid foundations for mathematics

Jaynes' first complaint is directed at the foundations of mathematics. His target is not mathematics as such, and certainly not calculation, experiment, modelling or honest abstraction. His target is a particular kind of mathematical culture: a culture in which formal permission is too easily confused with scientific usefulness, and in which the ability to write down a consistent-looking construction is mistaken for the ability to explain anything worth explaining.

The most visible symptom is what I will call infinite paradoxing. Modern mathematics is full of results that are technically precise and, at the same time, deliberately disorienting: Banach–Tarski decompositions, non-measurable Vitali sets, the Borel–Kolmogorov paradox, supertasks such as Thomson's lamp, and Ross–Littlewood-style urn puzzles [AK91; AK95; BT24; GHR17; Tho54; Vit05]. These examples do not all have the same status. Some are genuine theorems; some are philosophical stress tests; some are warnings about conditioning, measure or limiting procedures. Yet together they display the same family resemblance: infinity enters, ordinary intuition collapses, and mathematical sophistication is then measured by how gracefully one learns to live with the collapse.

Jaynes' temperament runs in the opposite direction. He repeatedly prefers finite problems, limiting procedures, explicit assumptions and calculational contact with experiment [Jay03]. The claim is not that every infinite construction is false, but that an infinite construction should earn its keep. If the point of mathematics in science is mathematical inference, then a method that converts finite observations into uncontrolled ideal objects should be treated with suspicion. It may still be useful, but usefulness is not conferred by grandeur.

The deeper objection is therefore methodological. Axiomatic set theory, especially in the Zermelo–Fraenkel tradition with choice, permits enormously powerful constructions. That power is one of the reasons for its success. But the same power can also hide the movement from finite information to ideal totality. The philosopher may admire the vertigo; the physicist may ask what was measured. Jaynes is, in this sense, closer to the physicist: start with what can be stated, counted, constructed or approximated, and move to limits only with a clear account of what the limiting process is doing.

I do not attempt to refute classical set theory here — it would be theatrical to pretend otherwise. The choice I make is narrower: for the formal reconstruction, the default mathematical environment will be constructive and finitary. The point is not purity but discipline. A framework that forces objects to be constructed, contexts to be declared and dependencies to be visible is better suited to the kind of scientific reasoning Jaynes wanted than a framework in which one may summon completed infinities too cheaply.

2.2 Solid foundations for statistics

Jaynes' second complaint is more direct and more devastating. In his telling, twentieth-century statistics had become a patchwork of procedures, tests, conventions and reputations. Some procedures worked well in practice; some were defended by tradition; some were inherited because nobody wanted to

2.2. Solid foundations for statistics

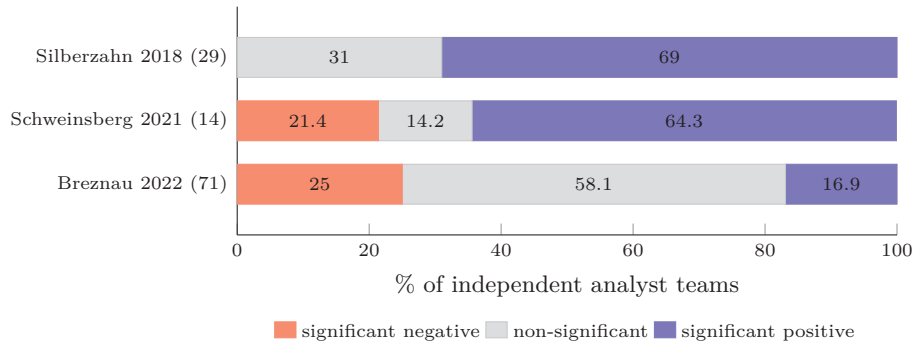


Figure 2.1: Three many-analyst projects in which independent teams analysed one dataset, classified by the *direction* of the reported result. Silberzahn et al. (29 teams): no significant-negative result, 31% non-significant, 69% significant-positive [Sil+18]; Schweinsberg et al. (Hypothesis 1, 14 analysts with known direction): 21.4% / 14.2% / 64.3% [Sch+21]; Breznau et al. (71 teams, estimate-level): about a quarter (25%) significant-negative, a little over half (58.1%) non-significant, 16.9% significant-positive [Bre+22]. Botvinik-Nezer et al.’s neuroimaging study is discussed in the text but omitted here, since its 70 teams answered nine yes/no hypotheses with no directional split [Bot+20].

reopen old battles. What was missing was a single account of why statistical inference should have the form it has.

Cox’s programme offered Jaynes exactly the kind of answer he wanted. If plausibility is represented numerically, if logically equivalent states of information are assigned equivalent plausibilities, and if reasoning is required to be consistent across different routes, then the familiar rules of probability can be recovered after a suitable rescaling [Cox46; Cox61; Jay03]. Probability is then not a bag of recipes. It is extended logic: the calculus of consistent reasoning under uncertainty.

This is the great attraction of the Jaynes–Cox view. It offers a way to say that statistical inference is neither a social convention, nor a frequentist habit, nor a computational convenience: it is what one must do if one wants uncertainty to obey the same demand for coherence that ordinary deductive logic obeys in the limiting case of certainty. Keynes and Ramsey had already made probability philosophically serious as a logic of partial belief; Cox and Jaynes gave the programme its sharpest mathematical form [Key21; Ram31].

The force of Jaynes’ criticism becomes clearer when placed next to modern examples of analytic instability. Independent teams can receive the same data and reach sharply different conclusions; many-analyst projects and large replication studies show that the path from data to claim is often less determinate than the published paper suggests [Bot+20; Bre+22; Kle+14; Ope15; Sch+21; Sil+18]. The point is not that every disagreement is fraud, incompetence or frequentism. The point is that inference is fragile when assumptions are hidden, workflows are underdescribed and technical choices are treated as harmless details.

I therefore read Jaynes’ solution as both important and incomplete — important because it gives a principled answer to a real problem (probability

2. First movement: Allegro con brio

as a norm of coherent inference), and incomplete because the derivation itself depends on assumptions that are not innocent, and because probability alone does not account for subjectivity, causality or institutional pressure. The Second movement will therefore keep the Jaynesian ambition while refusing to pretend that the Jaynesian derivation has already done all the work.

2.3 A good and honest attitude for doing these sciences

The third complaint is less formal but no less central. Jaynes does not only object to bad foundations; he objects to bad habits. His prose is full of irritation at gamesmanship, empty abstraction, prestige rituals and derivations that look impressive because the assumptions have been pushed into the furniture. This is more than a matter of tone. It is part of the argument.

The habits Jaynes attacks can be arranged under five headings. First, gamesmanship: the desire to win a local intellectual contest rather than to clarify a problem. Second, cutting corners: the production of polished results whose foundations are too thin to bear their own weight. Third, hidden assumptions: the silent transfer of what should have been stated into what is merely presumed. Fourth, hyperfocus: technical virtuosity that forgets the larger question. Fifth, theoretical display without experimental responsibility: the production of mathematics that neither explains, predicts nor helps anyone build or test anything.

These vices are not limited to statistics, and they are not limited to Jaynes' favourite opponents. They are temptations built into scholarly life. They appear whenever professional reward becomes too detached from intellectual responsibility. They also appear inside Jaynes' own work: some of his strongest claims about universality, subjectivity and completeness move faster than the derivations can justify. This is not a reason to discard Jaynes. It is a reason to take him seriously enough to apply his own standard to him.

That standard is simple: say what is assumed; say what follows; say what is only being interpreted afterwards. I would like the whole thesis to be read as an attempt to obey that standard. The probability calculus must be reconstructed in a setting where assumptions are explicit. Subjectivity must not be left as a slogan. Causality must not be bolted on as an afterthought. And scientific practice must not be treated as a decorative epilogue to formal work, because the formal work is only as useful as the practices in which it is used.

2.4 Mathematical manifesto

From these complaints a small mathematical manifesto follows. It is not a new foundation, nor a grand acronym, nor a shiny institutional programme. It borrows its form from the Agile Manifesto and from later workflow frameworks such as Scrum: a short set of comparative preferences, meant to guide practice rather than to settle every philosophical dispute in advance [Bec+01; SS20]. For the rest of the thesis, the discipline is this:

I am uncovering better ways of developing mathematical theories by doing it and helping others do it. Through this work I have come to value:

Algorithms over abstractions

Narrative over formulae

Explicit over implicit

Simple over complex

Algorithms over abstractions does not mean that abstraction is bad. It means that abstraction should remain connected to construction, computation or procedure. If a concept cannot in principle be implemented, checked, traced or approximated, then its scientific role should be handled with care.

Narrative over formulae does not mean that formulae are ornamental. It means that formulae should serve the movement of thought. A derivation is not a heap of symbols; it is a story about why one statement follows from another. When the story is hidden, the formulae may still be true, but they become harder to inspect.

Explicit over implicit is the most Jaynesian rule. Assumptions should be visible. Contexts should be named. Dependencies should be tracked. If a proof requires a finite domain, a decidable event, a classical logic or a causal graph, that requirement should not be smuggled in under the carpet.

Simple over complex is not a cult of simplicity. It is a preference for the smallest mechanism that does the job. Complexity is sometimes necessary, but it should be paid for consciously. The constructive framework used later is chosen in this spirit: not because it is the easiest language at first glance, but because it makes hidden commitments harder to hide.

CHAPTER 3

Second movement: Andante con moto

“A story is needed before one can set sail. Look at us. For the sake of the story, we are sitting here aboard a water taxi on our way to the Holy Grail, the unknown, and adventure. People think life becomes more enjoyable through stories. That is true enough, but it is not the point. Without stories, it becomes meaningless. Sense consists of sentences. The words that tell the story lay a reassuring relation of cause and effect over random facts and events. People crave plot because a plot reduces the unbearable and unmanageable chaos of the sublunary world to a human scale, and to a chain of initiatives and consequences that a person is able to comprehend. A plot gives a sense of control, provenance and destination, origin and direction”

Ilja Leonard Pfeijffer in *Grand Hotel Europa*

3.1 The Jaynes–Cox programme

The Jaynes–Cox programme begins with a deceptively modest question: what rules should govern reasonable degrees of belief? Cox’s answer is that plausibilities should behave consistently under logical operations; Jaynes’ answer is that this consistency requirement is not merely a technical convenience, but the very logic of science [Cox46; Cox61; Jay03].

Desiderata

The usual presentation starts with three desiderata. In Jaynes’ own words, in the compact formulation [Jay03]:

- (I) Representation of degrees of plausibility by real numbers;
- (II) Qualitative correspondence with common sense;
- (III) Consistency.

This is the short version. In the fuller discussion, Jaynes unpacks consistency into three demands: a conclusion should not depend on the route by which it is reached; relevant evidence should not be ignored; and equivalent states of knowledge should receive equivalent plausibility assignments. In the terminology of this thesis, the first desideratum is divisibility and comparability: plausibility should be represented on a single ordered numerical scale, so that one state of information can be compared with another. The second is qualitative agreement with common sense: ordinary Aristotelian logic should appear as the limiting case in which propositions are known with certainty, and small changes in the premises should not produce arbitrary jumps in the conclusions. The third is consistency: if a plausibility can be obtained by two legitimate routes, the two answers should agree. These desiderata are not yet probability theory. They are rules for how a calculus of plausible reasoning ought to behave.

One caveat carries through to the constructive reconstruction in the Fourth movement: the qualitative-agreement clause typically includes a complement rule of the form $\pi(\neg A \mid I) = S(\pi(A \mid I))$, which silently relies on excluded middle. In the constructive setting used later, that clause is not applied to arbitrary propositions; it is applied only inside decidable event algebras, such as finite urn events or the resolution-status events of Definition A.3.3. The desiderata are otherwise as in the standard Cox–Jaynes presentation [Cox46; Cox61; CW17; Jay03; Van03].

Derived calculus

Cox’s formal move is to apply these desiderata to the logical operations of negation and conjunction. If A and B are propositions and I is the background information, then the plausibility of $\neg A$ should be determined by the plausibility of A , and the plausibility of $A \wedge B$ should be determined by the plausibility of A and the plausibility of B once A is assumed. Under the usual regularity assumptions, one can rescale the plausibility function to a probability function P satisfying

$$P(A \wedge B \mid I) = P(A \mid I)P(B \mid A, I)$$

and

$$P(A \mid I) + P(\neg A \mid I) = 1.$$

The first equation is the product rule; the second is the sum rule for complements. Bayes' theorem then follows by applying the product rule in both orders:

$$P(A \mid B, I) = \frac{P(B \mid A, I)P(A \mid I)}{P(B \mid I)}$$

whenever $P(B \mid I)$ is non-zero.

In more familiar language, the end of the derivation is the ordinary probability calculus, whose finite fragment is fixed in Definitions A.3.7 and A.3.9. Probabilities are non-negative; the whole space receives probability one; disjoint alternatives add; conditioning obeys the product rule; Bayes' theorem follows from rearranging the product rule. In the full classical setting this is usually connected to Kolmogorov's axioms, including countable additivity. In the finite setting used in this thesis, countable additivity collapses to finite additivity: if $A_1, \dots, A_n$ are pairwise disjoint events, then

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i).$$

The philosophical gain is large: what looked like a specialised mathematical theory becomes a norm of coherent reasoning. Probability is more than a matter of repeated experiments. It is about what follows from information.

None of this means that Jaynes invented Bayesian reasoning. Keynes, Jeffreys, Ramsey and Cox are central ancestors of the project, and Jaynes explicitly inherits the Coxian line [Cox46; Cox61; Jef61; Key21; Ram31]. Jaynes' distinct contribution is the boldness of the synthesis: probability as extended logic, statistical inference as disciplined common sense, and scientific disagreement as often traceable to hidden assumptions or incoherent rules.

The attraction is obvious. If successful, the programme gives mathematical inference a spine. It tells us not only which formulae to use, but why those formulae are binding on anyone who wants to reason consistently. The remaining sections ask where this spine bends, where it breaks, and where it needs new vertebrae.

3.2 Objections

The objections to Jaynes do not all have the same force. Some are technical; some are philosophical; some concern scope. I will treat them in an order that moves from the derivation itself to the larger claims built on top of it: consistency, reasonableness, universality, subjectivity and causality.

The guiding distinction is the one already announced in the Prelude: what is derived, and what is claimed. Jaynes derives a powerful representation result under assumptions. He sometimes writes as if he has also derived a universal account of rationality, subjectivity and scientific method. The first achievement is real. The second needs repair.

Consistency

The first worry is whether Cox-style assumptions are even consistent in the form needed for Jaynes' conclusion. Halpern showed that naive formulations

3. Second movement: Andante con moto

of Cox’s theorem can fail — finite domains and insufficiently rich plausibility ranges can produce counterexamples to the intended representation [Hal99a; Hal99b]. This matters directly here, because finite domains are not a nuisance for me. They are the intended setting.

Later work partially repairs the situation. Arnborg and Sjödin treat finite models and argue for additional principles under which Bayesian rules can be recovered or justified [AS00; AS01; AS99]. Van Horn reconstructs the assumptions needed for a Cox-style theorem more carefully, and Snow defends finite-domain reasonableness from a related angle [Sno98; Van03].

The lesson is not that Cox is refuted; it is that Cox-style derivations are not magic. They require domain richness, regularity assumptions and a precise account of what counts as a permissible plausibility assignment. For this thesis, that is not an embarrassment but an instruction: if probability is to be rebuilt constructively, the needed assumptions must be stated inside the constructive setting rather than waved in from the classical background.

Reasonableness

A subtler worry is whether the assumptions are reasonable to begin with. An assumption can be consistent and still be philosophically unattractive. Shafer objects, among other things, to the idea that reasonable expectation must always be represented by a single real number [Sha04], and Van Horn’s reconstruction — although technically precise — does not by itself make every assumption compelling [Van03].

Clayton and Waddington are especially important here because they try to make the Coxian assumptions less mysterious. Instead of simply postulating technical conditions such as universality and strict monotonicity on large domains, they derive much of the needed structure from urn-like examples, symmetry and invariance [CW17]. The full details belong to their paper; for the storyline here, the relevant point is that the assumptions can be read concretely:

- *finite urn existence*: symmetric urns of every positive finite size are available;
- *logical equivalence*: equivalent decidable events receive equal plausibility;
- *qualitative coherence*: equality and strict comparison of plausibility behave like an ordered equivalence;
- *relabelling invariance*: permuting finite urn labels does not change plausibility;
- *strict monotonicity*: in a symmetric urn, more favourable cells means greater plausibility;
- *refinement invariance*: splitting every cell uniformly preserves plausibility;
- *product urn compatibility*: rectangular events in product urns behave like products.

Just as important is what the constructive proof below does not use: it does not rely on a completed continuum of plausibility values, on a universality or density condition over arbitrary domains, or on a complement rule for arbitrary propositions. Those stronger assumptions are part of the classical Cox–Van Horn route and of Clayton and Waddington’s repair of it. The reconstruction in this thesis keeps the finite urn principles, changes the order of derivation, and applies complementation only inside decidable event algebras.

I therefore do not take Jaynes' desiderata as sacred tablets. I treat them as a programme to be reconstructed, in a version closer in spirit to Clayton and Waddington: finite, symmetric, explicit and constructive. That makes the resulting probability types less grand, but also less evasive.

Universality

Then there is universality. Jaynes sometimes writes as if probability theory is the universal logic of science. Colyvan gives a useful warning against this overreach: Cox-style arguments presuppose a background logic, and the choice of that logic matters [Col04]. A theorem about plausible inference inside classical propositional logic is not automatically a theorem about all forms of uncertainty, all domains of discourse or all scientific reasoning.

The point becomes clearer once one distinguishes ontological and epistemological questions. An ontological logic asks what is the case; an epistemological logic asks what is known, knowable or justified from a given position [Hof23; SN25]. The question "Does God exist?" illustrates the contrast nicely. An ontological stance pushes towards truth or falsity; an epistemological stance may answer, with perfect seriousness, that the question is not settled from here.

Jaynes' framework is epistemological in temperament but classical in formal clothing. He wants probabilities to describe states of knowledge, yet his derivation usually assumes a classical propositional background in which every proposition is already the kind of thing that is true or false. That creates friction. Vagueness, fictional discourse, undecided mathematics, limited interest and forgotten information are not well represented by simply assigning a number to a fixed classical proposition.

The response is not to abandon probability, but to abandon the inflated claim that probability, derived once in one background logic, is the final logic of everything. The present thesis uses a constructive and epistemological setting precisely because Jaynes' slogan is better than Jaynes' formal environment. Probability remains central, but it is placed inside a richer account of contexts, modes and epistemic states.

Subjectivity

Jaynes calls probability subjective, but the subject is often strangely absent from the formalism. A prior may vary from one agent to another, but the structure of the agent's position is not itself represented. What questions are in scope? What information has been forgotten? Which distinctions are of no interest? Which alternative situations are being imagined? These are not merely psychological decorations. They affect what can be asked, what can be inferred and what counts as the same problem.

This is where the usual contrast between frequentism and Bayesianism is too crude. Jaynes rightly objects to treating probability as a property of objects in isolation. A coin does not carry a metaphysical label saying one half. A physical procedure, a state of information and a model of the experiment are doing work. But saying this in prose does not yet build subjectivity into the mathematics.

Interest is the simplest example. Classical logic lets every proposition sit in the same universe of discourse. Real agents do not work like that. A proposition

3. Second movement: Andante con moto

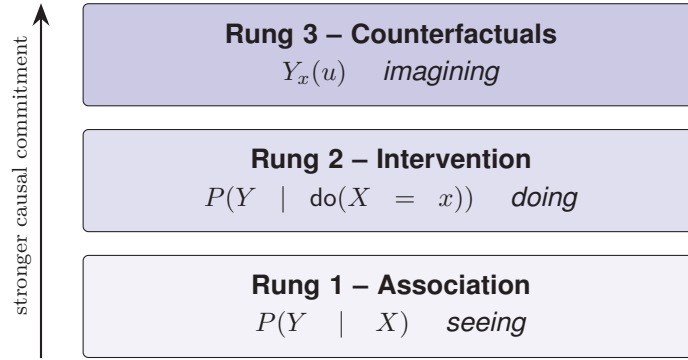


Figure 3.1: Pearl’s ladder of causation: each rung answers a strictly stronger class of question [PM18].

may be true, false, unknown, irrelevant, forgotten or outside the present domain of concern. One may not care whether Achilles was considered the fastest among mortals, while caring deeply about a medical diagnosis that shares only a small part of the same scientific vocabulary. A logic of subjectivity should be able to represent this asymmetry without pretending that every statement is waiting patiently for a probability.

My constructive response is to make epistemic position explicit. Contexts determine what is available; modes determine which epistemic state one occupies; transitions between modes represent learning, forgetting, hypothesising, intervening and changing the relevant domain. The formal vocabulary for this is fixed in Definitions A.1.2, A.2.1 and A.2.6. Subjectivity is then no longer a label placed on a Bayesian calculus after the fact. It becomes part of the machinery.

Causality

Last comes causality. Jaynes’ probability theory can express statistical dependence, but dependence is not causation. Wet grass and rain can be correlated in either direction; a barometer and a storm can be tightly associated without the barometer causing the storm. A theory that treats probability as the whole logic of science still needs to explain how intervention, mechanism and counterfactual structure enter the picture.

Pearl’s framework supplies the missing formal technology. It distinguishes three levels of causal inquiry. At the bottom sits association: observational questions such as $P(Y \mid X)$, where one asks how variables covary in the observed distribution. Above that comes intervention, with questions such as $P(Y \mid \text{do}(X = x))$, asking what would happen if X were externally set to x . At the top lives counterfactual reasoning — questions such as $Y_x(u)$, where one asks what the outcome for a particular unit u would have been under a different intervention [Pea09; PM18]. The corresponding definitions are collected in Definitions A.3.5, A.3.6, A.3.24, A.3.30 and A.3.35. These three levels are not stylistic variations. They require different assumptions, and they answer different questions.

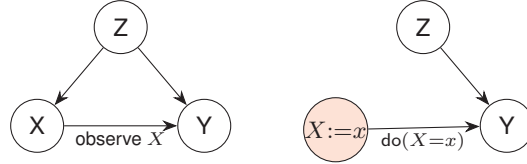


Figure 3.2: Observing X leaves the confounding path through Z intact; intervening on X severs its incoming edges (Definition A.3.25).

At the first two levels, causal Bayesian networks and causal graphs are enough for many purposes. A directed acyclic graph represents causal structure, and the causal Markov condition connects that graph to conditional independence: each variable is independent of its non-descendants once its parents are known. The graphical vocabulary used here is fixed in Definitions A.3.13 to A.3.16, A.3.18, A.3.19 and A.3.21. At the intervention level, Pearl introduces the do-operator to mark the difference between observing $X = x$ and setting $X = x$. Observing leaves the ordinary causal mechanism in place; intervening replaces the mechanism for X by an imposed value. This is exactly the distinction Jaynes’ probabilistic calculus cannot express by itself.

Do-calculus then gives a disciplined system for manipulating such interventional expressions: under graphically specified conditions, observations and actions may be inserted, deleted or exchanged, but only when the relevant graph-separation and graph-mutilation conditions license it. The exact rules are collected in Definitions A.3.23, A.3.25 and A.3.26. Completeness and identification results show that this is not merely convenient notation but a complete inferential system for a large class of causal questions [HV06; Pea12; SP06; TS09].

At the counterfactual level, structural causal models do the heavier work, through the standard abduction–action–prediction procedure made precise in Definitions A.3.28 and A.3.29. This is why Pearl’s theory equips an inferential system with something stronger than prediction: a formal language for asking not only what happened, but why it happened, and what would have happened under a different intervention.

I use counterfactuals here in this orienting Pearlian sense. The formal transfer proved later is narrower: it covers ordinary and conditional interventional distributions in the finite recursive fragment, not a general theory of counterfactual identification.

Pearl’s account is powerful, but it is not derived from the Jaynesian foundations. It is what one would call — charitably — engineering mathematics: precise, useful, and motivated by real inferential failures. What it does less fully is explain why causal reasoning is indispensable to human scientific practice.

The short version of that defence runs as follows. Human beings reason causally almost before they reason formally, and science can be described, at least in part, as the disciplined extension of this causal grammar [Hac83]. That extension is communal, historical and bounded: causal judgement is stabilised inside communities, inherited styles of thought and shared techniques [Fle79; Hui49], and the constructions retain an ironic, provisional character that should not be confused with reality itself [Era71; Kie80; Kie83; Kie89]. The middle position is therefore Feyerabend without “anything goes” [Fey75], Hume

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without metaphysical guarantees [Hum75; Hum78], and Kant without forcing the world/cognition dichotomy [Kan98]: causal talk is scientifically indispensable because science is a human practice of shared orientation, intervention and explanation. The full version of this argument is in Section B.1; what the main text needs from it is the conclusion that any formal theory of mathematical inference for science must give causal reasoning a home.

I therefore treat Pearl neither as an opponent nor as a mere add-on. Pearl-style causal models will be embedded in the constructive, epistemic setting developed later, which buys one extra freedom: causal structures need not be treated as eternally fixed. If epistemic states can change, and if modes can represent such states, then causal models themselves can become part of what changes when an agent learns, intervenes or imagines an alternative world.

CHAPTER 4

Third movement: Adagio

“A common mistake that people make when trying to design something completely foolproof is to underestimate the ingenuity of complete fools”

Douglas Adams in *Mostly Harmless*

4.1 Different types of logic

If Jaynes’ universality claim fails, the next question is not which logic wins. The next question is which logic is appropriate for which task. Classical logic is still magnificent when its assumptions fit: propositions are true or false; contradiction is fatal; excluded middle is accepted; and the inferential world is crisp. But scientific reasoning is not always crisp, and artificial intelligence rarely has the luxury of pretending otherwise.

Modal logic adds structure by allowing truth to vary across worlds, states or situations. In Kripke-style semantics, one does not merely ask whether a proposition is true; one asks whether it is true at a world, and how worlds are related. This is already close to the language needed later. An epistemic state can be treated like a world; moving between states can represent learning, forgetting or intervention; modal structure can record which movements are legitimate.

Fuzzy and many-valued logics respond to a different discomfort. Some predicates do not behave as if they have sharp borders. “Young”, “tall”, “similar” and “healthy” are not always best understood as binary switches. Probability is sometimes misused here: a degree of truth is not the same thing as a degree of belief. Van Horn makes this distinction important in the Coxian setting, because plausible inference concerns uncertainty about propositions, not the replacement of propositions by vague gradients [Van03].

A few nearby candidates matter as contrasts. Epistemic logic makes knowledge explicit, but in its usual modal form it is often too propositional and external for the dependent scientific structures needed later. Temporal and dynamic logics are useful for reasoning about change, action and programs, but they do not by themselves give the constructive judgemental discipline required here. Linear and other substructural logics track resources and structural rules, which is attractive when assumptions should not be copied or discarded silently. These neighbours explain the choice rather than replace it: the thesis needs a constructive, dependent and modal setting, not merely one more propositional logic with extra operators.

The contrast between ontological and epistemological logic is more important for this thesis. Ontological logic speaks as if it were describing what is. Epistemological logic speaks from a position: what is known, what is justified, what is in scope, what has been constructed [Hof23; SN25]. Jaynes’ language is deeply epistemological, but his formal derivation remains too close to a fixed classical background. That is the mismatch this thesis repairs.

Intuitionism is the relevant constructive tradition. It does not accept a proposition merely because its negation cannot be proved; it asks for construction. In this setting, the law of excluded middle is not a default rule. This does not make reasoning weaker in the pejorative sense. It makes reasoning more explicit. A claim is not a decorative token of truth; it is something that must be witnessed, built or derived [DP24].

The philosophical stance I take here is therefore agnostic about ontology and constructive about epistemology. I do not decide what reality is made of; I ask how finite agents, finite measurements and finite formal systems should represent and transform what is available to them.

The motivation comes from physics, where finiteness has practical force. Measurements always carry finite information; a measured value is a finite

Logic	Type theory
proposition	type
proof	program / term
$A \wedge B$	$A \times B$ (pair)
$A \vee B$	$A + B$ (sum)
$A \Rightarrow B$	$A \rightarrow B$ (function)
$\top, \perp$	1 , 0 (unit, empty)
$\forall x. P(x)$	Π -type
$\exists x. P(x)$	Σ -type

Figure 4.1: The Curry–Howard correspondence: proofs are programs, propositions are types.

construction with precision, procedure, instrument and context, not a completed real number handed down from above [Bar26; DG19; Gis20]. Spekkens’ toy theory shows that many features advertised as distinctively quantum — interference-like behaviour, no-cloning, no-broadcasting, epistemic update — arise from restrictions on knowledge alone [HS10; Spe07]. Pusey, Barrett and Rudolph then warn that one cannot keep a completely definite underlying ontic state and still expect such an epistemic view to do all the work [PBR12]. The natural conclusion is the one taken here: treat formal theories as epistemological vehicles for finite observations, refuse to fix a hidden ontology, and let the constructive reading carry the finiteness that physics already enforces. The fuller argument, including the comparison with Gisin’s intuitionism and the residual quantum phenomena it does not yet capture, is collected in Section B.2.

4.2 Type theory

Type theory is the natural technical home for this stance. It connects mathematics to computation through the Curry–Howard correspondence: propositions can be read as types, and proofs as terms inhabiting those types. To prove a proposition is, in this sense, to construct something. That is exactly the kind of discipline Jaynes wanted, but with a more suitable constructive engine.

A type may be understood as a domain of construction. A term $a : A$ says that a is an inhabitant of the type A . A context records the assumptions and variables currently available; the formal judgemental vocabulary is fixed in Definitions A.1.1 and A.1.2, where types and terms are presented as the basic judgemental forms. For example, a judgement of the form

$$\Gamma \vdash a : A$$

says that, in context Γ , the term a has type A . This already brings subjectivity nearer: what can be said depends on the context from which it is said.

Functions are terms that transform inhabitants of one type into inhabitants of another. Lambda abstraction constructs such functions, and application uses them. Substitution explains how a term depending on a variable changes when that variable is replaced by a particular term. Equality is not a purely metaphysical relation; it has computational content, for instance through beta-reduction, where applying a lambda abstraction to an argument computes by

4. Third movement: Adagio

substituting the argument into the body. Definitions A.1.4, A.1.5 and A.1.7 give the formal versions, with ordinary functions appearing as the non-dependent case.

Product types and sum types give the basic logical operations. A product type represents having both pieces of information; a sum type represents having one alternative or another, together with evidence of which alternative holds. Their dependent forms, which include the ordinary product and sum as non-dependent cases, are fixed in Definitions A.1.6 to A.1.8. This is already more explicit than ordinary informal mathematics. It is not enough to say that a disjunction is true. One must know which side has been constructed. This is the propositions-as-types discipline of Definitions A.1.19 and A.1.21.

Example 4.2.1 (A tenure-track offer as a type). Suppose a hiring committee is considering one candidate, Mira, for a tenure-track position. Classically, the proposition

$$\text{Offer}(\text{Mira})$$

is immediately subject to excluded middle:

$$\text{Offer}(\text{Mira}) \vee \neg \text{Offer}(\text{Mira}).$$

Moreover, classical reasoning validates double-negation elimination:

$$\neg \neg \text{Offer}(\text{Mira}) \rightarrow \text{Offer}(\text{Mira}).$$

In the propositions-as-types reading, by contrast, $\text{Offer}(\text{Mira})$ is a type. A proof that Mira receives the offer is a term

$$\ell : \text{Offer}(\text{Mira}),$$

which may be imagined as the constructed offer letter or the verified decision record. From such a witness one can always construct double negation:

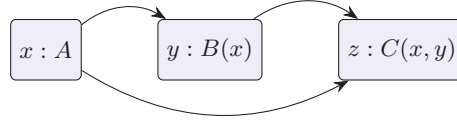
$$\lambda \ell. \lambda n. n(\ell) : \text{Offer}(\text{Mira}) \rightarrow \neg \neg \text{Offer}(\text{Mira}).$$

The converse is not available in Martin-Löf-style type theory without adding a classical principle. Knowing that a refutation of the offer would lead to contradiction is not the same as having the offer witness itself. This is the small formal difference that later becomes important for finite inspection: a file may be proved accepted, proved rejected or remain unresolved by the procedure at hand.

Type theory is computational by nature, but not automatically epistemological in the rich sense needed here. Plain type theory can record available variables and proofs, but the later theory must also distinguish epistemic states and transitions between them. For that, Martin-Löf-style type theory supplies the constructive core, and modal dependent type theory supplies the machinery for modes.

4.3 Martin-Löf-style type theory

Martin-Löf-style type theory, or intuitionistic type theory, develops type theory as a constructive foundation for mathematics [DP24; Mar84; NPS90]. Its



order matters: later entries may depend on earlier ones

Figure 4.2: A context $\Gamma = (x : A, y : B(x), z : C(x, y))$ as an ordered telescope of dependencies — the ordering that later mirrors a causal directed acyclic graph.

central habit is judgemental. One does not begin with a universe of sets and membership; one begins with judgements about contexts, types, terms and equality.

Dependent types are the decisive extension. A type may depend on a term. If A is a type and $B(x)$ is a type depending on $x : A$, then one can form dependent function types and dependent pair types, as in Definitions A.1.7 and A.1.8. The former represent functions that return an output type depending on the input; the latter represent pairs in which the second component depends on the first. This is crucial for scientific modelling, where the type of a later quantity often depends on earlier choices, measurements or assumptions.

Identity types internalise equality. They allow one to reason about when two terms of a type are equal, but in intensional versions of the theory this equality is not collapsed into external sameness. This matters later because the thesis needs to keep track of how things are known to be equal, not merely whether they are extensionally indistinguishable. The notation used here is fixed in Definition A.1.9.

Inductive types allow familiar mathematical objects to be generated by constructors: natural numbers from zero and successor, finite types from explicit labels, lists from empty lists and cons, and so on. Universes organise types into levels, preventing the system from treating the collection of all types too naively. The usual judgemental pattern is formation, introduction, elimination and computation: how a type is formed; how its inhabitants are introduced; how they are used; and how the resulting computation behaves. Definitions A.1.10 to A.1.12 and A.1.22 state these notions formally.

The law of excluded middle is not built in. This is not a bug to be patched later. It is part of the constructive discipline. If excluded middle is needed for a particular argument, it must be introduced as an assumption and paid for openly; the metatheoretic non-derivability claim used here is made explicit in Proposition B.3.1. This is exactly the attitude required by the mathematical manifesto.

There are many flavours of Martin-Löf-style type theory: intensional or extensional equality, different universe hierarchies, different inductive types, different structural rules, and different accounts of substitution and definitional equality. For the later modal work, I follow the style used by Gratzer and collaborators, because it provides a direct route from dependent type theory to modes, modalities and context-sensitive reasoning [Gra+21; Gra23; GSB19].

More precisely, I use a scoped Gratzer-style fragment. The modal notation follows the multimodal dependent type-theoretic architecture of modes,

4. Third movement: Adagio

modalities, modal context structure and modal type formers to organise judgemental reasoning across epistemic states. I do not reproduce the full proof theory, normalisation theorem or type-checking algorithm of Gratzer's systems. Identity types are read in the ordinary intensional way when they are mentioned, while extensional comparisons are used only when explicitly marked as metatheoretic, as with finite probability records or semantic countermodels.

This gives the promised mathematical setting. It is constructive enough to resist infinite paradoxing by default; computational enough to connect with proof checking; dependent enough to model scientific structure; and flexible enough, once modalised, to represent changes in epistemic state.

CHAPTER 5

Fourth movement: Allegro maestoso

*“Scientists have calculated that the chance of anything so
patently absurd actually existing are millions to one.
But magicians have calculated that million-to-one chances crop
up nine times out of ten”*

Terry Pratchett in *Mort*

5. Fourth movement: Allegro maestoso

The previous movements left three repairs to make. Subjectivity has to become part of the formal language rather than a slogan about priors. Probability has to be reconstructed over finite, decidable objects rather than arbitrary classical propositions. Causality has to enter as intervention and model change, not merely as statistical dependence. The Fourth movement takes these in order: contexts and modalities first, probability types next, and finite structural causal models last.

5.1 Context manipulation in Martin-Löf-style type theory

Contexts are the quiet machinery of subjectivity. A context records what is currently available: variables, assumptions, dependencies and previously constructed terms. In ordinary prose one says “given what we know”. In type theory one writes a context; Definitions A.1.2 and A.1.3 spell out the formal fragment used here.

Several operations on contexts have immediate epistemic readings. Extending a context by a new variable corresponds to learning or introducing a new assumption. Allowing a later variable to depend on an earlier one records directed dependence. Swapping variables is allowed only when dependencies permit it, which is already a small discipline against pretending that order never matters. Deleting an unused variable corresponds to forgetting or ignoring something that nothing else depends on.

These operations give a first, austere account of subjectivity. What is in scope can be used. What is out of scope cannot even be formed as a judgement. If a question depends on a variable absent from the current context, the appropriate answer is not false but unavailable — not in this context, not from here, not with this information.

This also gives a first account of intervention. To intervene is not merely to observe a value; it is to set a variable and sever or alter the dependencies that normally determine it. In a context, that means changing the dependency structure. To hypothesise is similar but more theatrical: one moves to an alternative context, changes something there, and asks what follows under that altered arrangement.

Context manipulation alone is not yet the full theory. It gives the raw operations, but not the classification of epistemic states as such. For that, the thesis treats certain context changes as transitions between modes. The point is not that ordinary type theory could not perform the operations. The point is that subjectivity becomes clearer when these operations are marked as movements between states of knowing.

5.2 Modalities inside Martin-Löf-style type theory

Modal dependent type theory supplies that marking. Before the apparatus, the motivation: ordinary dependent type theory already captures one half of subjectivity — what is in scope — by means of contexts. It does not capture the other half. Two agents may share the same context of available variables and still differ in which questions they treat as live, which mechanisms they take for granted, and which alternative situations they entertain. Plain context manipulation cannot mark these differences: if I add and then remove a variable

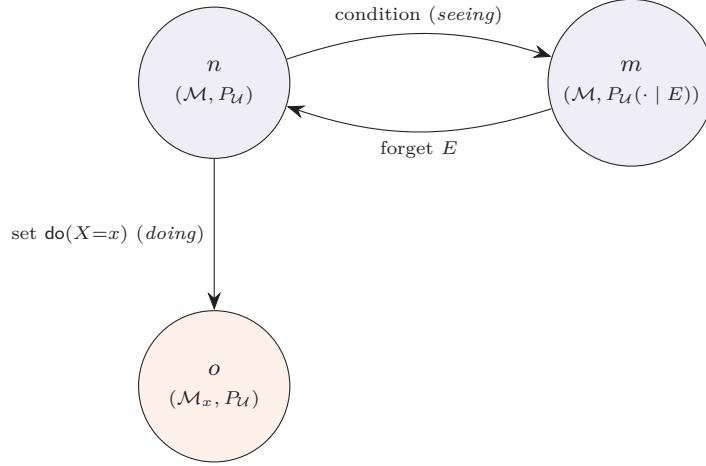


Figure 5.1: Modes as epistemic states; modalities as admissible transitions. Conditioning updates the exogenous distribution (abduction); setting replaces a mechanism (intervention). The fragment is introduction-only: modalities *mark* admissible moves rather than inspect the context (Definition A.2.4 and Remark A.2.8).

in the same context, the result is judgementally the same as never having added it, even though epistemically I have moved through a state where that variable was in scope. A mode theory is the smallest extra structure that records such epistemic location. It declares a small collection of named epistemic states (*modes*), together with the admissible transitions between them (*modalities*), and indexes every judgement by the mode at which it is made. The price is one extra index on every judgement; the benefit is that learning, forgetting, hypothesising and intervening become formal moves rather than informal annotations.

In Gratzer’s setting, one works not only with contexts and types, but with modes and modalities between modes [Gra+21; Gra23; GSB19]. The formal fragment is fixed in Definitions A.2.1 to A.2.6. A mode can be read, for present purposes, as an epistemic state. A modality can be read as a structured way of moving from one epistemic state to another.

This makes the earlier operations explicit. Learning, forgetting, changing attention, introducing a hypothesis and performing an intervention can all be treated as special transitions. Some transitions preserve more structure than others. Some are reversible; some are not. Some forget information; some add it. Some preserve causal mechanisms; some deliberately break them.

The modal apparatus also prevents a familiar confusion. If two agents reason in different epistemic states, it is not enough to say that they assign different probabilities. They may not even share the same context of questions. They may remember different facts, care about different distinctions or operate under different causal assumptions. A subjective theory should represent this before it assigns numbers.

Pfenning and Davies’ judgemental reconstruction of modal logic is useful background here because it shows how modal reasoning can be understood through judgements and contexts rather than merely through external possible-

5. Fourth movement: Allegro maestoso

world semantics [PD01]. Gratzer’s work then brings this style into dependent type theory in a form suited to the later construction.

For the purposes of this thesis, the technical ambition is limited. I do not need every possible modal type theory. I need enough structure to say: an epistemic state is a mode; admissible changes of epistemic state are modalities; and the formal language can track which judgements are available before and after such changes.

One restriction is essential. In Gratzer-style multimodal type theory, the mode theory is fixed independently of the local context: modes, modalities, composition and 2-cells are part of the ambient modal structure, not operations that inspect the current telescope and then decide what to do. Therefore the epistemic modalities used here should be read as *markers of admissible transitions*, not as context-dependent programs that themselves add, delete or rewrite variables. The actual work of learning, forgetting, conditioning, relating, setting or unsetting is an ordinary transformation of the finite context or structural-causal-model records at the endpoint modes. The modality records that this transformation is an admissible move between those modes. If two such moves are intended to cancel, this must be represented by an equation or 2-cell in the chosen mode theory, or proved propositionally as a separate record-level sanity theorem. The thesis implements this disciplined marking role for a small finite transition layer: fixed transition labels are paired with explicit endpoint records, preservation of well-formedness is checked, and the basic cancellation cases are proved. A full Gratzer syntactic formalisation and a complete algebra of epistemic transition operations remain outside the scope of the thesis.

Concretely, no result below depends on the full modal elimination rule, 2-cell action on terms or the full inter-mode substitution discipline. Those stronger tools remain in the cited modal metatheory; the main construction uses only the fragment needed to keep epistemic position visible. The finite implementation target for this reading is stated in Definition B.5.8; its preservation, conditioning and cancellation facts are Propositions B.5.9 to B.5.11, with twin networks treated as finite composites in Remark B.5.12.

5.3 Probability types

The next task is to place probability inside this constructive setting. The guiding idea is simple: a probability assignment should be an explicit object attached to a finite, decidable domain. Probability should not float above the formalism as an external numerical gloss. It should be represented by a type whose inhabitants carry the required structure, as in Definitions A.3.1, A.3.4 and A.3.7.

In Martin-Löf-style type theory this restriction is not cosmetic. An arbitrary proposition A does not automatically split the world into A and $\neg A$, because excluded middle is not available. The probabilistic object used here is therefore not a measure on all propositions. It is a probability assignment over decidable events in a finite sample type. When the question concerns an undecided proposition A , the event space is shifted to resolution status: proof found, refutation found, or neither found by the chosen finite inspection procedure.

These resolution-status events are decidable even when A itself is not; the definitions are fixed in Definitions A.3.2 and A.3.3.

For the tenure-track proposition of Example 4.2.1, this means that probability is not assigned directly to an arbitrary unresolved type as if excluded middle had already settled it. Instead, a finite procedure may inspect the file and return one of three decidable resolution statuses: an offer witness has been found, a rejection witness has been found, or the procedure has not resolved the file. Probability can then be assigned to these resolution-status events, while the distinction between unresolved and refuted remains intact.

Clayton and Waddington provide the bridge from Cox-style reasoning to a finite, constructive reconstruction [CW17]. The rules named in Section 3.2 are made formal in Definition A.3.8, but the order of use is deliberately different from the original Cox–Van Horn–Clayton–Waddington route. First, finite urn existence, logical equivalence, qualitative coherence, relabelling invariance, strict monotonicity and refinement invariance force the finite urn plausibilities to be represented by the rational ratio k/N . Product urn compatibility is then used only after this rescaling step, so rectangular product events acquire the expected multiplicative form. Universality, continuum regularity and arbitrary-proposition complementation are not used in the constructive proof.

Rule or ingredient	What it yields here	Where
Finite resolution-status events	Replaces probability on arbitrary unresolved propositions by a decidable three-way event partition.	Proposition B.4.1 and Corollary B.4.9
Finite urn existence, logical equivalence, qualitative coherence, relabelling invariance, strict monotonicity and refinement invariance	Gives the order-preserving rescaling $U_N(k) \mapsto k/N$; decidable urn complements then satisfy the finite complement rule.	Proposition B.4.2
Product urn compatibility	Gives multiplication for rectangular product events after the ratio rescaling, so the product rule is recovered after the sum/ratio step.	Proposition B.4.3
Record–urn equivalence	Shows that urn presentations and common-denominator finite rational records agree on every decidable event.	Proposition B.4.5
Finite event calculus	Proves non-negativity, normalisation and finite additivity, with monotonicity and inclusion–exclusion as corollaries.	Theorem B.4.6 and Corollaries B.4.7 and B.4.8
Conditional probability	Proves Bayes’ theorem and Bayesian update for decidable events with positive conditioning probability.	Definition A.3.9, Theorem B.4.10, and Propositions B.4.11 and B.4.12

Table 5.1: Route through the finite probability reconstruction used in this thesis.

5. Fourth movement: Allegro maestoso

The attraction of this route is that it is concrete. It does not begin by demanding a completed real interval of plausibility values and then asking the reader to accept technical regularity conditions. It begins with finite symmetric problems and shows how the probability scale is recovered. This is exactly the direction needed here: from finite construction to probabilistic calculus, not from ideal continuum to finite application.

In the formal development, the primitive sample space is the finite urn $\text{Fin}(N)$ rather than the boolean type. A probability type is generated by a finite draw map out of such an urn; equivalently, for finite value types, it induces a finite rational mass function. This is the familiar finite distribution construction in its concrete urn presentation. Non-negativity, normalisation and finite additivity are then theorems about these objects, and Bayes' theorem becomes a theorem about decidable events rather than a free-floating formula. The proof route is summarised in Table 5.1 and carried out in Section B.4. The main point is conceptual: probability is made into something constructible.

5.4 Structural causal models in Martin-Löf-style type theory

Pearl-style structural causal models can now be rebuilt using the same ingredients. A finite structural causal model consists of variables, exogenous noise, structural functions and a finite rational probability distribution over the exogenous variables [Pea09; PM18]. The formal definition used here is Definitions A.3.5, A.3.6 and A.3.29. In the constructive setting, each component can be represented explicitly: variables as finite types or indexed families, structural functions as terms, exogenous uncertainty as a probability type, and interventions as transformations of the structural functions.

The ordered shape of a type-theoretic context is doing real work here. After choosing a topological order for a finite acyclic causal graph, the structural term for V_i is typed only in a context containing the exogenous inputs and the earlier endogenous variables $V_1, \dots, V_{i-1}$. Later declarations may depend on earlier ones; earlier declarations cannot depend on later ones. Thus acyclicity is not merely an external graph condition in the restricted fragment: it is reflected in what the structural term is allowed to mention. This is the ordered-telescope reading of finite recursive structural causal models.

To see what such a model buys us, it is worth running a concrete example before the formal machinery does its work. The natural choice is to revisit Mira's situation from Example 4.2.1, but now from the hiring committee's side. Officially, the committee claims to select on research dossier strength alone; in practice, candidates from prestigious institutions tend to do better, fashionable topics tend to attract committee members with matching expertise, and funding lines have their own seasonality. The interesting question is not whether such a committee is biased — almost every committee is, in some direction — but whether the bias visible at the output reflects a stable causal mechanism or merely a hidden common cause. The model below is small enough to inspect line by line and rich enough to exhibit the standard causal phenomena, and it sits beneath Example 4.2.1: the type-level proposition $\text{Offer}(\text{Mira})$ now becomes the Boolean endogenous variable Off , and the three resolution statuses *proved*, *refuted* and *unresolved* from Definition A.3.3 become decidable events on the joint exogenous assignment.

Example 5.4.1 (Mira’s tenure-track search, as a structural causal model). Returning to Mira from Example 4.2.1, the running example is now a stylised tenure-track hiring decision viewed from the search committee’s side. The finite probabilities below should be read as a toy model for a candidate already under serious committee consideration, not as empirical base rates for all tenure-track applications. All endogenous variables are Boolean. Let Pr denote institutional prestige, Qu research dossier strength, Top topic fashionability, Com committee expertise, Fit committee fit, Short being shortlisted, Fund funding-line availability and Off receiving the tenure-track offer; the previous proposition $\text{Offer}(\text{Mira})$ is the event $\text{Off} = 1$ on Mira’s joint exogenous assignment. One finite structural causal model is given by independent Boolean exogenous variables $U_B, U_{\text{Pr}}, U_{\text{Qu}}, U_{\text{Top}}, U_{\text{Com}}, U_{\text{Fund}}$ and structural equations

$$\begin{aligned} \text{Pr} &:= U_B \vee U_{\text{Pr}}, & \text{Qu} &:= U_B \vee U_{\text{Qu}}, \\ \text{Top} &:= U_{\text{Top}}, & \text{Com} &:= U_{\text{Com}}, \\ \text{Fit} &:= \text{Top} \wedge \text{Com}, & \text{Short} &:= \text{Qu} \wedge (\text{Pr} \vee \text{Fit}), \\ \text{Fund} &:= \text{Top} \vee U_{\text{Fund}}, & \text{Off} &:= \text{Short} \wedge \text{Fund}. \end{aligned}$$

The directed structure contains

$$\begin{aligned} U_B &\rightarrow \text{Pr}, & U_B &\rightarrow \text{Qu}, & \text{Top} &\rightarrow \text{Fit} \leftarrow \text{Com}, \\ \text{Pr} &\rightarrow \text{Short} \leftarrow \text{Fit}, & \text{Qu} &\rightarrow \text{Short}, & \text{Top} &\rightarrow \text{Fund}, \\ & & \text{Short} &\rightarrow \text{Off} \leftarrow \text{Fund}. \end{aligned}$$

If U_B is not observed, the observed semi-Markovian projection (in the sense of Definition B.6.9) has the bidirected edge $\text{Pr} \leftrightarrow \text{Qu}$. Thus the example contains an unobserved common cause, colliders, ordinary directed chains and the graph predicates used later for d-separation and c-components. With the finite rational probabilities in Example A.4.1, one obtains, for instance,

$$P(\text{Qu} = 1 \mid \text{Pr} = 1) = \frac{3}{4}, \quad P(\text{Qu} = 1 \mid \text{do}(\text{Pr} = 1)) = \frac{1}{2}.$$

Seeing prestige gives evidence about the hidden background variable that also affects dossier strength. Setting prestige by intervention does not change that background variable and therefore does not by itself improve the dossier.

The same finite model also supports a representative counterfactual calculation. Suppose the actual evidence is that the candidate had no prestige, had a strong dossier, and received no offer:

$$E := \{\text{Pr} = 0, \text{Qu} = 1, \text{Off} = 0\}.$$

After conditioning the exogenous distribution on E , replacing the equation for Pr by the constant value 1, and evaluating the offer variable in the intervened model, one obtains

$$P(\text{Off}_{\text{Pr}=1} = 1 \mid E) = \frac{1}{2}.$$

This is a genuine abduction–action–prediction calculation for the concrete finite model, checked again in Example A.4.1; it is not a general counterfactual-identification theorem from a graph alone.

5. Fourth movement: Allegro maestoso

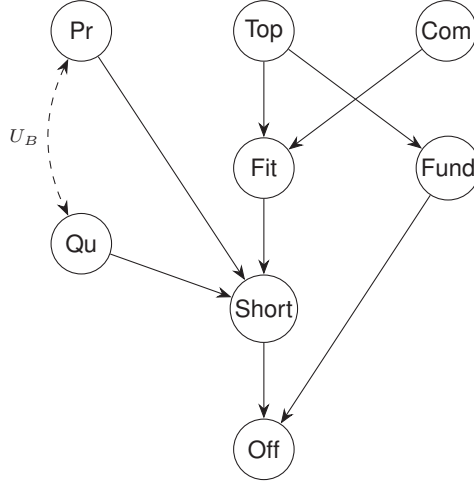


Figure 5.2: Mira's tenure-track structural causal model. The triples $\text{Top} \rightarrow \text{Fit} \leftarrow \text{Com}$, $\text{Pr} \rightarrow \text{Short} \leftarrow \text{Fit}$ and $\text{Short} \rightarrow \text{Off} \leftarrow \text{Fund}$ are colliders; the dashed edge is the latent common cause U_B inducing $\text{Pr} \leftrightarrow \text{Qu}$ (Example 5.4.1).

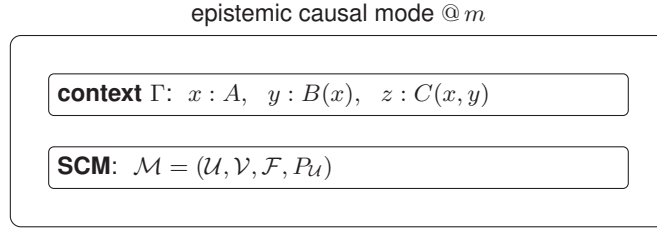


Figure 5.3: An epistemic causal mode bundles an ordered context with a finite structural causal model; moving between modes changes the model, the available information, or which interventions are in scope (Definition A.3.37).

The modal reading gives the construction its epistemic interpretation. A causal model is more than a graph in the sky. It is a state of causal knowledge: what variables are distinguished, what mechanisms are assumed, what interventions are meaningful, and which counterfactual questions can be asked. Moving from one causal model to another is therefore a change of epistemic state. I will call such a thesis-specific state an epistemic causal mode; Definition A.3.37 fixes this local term. Modes can represent these states; modalities can represent admissible transitions between them.

In the tenure-track example this means that the model is not treated as external background prose. Schematically, at the tenure-track mode one has a judgement

$$\Gamma_{\text{ten}} \vdash \mathcal{M}_{\text{ten}} : \text{FinSCM}@m_{\text{ten}},$$

where FinSCM is the restricted finite type-theoretic structural-causal-model fragment of Definition B.5.1. An intervention such as $\text{do}(\text{Pr} = 1)$ moves to an epistemic causal mode whose structural-causal-model record is $\mathcal{M}_{\text{ten}, \text{Pr}=1}$, obtained by replacing the structural equation for Pr by the constant value 1

5.4. Structural causal models in Martin-Löf-style type theory

and leaving the other mechanisms intact. A calculation such as

$$\Gamma_{\text{ten}} \vdash P_{\mathcal{M}_{\text{ten}, \text{Pr}:1}}(\text{Qu} = 1) = \frac{1}{2} @ m_{\text{ten}}$$

is therefore a judgement about a finite probability expression inside the current epistemic causal mode. The same expression can be transported to the encoded type-theoretic structural-causal-model record by the correspondence theorem.

This makes Pearl’s static formalism more flexible without discarding it. In an ordinary query, a structural causal model is held fixed; but scientific reasoning often changes the model — a mechanism is discovered, a variable is split into two, an intervention is reclassified, a hidden confounder is introduced, a previous assumption is withdrawn. Treating causal models as epistemic modes allows such changes to be part of the formal story.

Counterfactual reasoning fits this modal reading compositionally, not by adding a primitive “twin network” modality. In the finite layer below, the abduction step is represented by a **condition** transition: the exogenous distribution is replaced by a posterior distribution, either explicitly supplied with a normalisation proof or constructed by Bayesian conditioning from a decidable evidence event of positive probability. The posterior may be a point mass when the evidence fixes the exogenous assignment. The action step is represented by **set**, which changes the selected endogenous structural equation and leaves the exogenous distribution alone. Prediction is ordinary evaluation in the resulting finite record. A twin-network construction is then a specific finite composite: learn the copied endogenous variables, relate the actual and copied sectors to the shared exogenous references, set the counterfactual intervention in the copied sector, and evaluate the resulting record. This explains why no separate twin-network modality is introduced. It also marks the current limit: the worked tenure-track counterfactual is a model-level abduction–action–prediction calculation, while a full counterfactual identification theorem would still require a separate correspondence proof for multi-world or twin-network assignments.

The finite correspondence claim is modest but important. I am not claiming to rebuild all of Pearl’s theory from scratch. I am claiming something narrower: in the finite, discrete and recursive case, the ordinary ingredients of a structural causal model can be represented inside the type-theoretic setting used here. Variables become finite types; structural equations become terms; exogenous uncertainty becomes a finite rational probability type; interventions become transformations of the relevant structural equations; and the graph tests used by do-calculus remain finite combinatorial checks.

Once this bridge is in place, the classical finite results do not need to be reinvented. They can be transported. The appendix proves this in stages, moving from the restricted finite record fragment, through probability and graph transport, to the transferred identification and do-calculus results. The point is not to reprove Pearl–Tian–Shpitser completeness from scratch, but to show that the restricted type-theoretic fragment is an adequate target for the published finite results.

5. Fourth movement: Allegro maestoso

Step	What is established	Role in the argument	Where
1	Restricted finite causal records	Fixes the type-theoretic fragment that is meant to correspond to ordinary finite recursive structural causal models.	Definition B.5.1
2	Model correspondence	Shows that classical finite models and restricted type-theoretic records carry the same finite variables, values, structural equations, probabilities, interventions and graph data.	Theorem B.5.2
3	Expression and graph transport	Moves probability expressions and finite graph predicates across the correspondence, so that later causal arguments speak about the same finite objects on both sides.	Propositions B.5.4 and B.5.5
4	Query and derivation transport	Transfers interventional queries, identifiability claims and do-calculus derivations across the bridge.	Corollary B.5.13 and Propositions B.5.14 and B.5.15
5	Completeness transfer	Applies the published finite Pearl–Tian–Shpitser results on the classical side and transports the result back to the restricted type-theoretic fragment.	Section B.6

Table 5.2: Route through the finite causal correspondence and completeness transfer.

That transport is mostly bookkeeping, but it is precisely the kind of bookkeeping this thesis asks for: explicit, finite and checkable. What matters in the main text is the architecture: probability types supply uncertainty, modal dependent type theory supplies epistemic movement, and structural causal models supply intervention and counterfactual vocabulary.

5.5 Lean 4 implementation

The mathematical manifesto asks for explicitness, and Lean 4 [MU21] is where part of that demand becomes checking. Three Lean modules accompany this thesis: `lean/Thesis/Probability.lean` for the finite probability core, `lean/Thesis/SCM.lean` for finite structural causal models, and `lean/Thesis/Modalities.lean` for the finite epistemic transition layer. The aim is not to mechanise every philosophical claim of the text. The aim is to keep a small executable witness for the central finite constructions, so that hidden assumptions surface as type errors rather than as forgiving prose.

For orientation, the modules play the following roles.

Module	What it checks
<code>Thesis/Probability.lean</code>	Finite urns, common-denominator weighted records, the record–urn equivalence, finite Kolmogorov laws, Bayes calculations, the Clayton–Waddington rescaling steps and Bayesian-update normalisation.
<code>Thesis/SCM.lean</code>	Recursive finite structural causal models, observational and interventional probabilities, the tenure-track calculations, the model-level counterfactual helper and the local binary conditional-probability-table row construction.
<code>Thesis/Modalities.lean</code>	The finite transition layer: learning, forgetting, conditioning, relating, unrelating, setting, unsetting, preservation of well-formedness, the basic cancellation checks and the modal version of the tenure-track counterfactual.

Table 5.3: Lean modules and their checked finite constructions.

The boundary of the formalisation is just as important. The Lean development deliberately stays inside this finite, decidable fragment. It does not contain the full modal dependent type-theoretic apparatus of Definitions A.2.1, A.2.4 and A.2.5: `lean/Thesis/Modalities.lean` formalises a finite transition layer compatible with the Gratzner-style discipline, not Gratzner’s syntax, modal substitution calculus, normalisation theorem, or 2-categorical metatheory itself. It also does not mechanise the full do-calculus completeness transfer of Theorem B.6.11 or the causal-completeness transfer of Theorem B.6.13. Those broader results are stated and argued in the appendix; Lean checks the finite ingredients on which the construction rests.

Strictly speaking, a Lean proof is not a proof in raw intensional Martin-Löf type theory. Some standard-library facts used by the files, especially around rational arithmetic, quotients and list computation, inherit Lean kernel principles such as propositional extensionality, quotient soundness and occasional choice. The concrete tenure-track arithmetic theorems are finite Boolean/rational equalities discharged with `native_decide`; their mathematical content is the executable exhaustive enumeration, but their Lean axiom footprint includes the generated native-decision axioms. No thesis-specific classical axiom is postulated, and the mathematical arguments above avoid global excluded middle; nevertheless, the mechanisation should be read as a Lean 4 check of the finite constructions, not as a formalisation of Gratzner’s bare syntax.

The flavour of the Lean code is best seen in the record–urn equivalence theorem itself:

Listing 5.1: Record–urn extensionality (`lean/Thesis/Probability.lean`).

```

theorem urn_to_record_extensional (m : UrnProb 0) :
  forall E : Event 0, QProb.Equiv ((ofUrn m).probVal E) (m.probVal E) :=
  ofUrn_agrees m

theorem record_to_urn_extensional (R : FiniteProbRecord 0) :
  forall E : Event 0, QProb.Equiv (R.probVal E) (R.toUrn.probVal E) :=
  R.toUrn_agrees

```

5. Fourth movement: Allegro maestoso

Read informally, the two statements say that converting an urn to a finite record, or a finite record to an urn, does not change the probability assigned to any decidable event. The proof is not deep, and that is the point. It is the sort of finite bookkeeping that prose often hides: counts are preserved when atoms are expanded, unit-weight atoms count as one cell, and the resulting rational probabilities agree by cross multiplication. Lean is used here not for theorem-proving heroics, but to keep the finite construction honest.

5.6 Results

The formal result aimed at by the Fourth movement is a finite constructive language for mathematical inference. Its first component is probability: finite Jaynes–Cox-style updating represented by explicit probability types. Its second component is subjectivity: contexts, modes and modalities that represent what is known, remembered, ignored, hypothesised or changed. Its third component is causality: Pearl-style structural causal models embedded inside the same type-theoretic environment.

This does not produce a universal theory of everything. It produces a disciplined fragment. The fragment is finite, discrete, constructive and intended for proof checking. That is precisely why it is useful. It does not try to solve measure theory, continuous probability or all metaphysics of causality. It tries to build a clean machine for the cases where the assumptions can be stated and checked.

The formal scope should remain visible. The transferred causal results cover finite ordinary interventional distributions, and finite conditional effects when the relevant denominators are positive. They do not cover cyclic or non-recursive structural causal models, arbitrary type-theoretic causal records outside the decoded finite fragment, general counterfactual identification, transportability, meta-synthesis or path-specific effects.

The gain is conceptual as well as technical. Jaynes’ probability theory becomes less imperial but more honest. Subjectivity becomes formal rather than rhetorical. Causality becomes part of the same inferential architecture rather than an external supplement. And the mathematical manifesto is obeyed: algorithms over abstractions; narrative over formulae; explicit over implicit; simple over complex.

This answers the internal question: what a disciplined finite language for mathematical inference can look like. The external question remains. Even a clean inferential machine can be misused, ignored or made professionally irrational by the environment around it. The last movement therefore returns to the scientific pipeline in which such inferences are made.

CHAPTER 6

Fifth movement: Presto con fuoco

“For far too long economists have sought to define themselves in terms of their supposedly scientific methods. In fact, those methods rely on an immoderate use of mathematical models, which are frequently no more than an excuse for occupying the terrain and masking the vacuity of the content. Too much energy has been and still is being wasted on pure theoretical speculation without a clear specification of the economic facts one is trying to explain or the social and political problems one is trying to resolve”

Thomas Piketty in *Capital in the Twenty-First Century*

6.1 Funded research

Jaynes’ complaints about scientific attitude are symptoms of a larger institutional problem. McElreath’s reminder from the Prelude is worth recalling here [McE20]: inference is only one component of science, and even technically competent inference becomes unreliable inside a bad pipeline. Science can be imagined, charitably, in at least two noble ways — as the sincere search for truth, and as the organised attempt to solve important human problems. Modern research likes to advertise both ideals; its machinery increasingly rewards something rather different: funding, publication, visibility, speed and institutional survival.

This is why I think the word science should be used carefully. A system that rewards the appearance of discovery more reliably than discovery itself is not science with a few unfortunate incidents — it is funded research. Some funded research is excellent science, some is not. The distinction matters because it stops the institution from borrowing the moral prestige of science while organising itself around incentives that corrode scientific inference.

The pressure is visible from the start of a career. Young researchers need publications to obtain funding, funding to obtain positions, positions to obtain independence, and independence to obtain the time required for careful work. The resulting catch-22 has been noticed explicitly in the funding literature [Ibá99; NI12]: it favours well-resourced institutions, safe projects and researchers already close to the centre, and it does not naturally favour slow, strange or necessary work. The pressure is also visible from the other end of a career: Peter Higgs has remarked that, in today’s publish-or-perish climate, he would not have been considered productive enough to keep his post — the kind of long, slow, unfashionable thinking that produced the Higgs mechanism is not what modern incentives reward [Ait13; Mel14]. The system selects against the very work it then claims credit for.

The tenure-track model of Example 5.4.1 was deliberately small, but its ingredients are not arbitrary. Prestige, committee fit, fashionability, funding lines and committee expertise are precisely the sort of variables that make early-career selection hard to understand from outcomes alone. I do not claim the toy causal model is an empirical sociology of hiring — the point is institutional: the formal distinction between seeing and doing becomes relevant here, because observing that prestigious candidates are often selected is not the same as knowing what would happen if prestige were removed from the procedure.

The same pressure appears in dramatic fraud cases, but it would be too comforting to treat fraud as the whole problem. Stapel, Schön, Hwang, Obokata, Wakefield, Pandey, Raoult and Fujii are useful not because they are typical in every detail, but because their cases reveal failure modes that ordinary incentives also encourage: fabricated data, missing records, salami-sliced output, ethical shortcuts, prestige dependence, citation manipulation and inadequate review [BDK02; Cyr06; GSM11; Los11; MO15; OGr24; The16; The24].

Large-scale studies of questionable research practices, misconduct and paper mills make the same point without the melodrama of famous names [Fan09; FSC12; JLP12; Ric+25]. Bad science, then, is not only a matter of bad people — it is also, and perhaps mostly, a matter of bad selection pressures.

6. Fifth movement: Presto con fuoco

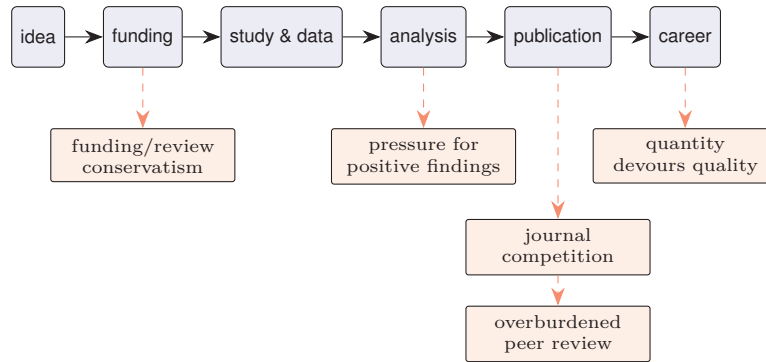


Figure 6.1: The funded-research pipeline and the five failure modes it breeds — the institutional shadow of the five vices of the First movement (Section 2.3).

6.2 Failure modes

The five failure modes treated in this section are best read as the institutional shadow of the five vices of Section 2.3 in the First movement. The pressure for positive findings is what *hidden assumptions* look like at scale, with researcher degrees of freedom hidden in the procedure. Journal competition is what *cutting corners* looks like once an editorial economy rewards volatility before reliability. Overburdened peer review is what *theoretical display without experimental responsibility* becomes when judgement is uncoupled from testing. Funding and review conservatism is the institutional form of *gamesmanship*: rationality at the micro level, mediocrity at the aggregate. And quantity devouring quality is *hyperfocus* after a career: training a person to do one job, then rewarding them for another. None of these is exotic. Each is the predictable institutional reflex of the corresponding vice.

Start with the most obvious one: pressure for positive, publishable findings. Flexible analysis, small samples, hidden researcher degrees of freedom and selective reporting all make results look cleaner than they are [But+13; GL13; GL14; SNS11]. This is the institutional version of a hidden assumption: a conclusion looks firm because the path that produced it has been made invisible.

Journal competition makes the picture worse. When journals compete for attention, surprising results become valuable in part because they are surprising — and surprise is not a synonym for truth. A publication market can reward the volatility of a claim long before the reliability of the evidence has caught up. Schekman has criticised prestige journals in exactly this spirit, and Eisen has made related arguments for open dissemination [Eis11; Sch13].

Then there is peer review, now an overburdened ritual. Reviewers are asked to judge novelty, correctness, significance, ethics, method, relevance and sometimes fraud — usually without enough time, training, data or reward. This is not a serious design for truth discovery; it is a heroic volunteer layer pasted onto an industrial publication system, and as the literature grows, the burden becomes worse [BHM21; Cam98; Lan16].

A subtler failure is conservatism. Funding and review systems can punish risk even when everyone in the room claims to value innovation [AS14; GB19; Lan+22]. The result is a strange inversion: radical fraud may be noticed because

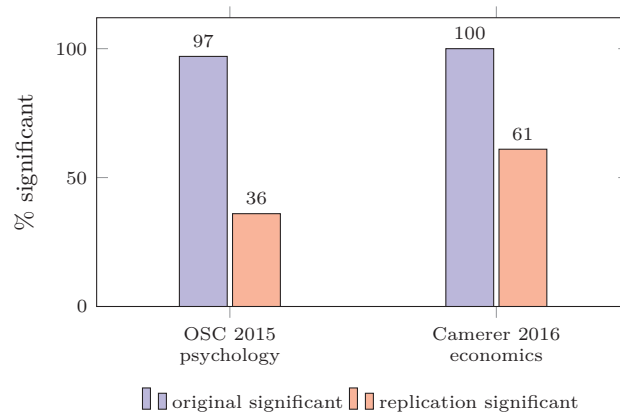


Figure 6.2: Two large replication projects. Of 100 psychology studies, 97% of originals reported a statistically significant result versus 36% of replications [Ope15]; of 18 economics experiments, all reported a positive original finding and 61% (11 of 18) replicated as a significant effect in the same direction [Cam+16].

it is too audacious, while timid mediocrity passes indefinitely because it looks like what everyone already expects.

Finally, quantity quietly devours quality. Career advancement rewards publication counts, citations, grants and institutional prestige more directly than careful reasoning, which makes hyperfocus rational, gamesmanship attractive and cutting corners tempting. It also trains researchers for one job — doing research — and then rewards them for another: managing output, personnel, funding, public relations and institutional survival.

6.3 Scientific manifesto

A scientific manifesto should answer the institutional problem in the same spirit that the mathematical manifesto answered the formal one. My aim is not purity, but to make the surrounding environment less hostile to good inference. In the same comparative spirit as the Agile and Scrum manifestos that already shaped Section 2.3’s mathematical preferences [Bec+01; SS20], the discipline I propose for funded research is this:

I am uncovering better ways of organising publicly funded research by doing it and helping others do it. Through this work I have come to value:

Science over funded research
 Pluralistic over monolithic funding
 Open over closed publication
 Honest over ritual review
 Researchers over operations

6. Fifth movement: Presto con fuoco

Each line is a comparative preference, not a denial of the right-hand term. Funded research is not the enemy of science; closed publication is not the same as fraud; reviewers are not the problem; operations are not optional. The point of the boxed form is the same as in the mathematical manifesto: I prefer the left-hand items, and I want the wider research environment to be designed around those preferences. The five proposals below elaborate the five lines.

My starting suggestion is to keep science and funded research as separate words. Publicly funded research should be judged by public standards — openness, accountability, reproducibility, usefulness and intellectual honesty — and where the system cannot meet those standards yet, it should not hide behind the romance of science. It should say what it is, and then improve.

Funding ought to be pluralistic. Peer judgement is necessary, but peer judgement alone drifts conservative. Citizen-directed funding can support local or neglected problems; democratic funding priorities can counterbalance the quiet drift towards corporate agendas. Researcher autonomy matters, of course, but autonomy without public structure does not float in neutral space — it is pulled by whoever has the clearest agenda and the strongest purse.

Publicly funded work should be openly published. The Council of the European Union has already called for transparent, equitable and open access to scholarly publications [Cou23]; I would treat that call not as a polite aspiration but as a design principle: if the public pays for research, the public should not pay again to read it. Raw data, code and materials should be deposited by default, with rare exceptions made explicit.

Review itself wants redesigning. Open review, post-publication review, recognised reviewing labour and reviewer training would make the system more honest about what review can and cannot do [Cen; DOR12; Mun+17; Nos+15; NSM12]. Anonymous review may still have a place, but anonymity should not mean invisibility of standards.

And then, finally, a plea for organisational modesty. Leading a research group and managing operations are not the same skill. A system that makes successful researchers automatically responsible for human resources, finance, public relations and institutional strategy should not be surprised when many of them do those tasks badly. Shared operational support, transparent funding, and separate research and operations roles would remove a great deal of unnecessary theatre.

None of these proposals will eliminate fraud, stupidity or vanity; no institutional design will. But they would reduce the reward for hiding assumptions, exaggerating claims and converting inquiry into career choreography. They are the practical counterpart of the formal thesis: good inference requires good rules, but also a world in which following those rules is not professionally irrational.

Acknowledgements

“Teleutias truly sailed home under the happiest of circumstances. When he went down to the sea to begin his journey home, there was not a soldier who did not clasp his hand. One placed a wreath on his head, another put a fillet around him, and those who came too late, when he was already putting out to sea, still threw wreaths into the water and wished him every good thing.

I realize that I am not describing any great expenditure, any danger, or any strategy worth mentioning; but by Zeus, I do think it is important for a man to consider what kind of conduct enabled Teleutias to evoke such a response from his subordinates, for that is, beyond any amount of money or danger, a man’s most noteworthy achievement”

Xenophon in *Hellenica*

Appendices

APPENDIX A

Definitions

“So if you are an expert and, as I said just now, have deceived me with your promise to give a demonstration about Homer, you are playing a trick on me. But if you are not an expert, but are possessed by Homer through a divine gift and, without knowledge, say many beautiful things about the poet, as I claimed is the case with you, then you are not playing a trick on me. So you need only choose how you wish us to regard you: as a man who plays tricks, or as a man governed by the gods”

Socrates in *Ion*

A.1 Type-theoretic definitions

The definitions in this appendix fix the fragment used in the main text. They are intentionally finite and judgemental. The type-theoretic notation follows the Martin-Löf tradition and the presentations used by Nordström–Petersson–Smith and Gratzner [Gra23; Mar84; NPS90].

Definition A.1.1 (Type-theoretic judgements). The basic judgemental forms used here are

$$\Gamma \text{ ctx}, \quad \Gamma \vdash A \text{ type}, \quad \Gamma \vdash a : A, \quad \Gamma \vdash \delta : \Delta,$$

together with definitional equality judgements

$$\Gamma \vdash A \equiv B \text{ type}, \quad \Gamma \vdash a \equiv b : A, \quad \Gamma \vdash \delta \equiv \epsilon : \Delta.$$

These are read as follows: Γ is a well-formed context; A is a type in context Γ , that is, a domain of possible constructions available in that context; a is a term of type A in context Γ (under Definition A.1.19, evidence for the proposition represented by A); and δ is a substitution from Γ to Δ . The symbol Γ is always a context; δ and ϵ are substitutions.

Definition A.1.2 (Context). A context is a finite list of typed variable declarations

$$\Gamma := x_1 : A_1, x_2 : A_2(x_1), \dots, x_n : A_n(x_1, \dots, x_{n-1}),$$

in which each type A_{i+1} may depend on the earlier variables $x_1, \dots, x_i$. Such a dependently-typed sequence is called a *telescope* in the de Bruijn–Jacobs tradition. Contexts are generated from the empty context by context extension:

$$\frac{\Gamma \text{ ctx} \quad \Gamma \vdash A \text{ type}}{\Gamma, x : A \text{ ctx}}.$$

Later declarations may depend on earlier declarations; earlier declarations may not depend on later ones.

Definition A.1.3 (Structural use of contexts). In the fragment used here, weakening, exchange and forgetting are admissible only when the dependency order permits them. Thus a fresh declaration $x : A$ may be added to a context when $\Gamma \vdash A \text{ type}$; two declarations may be exchanged only if neither dependency is broken; and a declaration $x : A$ may be forgotten from $\Gamma, x : A, \Delta$ only when neither Δ nor the target judgement contains a free occurrence of x .

Definition A.1.4 (Substitution). If $\Gamma, x : A, \Delta \vdash J$ is any judgement and $\Gamma \vdash a : A$, substitution replaces x by a :

$$\Gamma, \Delta[a/x] \vdash J[a/x].$$

In explicit-substitution presentations this is internalised as a judgement $\Gamma \vdash \delta : \Delta$ and an operation sending $\Delta \vdash b : B$ to $\Gamma \vdash b[\delta] : B[\delta]$.

Definition A.1.5 (Definitional equality). Definitional equality, written $\equiv$, is equality by computation, unfolding or stipulated judgemental equation. It is part of the judgemental structure of the theory, not a proposition internal to a type.

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Definition A.1.6 (Sum type). If $\Gamma \vdash A$ type and $\Gamma \vdash B$ type, then $\Gamma \vdash A + B$ type. Its introductions are $\text{inl}(a) : A + B$ for $a : A$ and $\text{inr}(b) : A + B$ for $b : B$. To define a term out of $A + B$, one must give a case for the left injection and a case for the right injection.

Definition A.1.7 (Dependent function type). A dependent type over A in context Γ is a family specified by a judgement $\Gamma, x : A \vdash B(x)$ type; it assigns a type $B(a)$ to each term $\Gamma \vdash a : A$ by substitution. Given such an A and B , the dependent function type is formed by

$$\Gamma \vdash \Pi_{x:A} B(x) \text{ type.}$$

Its introduction, elimination and computation rules are

$$\frac{\Gamma, x : A \vdash b(x) : B(x)}{\Gamma \vdash \lambda x. b(x) : \Pi_{x:A} B(x)}, \quad \frac{\Gamma \vdash f : \Pi_{x:A} B(x) \quad \Gamma \vdash a : A}{\Gamma \vdash f(a) : B(a)},$$

and $(\lambda x. b(x))(a) \equiv b(a)$. When B does not depend on x , this is the ordinary non-dependent function type $\Gamma \vdash A \rightarrow B$ type, with the same introduction, elimination and computation rules and the usual beta reduction $(\lambda x. b)(a) \equiv b[a/x]$.

Definition A.1.8 (Dependent pair type). Given $\Gamma \vdash A$ type and a dependent type $\Gamma, x : A \vdash B(x)$ type, the dependent pair type is formed by

$$\Gamma \vdash \Sigma_{x:A} B(x) \text{ type.}$$

Its canonical inhabitants are pairs (a, b) with $\Gamma \vdash a : A$ and $\Gamma \vdash b : B(a)$. The projections satisfy

$$\text{pr}_1(a, b) \equiv a, \quad \text{pr}_2(a, b) \equiv b.$$

When B does not depend on x , this is the ordinary non-dependent product type $\Gamma \vdash A \times B$ type, with pair inhabitants (a, b) and the same projections.

Definition A.1.9 (Identity type). If $\Gamma \vdash A$ type and $\Gamma \vdash a, b : A$, then

$$\Gamma \vdash \text{Id}_A(a, b) \text{ type.}$$

Its canonical introduction is reflexivity:

$$\frac{\Gamma \vdash a : A}{\Gamma \vdash \text{refl}_a : \text{Id}_A(a, a)}.$$

In intensional type theory, an inhabitant of $\text{Id}_A(a, b)$ is evidence of equality inside the theory; it is not automatically the same as external or definitional equality.

Definition A.1.10 (Universe). A universe level $\mathcal{U}_i$ is a type of codes for small types. In the Coquand-style notation used by Gratzer, the relevant operations are

$$\Gamma \vdash M : \mathcal{U}_i \Rightarrow \Gamma \vdash \text{El}(M) \text{ type}_i,$$

and

$$\Gamma \vdash A \text{ type}_i \Rightarrow \Gamma \vdash \text{Code}(A) : \mathcal{U}_i,$$

with definitional equations $\text{El}(\text{Code}(A)) \equiv A$ and $\text{Code}(\text{El}(M)) \equiv M$, within the chosen universe level.

Definition A.1.11 (Inductive type). An inductive type is specified by constructors and by an eliminator. If I is generated by constructors $c_k : C_k \rightarrow I$, then to define a dependent function

$$f : \prod_{x:I} P(x)$$

it suffices to give the corresponding constructor cases demanded by the eliminator for I . Natural numbers, lists and finite types are standard examples. The natural-number type $\mathbb{N}$ is the inductive type generated by $0 : \mathbb{N}$ and $\text{succ} : \mathbb{N} \rightarrow \mathbb{N}$. In particular, the type $\text{List}(A)$ of finite lists over a type A is the inductive type with constructors $\text{nil} : \text{List}(A)$ and $\text{cons} : A \rightarrow \text{List}(A) \rightarrow \text{List}(A)$; its eliminator yields ordinary list recursion. This is the List used in the weighted finite-support form of Definition A.3.7.

Definition A.1.12 (Finite type). For $n : \mathbb{N}$, the finite type $\text{Fin}(n)$ has exactly the canonical inhabitants $0, \dots, n-1$. Its eliminator says that to define $f : \prod_{i:\text{Fin}(n)} P(i)$, one gives terms $p_i : P(i)$ for each $i < n$.

Definition A.1.13 (Empty type). The empty type $\perp$ is the inductive type (Definition A.1.11) with no constructors; equivalently, $\perp \equiv \text{Fin}(0)$. Its eliminator says that to define

$$\text{absurd}_P : \perp \rightarrow P$$

for any type P , no constructor cases are needed: the empty type has no canonical inhabitants to dispatch on.

Definition A.1.14 (Negation). Negation is the type-theoretic function into the empty type:

$$\neg A := A \rightarrow \perp.$$

From $a : A$ and $n : \neg A$ one obtains $n(a) : \perp$. Double negation $A \rightarrow \neg\neg A$ is always derivable by $\lambda a. \lambda n. n(a)$; the converse $\neg\neg A \rightarrow A$ is a classical principle and is not available uniformly (cf. Proposition B.3.1).

Definition A.1.15 (Boolean type). The Boolean type Bool is the primitive inductive type with two constructors $\text{true}, \text{false} : \text{Bool}$. Its eliminator gives Boolean case analysis: to define $f : \text{Bool} \rightarrow P$, one provides values $p_{\text{true}}, p_{\text{false}} : P$ with computation rules $f(\text{true}) \equiv p_{\text{true}}$ and $f(\text{false}) \equiv p_{\text{false}}$. Equality on Bool is decidable, and $\text{true} \neq \text{false}$.

Definition A.1.16 (Decidable type and decidable predicate). A type A is *decidable* when the sum type $A + \neg A$ is inhabited, using Definitions A.1.6 and A.1.14; a term $d : A + \neg A$ is a *decision procedure* for A , returning either a witness of A or a refutation of A . A predicate $P : X \rightarrow \mathcal{U}$ is *decidable* when $P(x)$ is decidable for every $x : X$, equivalently when there is a function $X \rightarrow \text{Bool}$ in the sense of Definition A.1.15 together with a soundness proof. A type A has *decidable equality* when the predicate $\text{Id}_A(a, b)$ of Definition A.1.9 is decidable for every $a, b : A$. Finite types (Definition A.1.12) and the Boolean type have decidable equality by their eliminators, and finite sums and finite products of types with decidable equality have decidable equality.

Remark A.1.17 (Decidability conventions). Throughout this thesis, “decidable event”, “decidable predicate” and “decidable equality” are used in the sense of Definition A.1.16. The three role-tagged finite-type definitions used later — Definition A.3.1 for sample spaces, Definition A.3.5 for variable values,

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and Definition A.1.12 for the inductive families $\text{Fin}(n)$ — are all finite types with decidable equality; the distinct names mark distinct roles, not distinct constructions. “Decidable event” means the predicate form $E : \Omega \rightarrow \text{Bool}$ of Definition A.3.1, equivalently a finite subset of Ω .

Definition A.1.18 (Indicator). For a decidable predicate $E : X \rightarrow \text{Bool}$, the indicator function $\mathbb{1}_E : X \rightarrow \mathbb{N}$ is defined by Boolean case analysis: $\mathbb{1}_E(x) := 1$ when $E(x) = \text{true}$ and $\mathbb{1}_E(x) := 0$ when $E(x) = \text{false}$.

Definition A.1.19 (Propositions as types). The propositions-as-types reading identifies a proposition with a type and a proof with a term of that type:

$$A \text{ is proved in } \Gamma \iff \Gamma \vdash a : A \text{ for some term } a.$$

Remark A.1.20. This reading is used informally throughout the thesis but is not invoked as a rule inside the formal transfer of Theorem B.6.13; it serves as a background interpretation of the constructive judgemental discipline.

Definition A.1.21 (Logical connectives under propositions-as-types). Under Definition A.1.19, the standard logical connectives correspond to type formers:

$$\begin{aligned} A \wedge B &:= A \times B, & A \vee B &:= A + B, \\ A \Rightarrow B &:= A \rightarrow B, & \neg A &:= A \rightarrow \perp, \\ \top &:= \text{Fin}(1). \end{aligned}$$

The constructive content of a logical statement is then a term of the corresponding type. In particular, $A \wedge B$ requires both a witness of A and a witness of B ; $A \vee B$ requires a witness of one side together with evidence of which side; and $\neg A$ requires a procedure converting any witness of A into an inhabitant of the empty type.

Definition A.1.22 (Formation, introduction, elimination and computation). The rules for a type former are organised as follows: formation rules state when the type is well formed; introduction rules construct canonical inhabitants; elimination rules specify how inhabitants may be used; and computation rules state how eliminations applied to introductions reduce by definitional equality.

Remark A.1.23. This meta-schema is used as a structuring device for presenting individual type formers above; no proof in this thesis quantifies over rule-schemas as objects.

A.2 Modal definitions

The modal notation follows Gratzer’s multimodal dependent type theory: a mode theory is a small strict 2-category; objects are modes; morphisms are modalities; and 2-cells record transformations between modalities [Gra+21; Gra23; GSB19]. I use this notation locally; it is meant to track the relevant fragment, not to reproduce every rule of Gratzer’s full systems.

Definition A.2.1 (Mode theory). A mode theory is a small strict 2-category $\mathcal{M}$. Its objects $m, n, o \in \mathcal{M}$ are modes, its morphisms $\mu : n \rightarrow m$ are modalities, and its 2-cells $\alpha : \mu \Rightarrow \nu$ are transformations between modalities. No rule in this thesis exercises 2-cells; they are declared here to match the mode-theory definition of Gratzer and collaborators [Gra+21; Gra23].

Definition A.2.2 (Modal judgement). For each mode m , judgements are indexed by that mode:

$$\Gamma \text{ ctx } @m, \quad \Gamma \vdash A \text{ type } @m, \quad \Gamma \vdash a : A @m, \quad \Gamma \vdash \delta : \Delta @m.$$

The annotation $@m$ belongs to the judgement, not to the type expression A . Thus a judgement is not made in an unlocated universe of discourse; it is made at a mode. The full modal-substitution discipline of Gratzer's multimodal type theory [Gra+21; Gra23] is not reproduced here. For the purposes of this thesis, substitutions are used at a single mode at a time; inter-mode movement is mediated by Definition A.2.5 and Definition A.2.6 rather than by inter-mode substitutions.

Definition A.2.3 (Mode). A mode is an object m of a mode theory $\mathcal{M}$. In this thesis, modes are interpreted as epistemic states: structured positions from which certain variables, assumptions, questions and causal models are available.

Definition A.2.4 (Modality). A modality is a morphism $\mu : n \rightarrow m$ in a mode theory. The displayed arrow direction follows the Gratzer-style lock notation of Definition A.2.5: data judged at mode n may be marked by μ and used from mode m . In the intended interpretation, a modality is an admissible transition between epistemic states: learning, forgetting, changing attention, hypothesising or intervening.

Definition A.2.5 (Locked context). Let $\mu : n \rightarrow m$ be a modality in a mode theory $\mathcal{M}$. The locked context is the context extension

$$\frac{\Gamma \text{ ctx } @m \quad \mu : n \rightarrow m}{\Gamma.\{\mu\} \text{ ctx } @n}.$$

The lock entry $\{\mu\}$ is a context entry, not a postfix operator on contexts; it records that subsequent declarations are read at mode n rather than m . Composition of locks follows the modality 2-category: for $\mu : n \rightarrow m$ and $\nu : o \rightarrow n$,

$$\Gamma.\{\mu\}.\{\nu\} \equiv \Gamma.\{\mu \circ \nu\}, \quad \Gamma.\{1_m\} \equiv \Gamma,$$

where $\mu \circ \nu : o \rightarrow m$ is the composite modality and 1_m is the identity modality on m [Gra+21; Gra23]. An ordinary context extension under a lock takes the form $\Gamma.\{\mu\}, x : A$, adding the hypothesis $x : A$ in the locked context at mode n ; no separate "modal-hypothesis" entry is needed.

Definition A.2.6 (Modal type former). Let $\mu : n \rightarrow m$ be a modality. Given a context $\Gamma \text{ ctx } @m$, form the locked context $\Gamma.\{\mu\} \text{ ctx } @n$ of Definition A.2.5. If $\Gamma.\{\mu\} \vdash A \text{ type } @n$, then the modal type former gives

$$\Gamma \vdash \langle \mu \mid A \rangle \text{ type } @m.$$

If $\Gamma.\{\mu\} \vdash a : A @n$, then the modal introduction form gives

$$\Gamma \vdash \text{mod}_\mu(a) : \langle \mu \mid A \rangle @m.$$

This is the fragment needed here: a modality lets a type or term from mode n appear, under modal marking, at mode m .

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Remark A.2.7 (Modal fragment is introduction-only). Definition A.2.6 states only formation and introduction for $\langle \mu \mid A \rangle$. The corresponding modal elimination rule (the **letmod** or modal pattern-matching rule) of multimodal type theory [Gra+21; Gra23], the action of 2-cells $\alpha : \mu \Rightarrow \nu$ on modal terms, and the full inter-mode substitution discipline are deliberately not reproduced. No definition or proof in this thesis destructs a $\text{mod}_\mu(a)$, applies a 2-cell to a term, or transports a substitution across modes. The modal vocabulary used here is therefore the introduction-only fragment of multimodal type theory: enough to record that data lives at a mode and that an admissible modality lets it surface, under modal marking, at another mode. Results that genuinely depend on modal elimination, on 2-cell action or on inter-mode substitution lie outside the scope of this fragment and are deferred to the cited works.

Remark A.2.8 (Epistemic modalities mark record transitions). The intended epistemic transition names — learning, forgetting, conditioning, relating, unrelating, setting and unsetting — are not context-dependent modal operators. In a strict Gratzner-style reading, the mode theory first supplies modalities $\mu : n \rightarrow m$ between already specified modes. The finite context or structural-causal-model records at those endpoint modes may then differ: a fresh unused variable may have been added, an unused variable may have been forgotten, the exogenous distribution may have been conditioned on evidence, a structural equation may have been replaced by a constant for an intervention, or a relation may have been inserted or removed. Those changes are ordinary record or context transformations attached to the endpoint modes; the modality itself only witnesses that the transition is admissible in the mode theory.

Conditioning is deliberately separate from setting. Setting is the Pearl-style action step: it changes an endogenous mechanism and keeps the exogenous uncertainty fixed. Conditioning is the abduction or evidence-update step: it replaces the exogenous distribution by a posterior distribution and keeps the structural equations fixed. If the evidence determines the exogenous state, the posterior may be a point mass, but this is still conditioning rather than an intervention on an exogenous structural equation. Keeping the two labels separate prevents the formal layer from conflating seeing with doing.

This separation is important. If the modality inspected the current context and performed the transformation by itself, the system would move towards context-dependent modalities, which are substantially stronger than the fragment used here and would require new metatheory. In the present fragment, forward-only epistemic change is harmless: one declares two endpoint modes and a modality between them. Inverse-looking operations, such as adding a fresh variable and then forgetting exactly that unused variable, are not automatic. They require a stated equation in the mode theory or a propositional theorem saying that the composite record transformation returns a context equivalent to the starting one.

Remark A.2.9 (Narrative-only terms: subjectivity, interest, epistemic position). Several umbrella terms in the main narrative — *subjectivity*, *interest*, *epistemic position* — are not given formal definitions of their own. They are deliberately narrative-only: they name the conceptual target that the modal apparatus is built to serve, not separate objects of the calculus. Inside this fragment they are captured informally by the structural shape of the current judgement: *subjectivity* is captured by the dependence of every judgement on a mode and

a context (Definitions A.1.2 and A.2.2); *epistemic position* is captured by the current mode together with the current context (Definitions A.1.2 and A.2.3); and *interest* is captured by which variables, locks and modalities are present in Γ at the current mode — a question only enters scope when its variables are declared, an interest is forgotten when the corresponding declaration is removed, and a change of interest is a context manipulation or a modal transition. No proof in this thesis quantifies over *subjectivity*, *interest* or *epistemic position* as objects; they are names for the discipline that contexts, modes and modalities together impose.

A.3 Probabilistic and causal definitions

Three notation conventions are used throughout this section and the proofs that depend on it.

Independence. The symbol $\perp\!\!\!\perp$ has two readings. Probabilistic independence is always subscripted by the probability assignment, for example $A \perp\!\!\!\perp_{\mu} B$, $\mathcal{X} \perp\!\!\!\perp_{\mu} \mathcal{Y} \mid \mathcal{Z}$ or $\perp\!\!\!\perp_{P_{\mathcal{M}}}$. Graphical d-separation always carries a graph subscript on the surrounding expression, for example $(X \perp\!\!\!\perp Y \mid Z)_{\mathcal{G}}$. The full convention is fixed in Definition A.3.10.

Decidable events. Decidable events on a finite sample type Ω are predicates $E : \Omega \rightarrow \text{Bool}$, in the sense of Definitions A.1.15, A.1.16 and A.3.1. The notation $E \subseteq \Omega$ is used interchangeably with the predicate notation, with $\cap$, $\cup$, $\setminus$ and complement read pointwise on Booleans.

Intervention. The expression $\text{do}(X = x)$ always denotes Pearl’s intervention operator (Definition A.3.24). Its model-theoretic content is the intervened structural causal model $\mathcal{M}_x$ of Definition A.3.30; its graph-level content is the mutilated graph $\mathcal{G}_{\overline{X}}$ of Definition A.3.25.

Definition A.3.1 (Finite event type). A finite event type is a finite type Ω with decidable equality. An event is a decidable predicate $E : \Omega \rightarrow \text{Bool}$ in the sense of Definition A.1.15, equivalently a finite subset $E \subseteq \Omega$.

Definition A.3.2 (Resolution status). Let $A(s)$ be a proposition or question available in scenario $s : S$, where S is a finite event type of *scenarios* available to a finite inspection procedure. The word “scenario” is reserved here for the input type of the inspection procedure, in order to avoid confusion with Definition A.2.3, where the phrase “epistemic state” refers to a mode. The resolution-status type for A is the three-constructor finite type

Status_A generated by `proved`, `refuted`, `unresolved`.

A finite inspection procedure for A is a total finite procedure

$$\iota_A : S \rightarrow \text{Status}_A.$$

To read the tags as genuine proof status, the procedure must also be sound:

$$\text{sound}_{\text{proved}} : \Pi_{s:S} (\iota_A(s) = \text{proved}) \rightarrow A(s),$$

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and

$$\text{sound}_{\text{refuted}} : \Pi_{s:S} (\iota_A(s) = \text{refuted}) \rightarrow \neg A(s),$$

where $\neg A(s) := A(s) \rightarrow \perp$ as in Definition A.1.14. The value `unresolved` means only that neither kind of witness is found by the chosen finite procedure. It does not mean that no proof or refutation exists in principle.

Definition A.3.3 (Resolution-status events). Suppose S is a finite event type of scenarios in the sense of Definition A.3.2, and $\iota_A : S \rightarrow \text{Status}_A$ is a finite inspection procedure. The three resolution-status events associated with A are

$$\text{Prov}_A(s) := \begin{cases} \text{true}, & \iota_A(s) = \text{proved}, \\ \text{false}, & \text{otherwise}, \end{cases} \quad \text{Ref}_A(s) := \begin{cases} \text{true}, & \iota_A(s) = \text{refuted}, \\ \text{false}, & \text{otherwise}, \end{cases}$$

and

$$\text{Unr}_A(s) := \begin{cases} \text{true}, & \iota_A(s) = \text{unresolved}, \\ \text{false}, & \text{otherwise}. \end{cases}$$

Formally these are Boolean predicates $S \rightarrow \text{Bool}$ obtained by case analysis on the finite resolution-status type, with equality tests justified by decidable equality of Status_A . They form a partition of S : every $s : S$ lies in exactly one of them.

Definition A.3.4 (Urn-generated probability type). Let X be any type. An urn-generated finite probability assignment into X is

$$\text{Prob}_{\mathbb{Q}}^{\text{urn}}(X) := \Sigma_{N:\mathbb{N}} \Sigma_{h:N>0} (\text{Fin}(N) \rightarrow X).$$

If $\mu = (N, h, f) : \text{Prob}_{\mathbb{Q}}^{\text{urn}}(X)$ and $E : X \rightarrow \text{Bool}$ is a decidable predicate, then

$$P_{\mu}(E) := \frac{1}{N} \sum_{i:\text{Fin}(N)} \mathbf{1}_{E(f(i))},$$

using the indicator of Definition A.1.18. The finite type $\text{Fin}(N)$ is the urn of N equipossible cells. The map f pushes that urn forward to X . No finiteness of X is required for this definition: the finiteness lies in the source urn. Thus μ is a finite-supported assignment into X , and only explicitly decidable predicates on X are evaluated.

Definition A.3.5 (Value type). A value type for a variable X is a finite type $\text{Val}(X)$ whose inhabitants are the possible values of X . If $x : \text{Val}(X)$, then x is an admissible value assignment for X .

Definition A.3.6 (Finite variable). A finite variable is a pair consisting of a variable name X and a value type $\text{Val}(X)$. A finite family of variables $\mathcal{W}$ means a finite index type together with one finite variable at each index. For such a family, write

$$\text{Val}(\mathcal{W}) := \Pi_{W \in \mathcal{W}} \text{Val}(W)$$

for the type of joint assignments to all variables in $\mathcal{W}$. A joint event on $\mathcal{W}$ is a decidable predicate on $\text{Val}(\mathcal{W})$.

Definition A.3.7 (Finite probability type). Let Ω be a finite event type. In this thesis the coefficient field is fixed to the rational numbers $\mathbb{Q}$; $\mathbb{Q}_{\geq 0}$ is the

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subtype of rational numbers equipped with a proof of non-negativity. Finite sums over a finite event type are computed by the eliminator or by a chosen finite enumeration, and are invariant under finite reindexing. The record form of a probability type over Ω is

$$\text{Prob}_{\mathbb{Q}}(\Omega) := \Sigma_{p:\Omega \rightarrow \mathbb{Q}_{\geq 0}} \left(\sum_{\omega:\Omega} p(\omega) = 1 \right).$$

If $\mu = (p, h) : \text{Prob}_{\mathbb{Q}}(\Omega)$ and $E \subseteq \Omega$, then

$$P_{\mu}(E) := \sum_{\omega:\Omega} \mathbb{1}_E(\omega) p(\omega),$$

using the indicator of Definition A.1.18. If $E \cap F = \emptyset$, then $P_{\mu}(E \cup F) = P_{\mu}(E) + P_{\mu}(F)$ by finite summation. When only one probability assignment is in scope, the subscript is sometimes dropped: $P(E) \equiv P_{\mu}(E)$.

After a common denominator has been chosen, the same finite rational record can also be written in weighted finite-support form:

$$\begin{aligned} R &= (a, D, h, d), \\ a &: \text{List}(\Omega \times \mathbb{N}), \\ D &: \mathbb{N}, \\ h &: \sum_{(\omega, m) \in a} m = D, \\ d &: D > 0. \end{aligned}$$

Its value on a decidable event $E : \Omega \rightarrow \text{Bool}$ is

$$P_R(E) := \frac{\sum_{(\omega, m) \in a, E(\omega) = \text{true}} m}{D}.$$

This is the common-denominator form formalised in Lean.

When the codomain is itself a finite event type Ω , every urn-generated probability assignment $\nu = (N, h, f) : \text{Prob}_{\mathbb{Q}}^{\text{urn}}(\Omega)$ induces such a record by

$$p_{\nu}(\omega) := \frac{\#\{i : \text{Fin}(N) \mid f(i) = \omega\}}{N}.$$

Equivalently, in weighted finite-support form, ν is represented by the list of its urn cells with weight 1. Conversely, every weighted common-denominator record can be represented by an urn by replacing each pair (ω, m) by m urn cells labelled ω . Thus the urn and common-denominator record presentations are extensionally equivalent: they assign the same value to every decidable event. For ordinary finite rational records on finite event types, this applies after multiplying by a common denominator. This equivalence is not asserted for arbitrary rational-valued functions on arbitrary codomains X in Definition A.3.4; it uses finite support, and the usual record form over Ω additionally uses the finite enumeration of Ω .

Definition A.3.8 (Constructive Clayton–Waddington urn assumptions). For $N > 0$ and $0 \leq k \leq N$, let $U_N(k)$ denote the plausibility assigned to any k -cell decidable event in a symmetric N -cell urn. The constructive Clayton–Waddington assumptions used here are:

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A.3.8.1. finite urn existence: for every positive N , the symmetric urn $\text{Fin}(N)$ is available;

A.3.8.2. qualitative coherence: equality of plausibility is an equivalence relation, strict plausibility comparison respects this equivalence, and equivalent plausibilities are not strictly ordered;

A.3.8.3. logical equivalence: equivalent decidable events receive the same plausibility;

A.3.8.4. relabelling invariance: permuting $\text{Fin}(N)$ does not change plausibility;

A.3.8.5. strict monotonicity: if $k < \ell \leq N$, then $U_N(k)$ is strictly less plausible than $U_N(\ell)$;

A.3.8.6. refinement invariance: replacing every cell by $c > 0$ indistinguishable subcells preserves plausibility, so $U_N(k) = U_{cN}(ck)$.

These assumptions concern finite decidable urn events. They do not include a complement rule for arbitrary propositions. Product urn compatibility is used only after this qualitative rescaling step: if an M -cell urn and an N -cell urn are combined, their rectangular product is an MN -cell urn.

Definition A.3.9 (Conditional probability and Bayesian update). Let $\mu : \text{Prob}_{\mathbb{Q}}(\Omega)$ and let $A, B \subseteq \Omega$. If $P_{\mu}(B) > 0$, then

$$P_{\mu}(A \mid B) := \frac{P_{\mu}(A \cap B)}{P_{\mu}(B)}.$$

Bayesian update is the same operation read as a transformation on probability assignments: given $\mu = (p, h) : \text{Prob}_{\mathbb{Q}}(\Omega)$ and an event $E \subseteq \Omega$ with $P_{\mu}(E) > 0$, Bayesian update produces $\mu_E : \text{Prob}_{\mathbb{Q}}(\Omega)$ with mass function

$$p_E(\omega) := \frac{\mathbb{1}_E(\omega)p(\omega)}{P_{\mu}(E)}.$$

Both conditional probability and Bayesian update are partial: they are defined only when the conditioning event has positive probability under μ . Bayes' theorem is not part of the definition; it is the finite-event identity proved in Theorem B.4.10.

Definition A.3.10 (Independence and conditional independence of decidable events). Let $\mu : \text{Prob}_{\mathbb{Q}}(\Omega)$ and let $A, B, C \subseteq \Omega$ be decidable events.

Unconditional independence. The events A and B are *independent under μ* , written $A \perp_{\mu} B$, when

$$P_{\mu}(A \cap B) = P_{\mu}(A) \cdot P_{\mu}(B).$$

Equivalently, when $P_{\mu}(B) > 0$, $P_{\mu}(A \mid B) = P_{\mu}(A)$; and when $P_{\mu}(A) > 0$, $P_{\mu}(B \mid A) = P_{\mu}(B)$.

Conditional independence. The events A and B are *conditionally independent given C under μ* , written $A \perp_{\mu} B \mid C$, when

$$P_{\mu}(A \cap B \cap C) \cdot P_{\mu}(C) = P_{\mu}(A \cap C) \cdot P_{\mu}(B \cap C).$$

Equivalently, when $P_\mu(C) > 0$, $P_\mu(A \cap B \mid C) = P_\mu(A \mid C) \cdot P_\mu(B \mid C)$; and when additionally $P_\mu(B \cap C) > 0$, $P_\mu(A \mid B \cap C) = P_\mu(A \mid C)$. The product form is the primary definition because it does not require a positivity side condition; the conditional forms are recorded as equivalent characterisations when the relevant denominators are realised.

Variable families. Suppose $\Omega = \text{Val}(\mathcal{V})$ for a finite family of variables $\mathcal{V}$, and $\mathcal{X}, \mathcal{Y}, \mathcal{Z} \subseteq \mathcal{V}$. A joint event on $\mathcal{X}$ is read under μ as the corresponding cylinder event on Ω , obtained by precomposition with the projection $\text{Val}(\mathcal{V}) \rightarrow \text{Val}(\mathcal{X})$. The families $\mathcal{X}$ and $\mathcal{Y}$ are independent under μ when every joint event on $\mathcal{X}$ is independent of every joint event on $\mathcal{Y}$ in the product sense. They are conditionally independent given $\mathcal{Z}$, written $\mathcal{X} \perp_\mu \mathcal{Y} \mid \mathcal{Z}$, when, for every joint value assignment z to $\mathcal{Z}$, every joint event on $\mathcal{X}$ is conditionally independent of every joint event on $\mathcal{Y}$ given the cylinder event $\mathcal{Z} = z$ in the product sense.

Notation conventions. The symbol $\perp$ is used in two readings throughout this thesis: *probabilistic* independence, always subscripted by the probability assignment (e.g. $A \perp_\mu B$, $\mathcal{X} \perp_\mu \mathcal{Y} \mid \mathcal{Z}$, and $\perp_{P_\mathcal{M}}$ when the assignment is the observational distribution induced by a structural causal model $\mathcal{M}$ in the sense of Definition A.3.29); and *graphical* d-separation, always carrying a graph subscript on the surrounding expression, e.g. $(X \perp Y \mid Z)_\mathcal{G}$, as in Definitions A.3.23 and A.3.26. Every occurrence of $\perp$ in this thesis carries one of these two subscripts; the probabilistic subscript is the distribution under discussion at that point, as in Definition A.3.19.

Remark A.3.11. The equivalence between the product and conditional forms is the one-line calculation $P_\mu(A \mid B) \cdot P_\mu(B) = P_\mu(A \cap B)$ for the unconditional case, and the analogous $P_\mu(A \cap B \mid C) \cdot P_\mu(C) = P_\mu(A \cap B \cap C)$ for the conditional case, valid when the relevant denominator is positive by Definition A.3.9. Because the product form imposes no positivity side condition, it is the form used inside the SCM proofs of Sections B.5 and B.6, including Lemma B.6.1. In those proofs, conditioning events on observed variables can have zero rational mass when the structural functions only realise a proper subset of the declared value type; the conditional form $P_\mu(A \mid B) = P_\mu(A)$ is then undefined, while the product form $P_\mu(A \cap B) = P_\mu(A) \cdot P_\mu(B)$ remains a well-defined equality, with both sides equal to zero whenever B is unreachable.

Definition A.3.12 (Marginalisation). *Event form.* Let $\mu : \text{Prob}_\mathbb{Q}(\Omega)$, and let $\{F_y\}_{y:Y}$ be a finite family of decidable events on Ω indexed by a finite type Y that is pairwise disjoint and jointly exhaustive (every $\omega : \Omega$ lies in exactly one F_y). For any decidable event $A \subseteq \Omega$, the *marginal of A over Y* is the finite sum

$$P_\mu(A) = \sum_{y:Y} P_\mu(A \cap F_y).$$

This is finite additivity (Theorem B.4.6) applied to the disjoint decomposition $A = \bigcup_{y:Y} (A \cap F_y)$.

Variable form. If $\Omega = \text{Val}(\mathcal{V})$ for a finite family of variables $\mathcal{V}$ in the sense of Definition A.3.6, and $\mathcal{Y} \subseteq \mathcal{V}$ is a subset of variables, the marginal of μ over the complement $\mathcal{V} \setminus \mathcal{Y}$ is the probability assignment on $\text{Val}(\mathcal{Y})$ obtained by finite summation over the remaining variables:

$$P_\mu(\mathcal{Y} = y) := \sum_{z:\text{Val}(\mathcal{V} \setminus \mathcal{Y})} P_\mu(\mathcal{Y} = y, \mathcal{V} \setminus \mathcal{Y} = z).$$

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The summation is over a *finite fibre*: the fibre of the projection $\text{Val}(\mathcal{V}) \rightarrow \text{Val}(\mathcal{Y})$ over y is the finite type $\text{Val}(\mathcal{V} \setminus \mathcal{Y})$. Both forms are special cases of finite additivity; the variable form is the version used inside the do-calculus derivation steps of Definition A.3.27 and in the SCM proofs of Sections B.5 and B.6.

Definition A.3.13 (Directed acyclic graph). A directed graph is a pair $\mathcal{G} = (V, E)$ with finite node type V and decidable edge relation $E : V \times V \rightarrow \text{Bool}$. Write $X \rightarrow Y$ when $E(X, Y) = \text{true}$. A directed path is a sequence $X_0 \rightarrow X_1 \rightarrow \dots \rightarrow X_k$. A graph path, used for d-separation, is a sequence of adjacent nodes in which each consecutive pair is connected by an edge in either direction. The directed graph is acyclic when there is no directed path with $k > 0$ and $X_0 = X_k$; a directed acyclic graph is such an acyclic directed graph.

Definition A.3.14 (Causal graph). A causal graph is a graph whose nodes are variables and whose edges encode the direct causal structure assumed by a model. Two cases are distinguished:

- *Markovian (finite) case*: $\mathcal{G}$ is a directed acyclic graph with no bidirected edges. All causal dependence between observed variables is mediated by declared parents in $\mathcal{G}$.
- *Semi-Markovian case*: $\mathcal{G}$ is a finite directed-and-bidirected graph over observed variables, with a decidable directed edge relation and a decidable symmetric bidirected edge relation. Directed arrows represent direct causal dependence between observed variables; a bidirected edge $X \leftrightarrow Y$ represents an unobserved common cause of X and Y . In this setting, a graph path may use directed edges in either orientation and bidirected edges as adjacencies. The directed part of the graph is acyclic. This is the usual graphical setting for c-components and hedges in finite identification theory.

Definition A.3.15 (Parents, ancestors and descendants). In a directed graph $\mathcal{G}$, the parents, ancestors and descendants of Y are

$$\text{Pa}_{\mathcal{G}}(Y) := \{X \mid X \rightarrow Y\},$$

$$\text{An}_{\mathcal{G}}(Y) := \{X \mid X = Y \text{ or there is a directed path } X \rightarrow \dots \rightarrow Y\},$$

and

$$\text{De}_{\mathcal{G}}(Y) := \{X \mid X = Y \text{ or there is a directed path } Y \rightarrow \dots \rightarrow X\}.$$

For a semi-Markovian directed-and-bidirected graph, these three notions are computed in the directed part of the graph; bidirected edges are used for c-components, hedges and the latent-variable reading of d-separation, but they do not themselves create directed parents, ancestors or descendants.

Definition A.3.16 (Chains, forks and colliders). On a path through three variables, a chain has the form $A \rightarrow B \rightarrow C$ or $A \leftarrow B \leftarrow C$, a fork has the form $A \leftarrow B \rightarrow C$, and a collider has the form $A \rightarrow B \leftarrow C$. Chains and forks are non-collider configurations; colliders behave differently under conditioning.

Remark A.3.17. This vocabulary is used to read d-separation (Definition A.3.23) informally; the formal d-separation predicate quantifies over paths and conditioning sets directly, so these three configurations appear inside its non-collider/collider clauses rather than as standalone objects.

Definition A.3.18 (Causal Bayesian network). A causal Bayesian network is a directed acyclic graph $\mathcal{G}$ over variables $V_1, \dots, V_n$ together with a finite rational probability distribution P that factorises as

$$P(v_1, \dots, v_n) = \prod_{i=1}^n P(v_i \mid \text{pa}_i),$$

where pa_i is the tuple of values of $\text{Pa}_{\mathcal{G}}(V_i)$. The intended causal reading is that each variable is independent of its non-descendants conditional on its parents.

Definition A.3.19 (Causal Markov condition). Let $\mathcal{V}$ be the full variable set of a directed acyclic graph $\mathcal{G}$. A finite rational probability distribution P satisfies the causal Markov condition with respect to $\mathcal{G}$ when, for every variable $X \in \mathcal{V}$,

$$X \perp\!\!\!\perp_P (\mathcal{V} \setminus (\text{De}_{\mathcal{G}}(X) \cup \text{Pa}_{\mathcal{G}}(X))) \mid \text{Pa}_{\mathcal{G}}(X).$$

This is the bridge from the graph to ordinary conditional independence claims [Pea09].

Remark A.3.20. This condition is recorded for completeness and for comparison with classical presentations; the formal correspondence theorem of Theorem B.5.2 works directly with the factorisation of Definition A.3.18 rather than with this independence form.

Definition A.3.21 (Faithfulness). A distribution P is faithful to a causal graph $\mathcal{G}$ when every conditional independence in P is entailed by d-separation in $\mathcal{G}$ and every d-separation in $\mathcal{G}$ corresponds to a conditional independence in P . Markovness says that the graph licenses certain independencies; faithfulness says that the distribution contains no extra independencies produced by accidental parameter cancellation.

Remark A.3.22. Faithfulness is not assumed by any result in this thesis. It is recorded as background context for readers familiar with the causal-discovery literature, where faithfulness is often used to infer graph structure from observed independencies. The identification and do-calculus completeness results transferred in Section B.6 do not rely on faithfulness; they work with the graph and model class as given.

Definition A.3.23 (d-separation). d-separation is the graphical criterion for reading conditional independence from a directed acyclic graph. Here a path means a graph path in the sense of Definition A.3.13, not necessarily a directed path. A path is blocked by a conditioning set Z if it contains a non-collider on the path (in the sense of Definition A.3.16) that is in Z , or if it contains a collider such that neither the collider nor any of its descendants is in Z . If every path between variables in X and variables in Y is blocked by Z , then X and Y are d-separated by Z , written $(X \perp\!\!\!\perp Y \mid Z)_{\mathcal{G}}$ in the graph [FK23; Pea09]. In a semi-Markovian graph with bidirected edges, this notation is read through the canonical finite latent-variable directed acyclic graph obtained by replacing each bidirected edge $A \leftrightarrow B$ by a finite latent root with arrows into A and B .

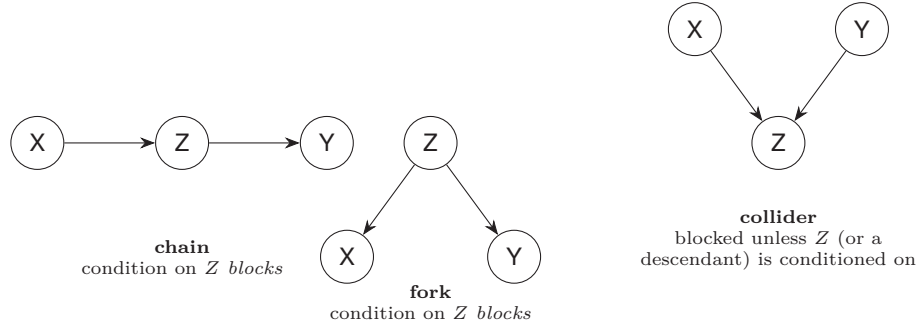


Figure A.1: The three path configurations behind d-separation (Definitions A.3.16 and A.3.23): chains and forks are non-colliders (conditioning on the middle node blocks the path); a collider blocks the path by default and is *opened* by conditioning on it or any of its descendants.

Definition A.3.24 (Do-operator). The expression $\text{do}(X = x)$ denotes an intervention that sets X to x . It differs from the observation $X = x$ because it changes the causal mechanism for X rather than only conditioning on the event that X has value x . The model-theoretic content of $\text{do}(X = x)$ is the intervened structural causal model of Definition A.3.30; its graph-level content is the mutilated graph of Definition A.3.25.

Definition A.3.25 (Graph mutilation). Graph mutilation is Pearl’s graphical representation of intervention. For a graph $\mathcal{G}$ and a set of variables X , the mutilated graph $\mathcal{G}_{\overline{X}}$ is obtained by deleting all directed arrows into X , because an intervention on X replaces its usual causes. The graph $\mathcal{G}_{\underline{X}}$ is obtained by deleting all directed arrows out of X . In the semi-Markovian reading, these operations are applied after passing to the canonical finite latent-variable directed acyclic graph of Definition A.3.23; equivalently, a bidirected edge incident with an intervened variable no longer contributes a latent arrowhead into that variable in $\mathcal{G}_{\overline{X}}$. These operations are combined when several marks occur together.

Definition A.3.26 (Do-calculus rules). Let $\mathcal{G}$ be a causal graph. All probability expressions below are finite conditional expressions in the sense of Definition A.3.9, so any displayed conditioning event carries the required positive-denominator side condition. Pearl’s do-calculus consists of three transformation rules [Pea09; Pea12]:

A.3.26.1. insertion and deletion of observations:

$$P(y \mid \text{do}(x), z, w) = P(y \mid \text{do}(x), w)$$

whenever $(Y \perp\!\!\!\perp Z \mid X, W)_{\mathcal{G}_{\overline{X}}}$;

A.3.26.2. action and observation exchange:

$$P(y \mid \text{do}(x), \text{do}(z), w) = P(y \mid \text{do}(x), z, w)$$

whenever $(Y \perp\!\!\!\perp Z \mid X, W)_{\mathcal{G}_{\overline{X}, \underline{Z}}}$;

A.3.26.3. insertion and deletion of actions:

$$P(y \mid \text{do}(x), \text{do}(z), w) = P(y \mid \text{do}(x), w)$$

whenever $(Y \perp\!\!\!\perp Z \mid X, W)_{\mathcal{G}_{\overline{X}, \overline{Z(W)}}$, where $Z(W)$ contains the variables in Z that are not ancestors of any variable in W in $\mathcal{G}_{\overline{X}}$.

Definition A.3.27 (Do-calculus derivation). Let $\mathcal{G}$ be a finite causal graph in the Markovian or semi-Markovian sense of Definition A.3.14. A *do-calculus derivation* of an equality $E_0 = E_n$ between finite probability expressions over $\mathcal{G}$ is a finite sequence

$$E_0 \longrightarrow E_1 \longrightarrow \cdots \longrightarrow E_n$$

in which each step $E_i \longrightarrow E_{i+1}$ is one of the following finite rule applications:

A.3.27.1. *Reflexivity step*: $E_i = E_{i+1}$ as a definitional equality of finite rational expressions.

A.3.27.2. *Model-evaluation step*: E_i contains a sub-expression $P_{\mathcal{M}_x}(y)$ for an intervened structural causal model $\mathcal{M}_x$ (Definition A.3.30) whose recursive evaluation determines that probability as a finite rational expression r in the exogenous distribution and structural functions of $\mathcal{M}_x$; E_{i+1} replaces $P_{\mathcal{M}_x}(y)$ by r in E_i . This step is used in finite countermodel arguments (such as the proof of Lemma B.6.2) where a concrete numerical value of $P_{\mathcal{M}_x}(y)$ is needed. Each such step is licensed by Theorem B.5.2.

A.3.27.3. *Probability step*: E_{i+1} is obtained from E_i by a finite rewrite using one of finite additivity (Theorem B.4.6), the definition of conditional probability (Definition A.3.9) with explicit positive denominator, or marginalisation in the sense of Definition A.3.12.

A.3.27.4. *Do-calculus step*: E_{i+1} is obtained from E_i by one application of the insertion-and-deletion-of-observations rule, the action-and-observation-exchange rule, or the insertion-and-deletion-of-actions rule of Definition A.3.26, together with a finite proof that the licensing d-separation condition holds in the appropriate mutilated graph.

A derivation is therefore a finite sequence of finite rule applications, each carrying its own finite side conditions. The set of do-calculus derivations is closed under composition: if $E_0 \rightarrow \cdots \rightarrow E_n$ and $E_n \rightarrow \cdots \rightarrow E_m$ are derivations, their concatenation is a derivation of $E_0 = E_m$. This is the formal object on which the induction in the proof of Proposition B.5.15 runs.

Definition A.3.28 (Endogenous and exogenous variables). In a structural causal model, endogenous variables are the variables whose values are determined inside the model by structural functions. Exogenous variables are background or noise variables whose values are not explained inside the model but are assigned a finite rational probability distribution. The split between observed and unobserved variables is orthogonal: endogenous variables are observed by convention here, while exogenous variables may be either observed or unobserved. Exogenous variables that are unobserved are called *latent*; the projection onto the observed variables that records latent common causes as bidirected edges is the finite latent projection of Definition B.6.9.

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Definition A.3.29 (Structural causal model, solution map and observational distribution). A finite structural causal model is a tuple

$$\mathcal{M} = (\mathcal{U}, \mathcal{V}, \mathcal{F}, P_{\mathcal{U}})$$

where $\mathcal{U}$ is a finite family of exogenous finite variables, $\mathcal{V}$ is a finite family of endogenous finite variables, $P_{\mathcal{U}}$ is a finite rational probability distribution over $\text{Val}(\mathcal{U})$, and $\mathcal{F}$ is the structural package consisting of a decidable declared endogenous-parent relation $\text{Pa}(V) \subseteq \mathcal{V}$, finite declared exogenous-input families $\mathcal{U}_V \subseteq \mathcal{U}$, and structural functions

$$f_V : \text{Val}(\text{Pa}(V)) \times \text{Val}(\mathcal{U}_V) \rightarrow \text{Val}(V) \quad (V \in \mathcal{V}).$$

Thus each endogenous variable is assigned from its declared endogenous parents and relevant exogenous inputs. The exogenous-input declaration is the exogenous-incidence relation used in Definition B.5.1. The *induced causal graph* $\mathcal{G}$ (in the sense of Definition A.3.14) is read off the declared parent relation and this incidence relation: its nodes are the endogenous variables $\mathcal{V}$; there is a directed edge $X \rightarrow V$ exactly when X is a declared endogenous parent of V , i.e. $X \in \text{Pa}(V)$; and, in the semi-Markovian reading obtained after the finite latent projection of Definition B.6.9, there is a bidirected edge $A \leftrightarrow B$ exactly when some unobserved exogenous variable lies in both $\mathcal{U}_A$ and $\mathcal{U}_B$.

All structural causal models in this thesis are *recursive*: the directed part of the induced causal graph $\mathcal{G}$ is acyclic, and the endogenous variables admit a topological order in which each structural equation depends only on earlier endogenous variables and on its exogenous inputs. In this case the structural equations have a unique solution for each exogenous assignment $u : \text{Val}(\mathcal{U})$ [Pea09].

The *solution map* of $\mathcal{M}$ is the total function

$$\text{sol}_{\mathcal{M}} : \text{Val}(\mathcal{U}) \rightarrow \text{Val}(\mathcal{V})$$

defined by recursion over the topological order: for each endogenous variable V_i ,

$$\text{sol}_{\mathcal{M}}(u)_i := f_{V_i}(\text{sol}_{\mathcal{M}}(u)_{<i} \upharpoonright_{\text{Pa}(V_i)}, u \upharpoonright_{\mathcal{U}_{V_i}}),$$

where $\text{sol}_{\mathcal{M}}(u)_{<i}$ denotes the already-computed values of the earlier endogenous variables and the restriction symbols pick out the declared parents and exogenous inputs of V_i . The recursion terminates after $|\mathcal{V}|$ steps by recursiveness.

The *observational distribution induced by $\mathcal{M}$* is the pushforward of $P_{\mathcal{U}}$ along $\text{sol}_{\mathcal{M}}$. For any decidable event $E \subseteq \text{Val}(\mathcal{V})$,

$$P_{\mathcal{M}}(E) := \sum_{u : \text{Val}(\mathcal{U})} \mathbb{1}_E(\text{sol}_{\mathcal{M}}(u)) P_{\mathcal{U}}(u),$$

using the indicator of Definition A.1.18 and the finite probability assignment of Definition A.3.7. Marginal and conditional probabilities derived from $P_{\mathcal{M}}$ are taken in the sense of Definitions A.3.9 and A.3.12; (conditional) independence under $P_{\mathcal{M}}$ is written $\perp\!\!\!\perp_{P_{\mathcal{M}}}$ in the notation convention of Definition A.3.10.

Definition A.3.30 (Intervention and interventional distribution). For a structural causal model $\mathcal{M} = (\mathcal{U}, \mathcal{V}, \mathcal{F}, P_{\mathcal{U}})$ in the sense of Definition A.3.29 and a

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joint assignment $x : \text{Val}(X)$ to a finite set of endogenous variables $X \subseteq \mathcal{V}$, the *intervened model* is

$$\mathcal{M}_x := (\mathcal{U}, \mathcal{V}, \mathcal{F}_x, P_{\mathcal{U}}),$$

where $\mathcal{F}_x$ replaces, for each $V \in X$, the declared endogenous-parent and exogenous-input families of V by empty families and replaces f_V by the constant value $x_V : \text{Val}(V)$; all other declarations and structural functions are left unchanged. The intervened model is again recursive over the same topological order; in particular, the solution map $\text{sol}_{\mathcal{M}_x}$ and the observational distribution $P_{\mathcal{M}_x}$ are defined as in Definition A.3.29. The *interventional distribution* associated with $\text{do}(X = x)$ is then

$$P_{\mathcal{M}_x}(E) = \sum_{u: \text{Val}(\mathcal{U})} \mathbf{1}_E(\text{sol}_{\mathcal{M}_x}(u)) P_{\mathcal{U}}(u)$$

for any decidable event $E \subseteq \text{Val}(\mathcal{V})$. The interventional distribution $P_{\mathcal{M}_x}$ is the model-theoretic content of the intervention $\text{do}(X = x)$; its graph-level content is the mutilated graph $\mathcal{G}_{\overline{X}}$ of Definition A.3.25.

Definition A.3.31 (Compatibility and identification). Let $\mathcal{G}$ be a finite causal graph in the Markovian or semi-Markovian sense of Definition A.3.14, and let $\mathcal{M}$ be a finite recursive structural causal model in the sense of Definition A.3.29. The model $\mathcal{M}$ is *compatible with* $\mathcal{G}$ when the induced causal graph of $\mathcal{M}$ agrees with $\mathcal{G}$ on its directed edges and, in the semi-Markovian case, on its bidirected edges as induced by the exogenous-incidence relation, and when $\mathcal{M}$ satisfies the finite recursive structure required by the fragment. Write $\text{Mod}(\mathcal{G})$ for the class of finite rational recursive models compatible with $\mathcal{G}$.

A finite interventional query is an expression $Q = P_x(E)$, often written $P_x(y)$ for the event $Y = y$, where $P_x(E)$ abbreviates $P_{\mathcal{M}_x}(E)$ for the intervened model of Definition A.3.30. Such a query is *identifiable from $\mathcal{G}$ and a finite rational observational distribution P^{obs}* when, for every $\mathcal{M} \in \text{Mod}(\mathcal{G})$ whose observational distribution $P_{\mathcal{M}}$ in the sense of Definition A.3.29 equals P^{obs} , the corresponding interventional distribution $P_{\mathcal{M}_x}$ returns the same value on the target event E . Equivalently, any two compatible models in $\text{Mod}(\mathcal{G})$ with the same observational distribution give the same value to the target interventional query. In that case the effect can, in principle, be written as a formula involving only observational quantities [Pea09; SP06; TS09].

Definition A.3.32 (c-component). In a semi-Markovian causal graph, a c-component is a maximal set of observed variables connected by bidirected paths, where a bidirected path is a graph path using only bidirected edges. The bidirected edges represent shared unobserved causes, so c-components mark the parts of the graph tied together by latent confounding [TP02; TS09].

Definition A.3.33 (c-forest). A c-forest is a directed-and-bidirected subgraph whose observed variables form one c-component and in which each observed variable has at most one directed child inside the subgraph. The roots of the c-forest are the variables with no directed children inside it.

Definition A.3.34 (Hedge). A hedge for the effect of X on Y is a pair of c-forests F and F' with the same roots such that $F' \subset F$, F contains at least one variable in X , F' contains no variable in X , and the roots lie among the ancestors of Y in the graph where arrows into X have been deleted. The existence of a

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hedge witnesses that the causal effect is not identifiable from the available finite rational observational distribution [TS09].

Definition A.3.35 (Counterfactual query). Given a structural causal model $\mathcal{M}$ in the sense of Definition A.3.29, an exogenous assignment u , and an intervention $X = x$ in the sense of Definition A.3.30, the counterfactual $Y_x(u)$ is the value of Y in the solution of the intervened model $\mathcal{M}_x$ under u , that is, $\text{sol}_{\mathcal{M}_x}(u)$ projected to Y . With observed evidence, Pearl’s evaluation procedure is abduction, action and prediction: update the finite rational probability distribution over u , replace the relevant equations, and compute the resulting value or finite rational probability distribution.

Concretely, if $E \subseteq \text{Val}(\mathcal{V})$ is a decidable observed event with positive probability, write

$$p_E(u) := \frac{\mathbb{1}_E(\text{sol}_{\mathcal{M}}(u))P_{\mathcal{U}}(u)}{P_{\mathcal{M}}(E)}.$$

Then the model-level counterfactual probability of a decidable target event $F \subseteq \text{Val}(\mathcal{V})$ under the action $X = x$ is

$$P(F_x \mid E) := \sum_{u: \text{Val}(\mathcal{U})} \mathbb{1}_F(\text{sol}_{\mathcal{M}_x}(u)) p_E(u).$$

This is a calculation inside the given structural causal model. It is not, by itself, an identifiability claim saying that the value is determined by an observational distribution or a graph alone.

Remark A.3.36. Counterfactual queries are placed here, after the identification machinery of Definitions A.3.31 to A.3.34, because they sit on Pearl’s third level of inquiry: the unconditional and conditional interventional fragment (Pearl’s second level) is closed by the identification definitions above, and counterfactuals add an extra step (abduction) on top of intervention. They are recorded for orientation; the formal completeness transfer of Theorem B.6.13 is restricted to ordinary and conditional interventional distributions, and does not cover counterfactual identification.

Definition A.3.37 (Epistemic causal mode). In this thesis, an epistemic causal mode is a mode whose content includes a finite structural causal model. This is a thesis-specific construction: it uses Gratzner-style modes to organise Pearl-style causal models as epistemic states. Moving between such modes represents changing the causal model, changing the available information, or changing which interventions and counterfactuals are in scope.

A.4 Examples

The example collected here is the worked-out finite calculation supporting the running tenure-track example of Example 5.4.1. It is placed in this appendix rather than alongside the definitions because it is a numeric illustration, not a definitional commitment of the theory. The narrative version in the Fourth movement carries the conceptual weight; the calculations below carry the rational arithmetic.

Example A.4.1 (Mira’s tenure-track search, finite calculation). This is the fully finite version of the tenure-track model used in Example 5.4.1. All endogenous

value types are the two-element Boolean type. The endogenous variables are

$$\mathcal{V} = \{\text{Pr}, \text{Qu}, \text{Top}, \text{Com}, \text{Fit}, \text{Short}, \text{Fund}, \text{Off}\}.$$

The exogenous variables are independent Boolean variables

$$U_B, U_{\text{Pr}}, U_{\text{Qu}}, U_{\text{Top}}, U_{\text{Com}}, U_{\text{Fund}},$$

with finite rational masses

$$P(U_B = 1) = \frac{1}{3}, \quad P(U_{\text{Pr}} = 1) = P(U_{\text{Qu}} = 1) = P(U_{\text{Fund}} = 1) = \frac{1}{4},$$

and

$$P(U_{\text{Top}} = 1) = P(U_{\text{Com}} = 1) = \frac{1}{2}.$$

The structural functions are

$$\begin{aligned} \text{Pr} &:= U_B \vee U_{\text{Pr}}, & \text{Qu} &:= U_B \vee U_{\text{Qu}}, \\ \text{Top} &:= U_{\text{Top}}, & \text{Com} &:= U_{\text{Com}}, \\ \text{Fit} &:= \text{Top} \wedge \text{Com}, & \text{Short} &:= \text{Qu} \wedge (\text{Pr} \vee \text{Fit}), \\ \text{Fund} &:= \text{Top} \vee U_{\text{Fund}}, & \text{Off} &:= \text{Short} \wedge \text{Fund}. \end{aligned}$$

In the full finite directed acyclic graph, U_B is a common cause of Pr and Qu . If U_B is hidden, the observed semi-Markovian graph has the bidirected edge $\text{Pr} \leftrightarrow \text{Qu}$, so $\{\text{Pr}, \text{Qu}\}$ is a c-component. The graph also contains the colliders

$$\text{Top} \rightarrow \text{Fit} \leftarrow \text{Com}, \quad \text{Pr} \rightarrow \text{Short} \leftarrow \text{Fit}, \quad \text{Short} \rightarrow \text{Off} \leftarrow \text{Fund}.$$

Thus Top and Com are d-separated by the empty conditioning set, but the path through Fit is opened when one conditions on Fit or one of its descendants. Numerically, the deterministic equation $\text{Fit} := \text{Top} \wedge \text{Com}$ gives

$$P(\text{Top} = 1 \mid \text{Com} = 1) = \frac{1}{2} = P(\text{Top} = 1),$$

while conditioning on the collider event $\text{Fit} = 0$ changes the relation:

$$P(\text{Top} = 1 \mid \text{Fit} = 0) = \frac{1}{3}, \quad P(\text{Top} = 1 \mid \text{Com} = 1, \text{Fit} = 0) = 0.$$

This is the collider phenomenon in a finite deterministic form.

The distinction between observation and intervention is also visible by finite summation. First,

$$P(\text{Pr} = 1) = 1 - P(U_B = 0, U_{\text{Pr}} = 0) = 1 - \frac{2}{3} \cdot \frac{3}{4} = \frac{1}{2}.$$

Similarly $P(\text{Qu} = 1) = 1/2$, and

$$P(\text{Pr} = 1, \text{Qu} = 1) = 1 - \frac{1}{2} - \frac{1}{2} + \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{3}{8}.$$

Hence

$$P(\text{Qu} = 1 \mid \text{Pr} = 1) = \frac{P(\text{Pr} = 1, \text{Qu} = 1)}{P(\text{Pr} = 1)} = \frac{3/8}{1/2} = \frac{3}{4}.$$

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But the intervention $\text{do}(\text{Pr} = 1)$ replaces the equation for Pr by the constant value 1 and leaves the mechanism for Qu unchanged, so

$$P(\text{Qu} = 1 \mid \text{do}(\text{Pr} = 1)) = P(\text{Qu} = 1) = \frac{1}{2}.$$

A second calculation shows why observing committee fit is not the same as forcing committee fit. Since $\text{Fit} = 1$ implies $\text{Top} = 1$ and $\text{Com} = 1$, it also implies $\text{Fund} = 1$. Under $\text{Fit} = 1$, the shortlist equation reduces to $\text{Short} = \text{Qu}$, and therefore

$$P(\text{Off} = 1 \mid \text{Fit} = 1) = P(\text{Qu} = 1) = \frac{1}{2}.$$

By contrast, in the intervened model $\text{do}(\text{Fit} = 1)$, the variable Top is not forced. The offer requires both $\text{Qu} = 1$ and $\text{Fund} = 1$, and these are independent because they depend on disjoint exogenous variables. Since

$$P(\text{Fund} = 1) = 1 - P(\text{Top} = 0, U_{\text{Fund}} = 0) = 1 - \frac{1}{2} \cdot \frac{3}{4} = \frac{5}{8},$$

one gets

$$P(\text{Off} = 1 \mid \text{do}(\text{Fit} = 1)) = \frac{1}{2} \cdot \frac{5}{8} = \frac{5}{16}.$$

Finally, the same finite model permits a model-level counterfactual calculation. Let

$$E := \{\text{Pr} = 0, \text{Qu} = 1, \text{Off} = 0\}.$$

Under the structural equations, $\text{Pr} = 0$ forces $U_B = 0$ and $U_{\text{Pr}} = 0$. Together with $\text{Qu} = 1$, this forces $U_{\text{Qu}} = 1$. Moreover, when $\text{Pr} = 0$ and $\text{Qu} = 1$, the equation for Short reduces to $\text{Short} = \text{Fit}$. Since $\text{Fit} = 1$ implies $\text{Top} = 1$, hence $\text{Fund} = 1$, the additional evidence $\text{Off} = 0$ forces $\text{Fit} = 0$. Thus

$$P(E) = \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{32},$$

so the abduction step is well defined.

Condition on E , then perform the action $\text{do}(\text{Pr} = 1)$. In the intervened model, $\text{Short} = 1$, because $\text{Qu} = 1$ is retained from the posterior evidence and Pr has been set to 1. Therefore the counterfactual offer depends only on

$$\text{Fund} = \text{Top} \vee U_{\text{Fund}}.$$

The evidence about Pr and Qu is independent of $U_{\text{Top}}, U_{\text{Com}}, U_{\text{Fund}}$, and the remaining relevant information is $\text{Fit} = 0$, i.e. not both Top and Com . Hence

$$P(\text{Fund} = 1 \mid E) = P(\text{Fund} = 1 \mid \text{Fit} = 0) = \frac{1/4 + 1/8}{3/4} = \frac{1}{2}.$$

Consequently

$$P(\text{Off}_{\text{Pr}=1} = 1 \mid E) = \frac{1}{2}.$$

The same posterior also gives

$$P(\text{Off}_{\text{Fit}=1} = 1 \mid E) = \frac{1}{2},$$

because setting $\text{Fit} = 1$ likewise makes $\text{Short} = 1$ while leaving the same posterior uncertainty about Fund . This is Pearl’s abduction–action–prediction procedure in the finite model: first condition the exogenous distribution, then change the structural equation, then evaluate the target. It is not a proof that arbitrary counterfactuals are identifiable from the observational distribution alone.

Internally to the thesis framework, this example is a finite structural-causal-model record at an epistemic causal mode:

$$\Gamma_{\text{ten}} \vdash \mathcal{M}_{\text{ten}} : \text{FinSCM}@m_{\text{ten}},$$

where FinSCM abbreviates the restricted finite type-theoretic structural-causal-model fragment of Definition B.5.1. The structural equations are terms in the Boolean value types; the exogenous distribution is a finite rational probability type; and the displayed probability calculations are equalities of finite sums and quotients with positive denominators. In the restricted Lean core of `lean/Thesis/SCM.lean`, the same shape is represented by a recursive finite record whose evaluation map pushes the exogenous distribution forward to observational and interventional probabilities.¹

¹The Lean module is called `Thesis.SCM` for backward compatibility with the existing formalisation.

APPENDIX B

Proofs and arguments

“The important thing is life — life and nothing else! What is any ‘discovery’ whatever compared with the incessant, eternal discovery of life?”

Hippolyte in *Idiot*

B.1 Causality argument

Human beings reason causally almost before they reason formally. We ask what made the glass fall, why the patient recovered, what caused the model to fail, which policy changed the outcome, and what would have happened had the intervention been different. This is not an exotic philosophical habit. It is one of the ordinary grammars of human orientation.

Science can be described, at least in part, as the disciplined extension of this causal grammar. It discovers, tests, rejects and stabilises causal relations in the material world, the biological world and the social world. A science without causality would not simply be incomplete in a technical sense. It would fail to speak the language in which many scientific questions are actually asked.

None of which is to say that any causal story will do. Scientific inquiry is communal, historical and bounded. With Huizinga, one may say that it contains a playful and creative element: scientists construct models, concepts and experimental situations together [Hui49]. With Erasmus, one may add that such constructions retain an ironic or provisional character: they are humanly made, and therefore should not be confused with reality itself [Era71]. With Kierkegaard, one may add that irony is not limitless: it is bounded by what has already been inherited [Kie80; Kie83; Kie89]. Fleck sharpens this social point: causal judgement is stabilised inside communities, inherited styles of thought, shared techniques and historically formed expectations [Fle79]. Hacking sharpens the practical point: science is not only representational, but interventionist, because scientists isolate, manipulate, produce, measure and stabilise phenomena [Hac83]. To this I add that scientific construction is bounded by intersubjective accountability: claims must pass through other people, practices, instruments, institutions and traditions of criticism.

The result is a middle position. Against a crude Feyerabendian reading, not anything goes [Fey75]. Against a crude Humean scepticism, causal talk need not be metaphysically guaranteed in order to be scientifically indispensable [Hum75; Hum78]. Against a purely Kantian framing, the question whether causality originates in the world or in human cognition becomes secondary once science is understood as a human practice of shared orientation, intervention and explanation [Kan98].

Pearl's framework is therefore justified pragmatically before it is justified metaphysically [Pea09; PM18]. It gives a rigorous language for something science already needs to do: distinguish seeing from doing, association from intervention, and prediction from counterfactual explanation. This thesis uses Pearl not because Pearl solves every philosophical problem about causality, but because any formal theory of mathematical inference for science must give causal reasoning a home.

B.2 Physics argument

The physics argument begins with a modest observation: measurements contain finite information. A measured value is not an actually completed real number handed down from the heavens. It is a finite construction, with precision, procedure, instrument and context. Gisin's work is valuable because it presses

this point against the habitual use of completed real numbers in physics [Bar26; BDG25; DG19; Gis20].

On a finite-information picture, measurement can actualise new information and the future can be genuinely open. One can recover analogues of quantum-style indeterminism, measurement-problem-like tensions, Wigner’s-friend-like situations and no-cloning phenomena. This is a helpful direction. Yet it does not recover the whole quantum core: incompatible observables, Bell violations, contextuality, the precise Born-rule structure and full quantum entanglement remain outside the classical indeterministic reconstruction.

The difficulty is that Gisin’s proposed numbers still risk importing infinity through the back door. If an observed number is treated as an infinite sequence of propensities, then the number may look constructive while depending on an indefinitely extended object. That is not the constructivist instinct taken seriously enough. A genuinely constructive reading should say that measurement constructs a value up to the precision of the measurement, not that it reveals part of a hidden infinite propensity sequence.

Spekkens’ toy theory supports the epistemological half of the argument [HS10; Spe07]. Many features often advertised as distinctively quantum can arise from restrictions on knowledge: interference-like behaviour, noncommuting measurements, no-cloning, no-broadcasting, teleportation-like structure and epistemic update. Yet Spekkens also does not recover full quantum mechanics. Pusey, Barrett and Rudolph then sharpen the warning: one cannot keep a completely definite underlying ontic state and expect the epistemic view to do all the work [PBR12].

The natural conclusion is agnostic and constructive. Theories should be treated as epistemological vehicles for finite observations. There is no need, for the purposes of this thesis, to decide what the ultimate ontology of physics is. Measurements are finite; formal reasoning about them should therefore begin constructively. The epistemological reading gives room for Spekkens-style insight; the constructive reading gives room for Gisin-style finite information; and the refusal to fix a hidden ontology leaves space for the quantum phenomena that neither classical finite information nor purely epistemic toy theory fully captures.

B.3 Excluded middle is not derivable

The claim used in the main text is modest. It is not that Martin-Löf-style type theory proves the negation of excluded middle as an internal theorem. The claim is metatheoretic: global excluded middle is not derivable in the bare constructive theory. If one wants to use it, one must add it as an additional classical principle.

Proposition B.3.1 (Global excluded middle is not derivable). *In the Martin-Löf-style fragment used in this thesis, the uniform principle, read over small propositions or proposition-like types in the universe,*

$$\Pi_{A:\mathcal{U}}(A + \neg A)$$

is not derivable without adding a classical axiom or restricting A to a decidable class of types.

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Proof. The argument is by soundness with respect to standard Kripke/presheaf semantics for dependent type theory [AR11; Gra23]. The model used here validates the underlying non-modal fragment needed in this argument: contexts and substitution, dependent function types, finite product and sum types, the empty type, negation as function into the empty type and a small universe containing the proposition-like type used below. It is not a Boolean model, and it does not validate uniform excluded middle.

The countermodel is used only for the propositional fragment of the argument, so the intensional/extensional distinction for identity types does not affect the failure of uniform excluded middle.

Consider the two-stage Kripke frame

$$w_0 \leq w_1,$$

and let P be an atomic proposition which is not forced at w_0 but is forced at w_1 . Forcing is persistent along extensions: if a proposition is forced at an earlier stage, it remains forced at later stages. Thus information may grow from w_0 to w_1 , so it is coherent for P not to be forced at w_0 while being forced at w_1 .

At w_0 , P is not forced by construction. Moreover, $\neg P$ is not forced at w_0 , because forcing $\neg P$ would mean that no extension of w_0 could force P ; but w_1 is an extension of w_0 where P is forced. Hence neither disjunct of

$$P + \neg P$$

is forced at w_0 . Thus excluded middle is not valid in this sound model of the constructive propositional fragment.

If the displayed uniform excluded-middle term were derivable in bare Martin-Löf-style type theory for a universe containing this small propositional type P , its interpretation would give, at every stage and for every such proposition, either a witness of the proposition or a witness of its negation. Instantiating it with P at w_0 would force $P + \neg P$, contradicting the model above. Therefore no such term is derivable in the base theory. ■

The same warning applies to double-negation elimination as a global principle. Constructively one always has

$$A \rightarrow \neg\neg A,$$

by sending $a : A$ and $n : A \rightarrow \perp$ to $n(a) : \perp$. The converse

$$\neg\neg A \rightarrow A$$

is a classical principle and is not available uniformly in the base theory.

This does not conflict with the finite probability constructions used in this thesis. For particular finite types, equality, membership and Boolean predicates are decidable by construction. Thus one may use $E + \neg E$ for a finite decidable event E , and one may take complements inside finite event algebras. What is rejected is the global move from an arbitrary proposition A to $A + \neg A$ without a decision procedure or an explicit classical assumption.

B.4. Clayton–Waddington probability proof

Classical Cox / Van Horn route	This thesis (finite, constructive)
desiderata: real-valued, common sense, consistency	finite urns + relabelling, refinement, strict monotonicity
associativity equation $\Rightarrow$ product rule first	urn rescaling $U_N(k) \mapsto k/N \Rightarrow$ ratio and sum rule first
negation function $S \Rightarrow$ sum rule second	product urns $\Rightarrow$ product rule second
<i>universality</i> (R4) density assumed/derived	universality <i>eliminated</i> : values are k/N by construction
continuity on $[0, 1]$	no continuum; complements only on decidable events
construct rescaling p	record–urn equivalence $\rightarrow$ Kolmogorov laws $\rightarrow$ Bayes

Figure B.1: Where this thesis’s finite derivation departs from the Cox–Jaynes–Van Horn argument refined by Clayton and Waddington [CW17]: the sum and product rules are derived in the opposite order, and the universality assumption is dropped rather than patched.

B.4 Clayton–Waddington probability proof

This proof is deliberately finite. It recovers the ordinary finite probability calculus from urn-generated probability types, not the full measure-theoretic theory of countably additive probability. The constructive point is also important: no global law of excluded middle is used. Complements appear only inside finite decidable event algebras. The urn rescaling below is a representation result for the stated Clayton–Waddington-style assumptions; it does not claim that every possible plausibility calculus must satisfy those assumptions. The corresponding Lean 4 formalisation is in the module `Thesis.Probability`, in `lean/Thesis/Probability.lean`.

Proposition B.4.1 (Resolution-status partition). *Let S be a finite event type and let $\iota_A : S \rightarrow \text{Status}_A$ be a finite inspection procedure as in Definition A.3.2. Then the events Prov_A , Ref_A and Unr_A of Definition A.3.3 are decidable, pairwise disjoint and jointly exhaustive.*

Proof. Decidability follows because Status_A is the finite three-element type

$$\{\text{proved}, \text{refuted}, \text{unresolved}\}$$

and equality in a finite type is decidable. For any $s : S$, the value $\iota_A(s)$ is exactly one of these three constructors. If it is `proved`, then s lies in Prov_A and in neither of the other two events. The cases `refuted` and `unresolved` are identical. Thus the three events are pairwise disjoint and their union is S . ■

Proposition B.4.2 (Qualitative urn rescaling). *Assume the constructive Clayton–Waddington urn assumptions of Definition A.3.8. Then the plausibility scale for finite symmetric urn events can be rescaled so that*

$$p(U_N(k)) = \frac{k}{N}$$

for every $N > 0$ and $0 \leq k \leq N$. This rescaling is well-defined and order-preserving: equal ratios give the same qualitative plausibility class, and strictly

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smaller ratios give strictly smaller plausibility. Under this rescaling, decidable complements in an N -cell urn satisfy

$$p(E^c) = 1 - p(E).$$

Proof. Relabelling invariance implies that the plausibility of a k -cell event in $\text{Fin}(N)$ depends only on k and N , not on which particular k cells were chosen. Thus $U_N(k)$ is well-defined.

Refinement invariance gives

$$U_N(k) = U_{cN}(ck)$$

for every $c > 0$. If two fractions agree, $k/N = k'/N'$, then refinement to the common urn of size NN' gives

$$U_N(k) = U_{NN'}(kN') = U_{N'N}(k'N) = U_{N'}(k').$$

Hence the rational number k/N determines a single plausibility class. If $k/N < \ell/M$, refinement to the common urn of size NM gives $kM < \ell N$, and strict monotonicity makes the first refined event strictly less plausible than the second. Therefore the assignment

$$U_N(k) \mapsto \frac{k}{N}$$

is a well-defined order-preserving rescaling of the finite urn plausibilities, using qualitative coherence to transport strict comparison across plausibility equivalence.

In a decidable N -cell urn, the complement of a k -cell event has $N - k$ cells. Therefore

$$p(E^c) = \frac{N - k}{N} = 1 - \frac{k}{N} = 1 - p(E).$$

This is a statement about decidable urn complements, not about arbitrary propositions in Martin-Löf-style type theory. ■

Proposition B.4.3 (Product urns after rescaling). *After the rescaling of Proposition B.4.2, product urn compatibility gives the expected multiplication law for rectangular decidable events. If E has k cells in an M -cell urn and F has ℓ cells in an N -cell urn, then*

$$p(E \times F) = p(E)p(F).$$

Proof. The product event $E \times F$ has $k\ell$ cells in the product urn of size MN . Therefore the rescaling of Proposition B.4.2 gives

$$p(E \times F) = \frac{k\ell}{MN} = \frac{k}{M} \frac{\ell}{N} = p(E)p(F).$$

■

Proposition B.4.4 (Finite rational records admit a common denominator). *Let Ω be a finite event type and let*

$$\mu = (p, h) : \text{Prob}_{\mathbb{Q}}(\Omega)$$

B.4. Clayton–Waddington probability proof

be a finite rational probability record in the sense of Definition A.3.7. Then μ has an extensionally equal weighted common-denominator presentation: there are a positive natural number D and natural numbers m_ω indexed by $\omega : \Omega$ such that

$$p(\omega) = \frac{m_\omega}{D} \quad \text{for every } \omega : \Omega,$$

and

$$\sum_{\omega : \Omega} m_\omega = D.$$

Proof. Choose a finite enumeration $\omega_1, \dots, \omega_r$ of Ω . Since each $p(\omega_i)$ is a non-negative rational number, write it as

$$p(\omega_i) = \frac{a_i}{b_i}$$

with $a_i, b_i : \mathbb{N}$ and $b_i > 0$. Let

$$D := \prod_{i=1}^r b_i,$$

taking the empty product to be 1. For each i , define

$$m_i := a_i \prod_{j \neq i} b_j.$$

Then $m_i : \mathbb{N}$ and

$$\frac{m_i}{D} = \frac{a_i}{b_i} = p(\omega_i).$$

Summing over the finite enumeration and using the normalisation proof $h : \sum_{\omega : \Omega} p(\omega) = 1$ gives

$$\frac{1}{D} \sum_{i=1}^r m_i = \sum_{i=1}^r \frac{m_i}{D} = \sum_{i=1}^r p(\omega_i) = 1,$$

hence $\sum_i m_i = D$. This construction uses only the finite enumeration of Ω and the numerator-denominator data of the finitely many rational masses. It is therefore constructive in the finite rational fragment. $\blacksquare$

Proposition B.4.5 (Common-denominator record–urn equivalence). *Let X be any type. For weighted finite-support records in the common-denominator form of Definition A.3.7, urn presentations and record presentations are extensionally equivalent. More precisely:*

B.4.5.1. *every urn-generated assignment $\mu : \text{Prob}_{\mathbb{Q}}^{\text{urn}}(X)$ determines a weighted record R_μ such that*

$$P_{R_\mu}(E) = P_\mu(E)$$

for every decidable event $E : X \rightarrow \text{Bool}$;

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B.4.5.2. *every weighted common-denominator record R determines an urn-generated assignment μ_R such that*

$$P_R(E) = P_{\mu_R}(E)$$

for every decidable event $E : X \rightarrow \text{Bool}$.

Consequently, after choosing a common denominator, finite rational records on finite event types may be treated as urn presentations without changing any event probabilities.

Proof. For the first direction, let $\mu = (N, h, f)$ be an urn-generated assignment. Form the weighted record whose denominator is N and whose weighted list is

$$[(f(i), 1) \mid i : \text{Fin}(N)].$$

For a decidable event $E : X \rightarrow \text{Bool}$, the record numerator is the number of urn cells i for which $E(f(i)) = \text{true}$. This is exactly the numerator in the definition of $P_\mu(E)$.

For the second direction, let $R = (a, D, h, d)$ be a weighted common-denominator record, where

$$a = [(x_1, m_1), \dots, (x_r, m_r)] \quad \text{and} \quad D : \mathbb{N} \quad \text{with} \quad \sum_j m_j = D.$$

The proof $d : D > 0$ supplies the positivity side condition for the urn denominator. Construct an urn with D cells by expanding each weighted atom (x_j, m_j) into m_j cells labelled x_j . For any decidable event E , the number of expanded cells satisfying E is

$$\sum_{j, E(x_j)=\text{true}} m_j,$$

which is precisely the numerator of $P_R(E)$. The denominators are both D , so the probabilities agree for every decidable event.

The equality is extensional rather than literal equality of data structures: permuting cells, splitting cells into repeated equal labels or choosing a different but equivalent support list may produce a different urn or record with the same event probabilities. The Lean formalisation proves these two directions as `FiniteProbRecord.urn_to_record_extensional` and `FiniteProbRecord.record_to_urn_extensional`. ■

Theorem B.4.6 (Finite Kolmogorov laws). *Let $\mu = (N, h, f) : \text{Prob}_{\mathbb{Q}}^{\text{urn}}(X)$, and let $E, F : X \rightarrow \text{Bool}$ be decidable events. Then:*

B.4.6.1. $P_\mu(E) \geq 0$;

B.4.6.2. $P_\mu(\top_X) = 1$, where $\top_X(x) := \text{true}$ is the total event;

B.4.6.3. if E and F are disjoint, then

$$P_\mu(E \cup F) = P_\mu(E) + P_\mu(F).$$

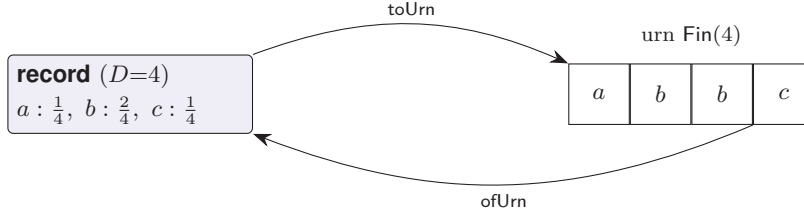


Figure B.2: Record and urn presentations of a finite rational distribution agree on every decidable event (Proposition B.4.5); the maps `toUrn/ofUrn` are the computable witnesses checked in Lean.

More generally, every finite pairwise disjoint family of decidable events is finitely additive.

Proof. For a decidable event E , write

$$\chi_E(i) := \mathbf{1}_{E(f(i))}$$

for the indicator of E after pulling it back along $f : \text{Fin}(N) \rightarrow X$. By Definition A.3.4,

$$P_\mu(E) = \frac{1}{N} \sum_{i:\text{Fin}(N)} \chi_E(i).$$

Each $\chi_E(i)$ is either 0 or 1, so the finite sum is non-negative. Since $N > 0$, division by N preserves non-negativity. This proves the first law.

For the total event $\top_X$, every indicator is 1. Hence

$$P_\mu(\top_X) = \frac{1}{N} \sum_{i:\text{Fin}(N)} 1 = \frac{N}{N} = 1.$$

This proves normalisation.

If E and F are disjoint, then for each $i : \text{Fin}(N)$ one has

$$\chi_{E \cup F}(i) = \chi_E(i) + \chi_F(i),$$

because at most one of $E(f(i))$ and $F(f(i))$ can hold. Finite summation distributes over addition, so

$$\begin{aligned} P_\mu(E \cup F) &= \frac{1}{N} \sum_i (\chi_E(i) + \chi_F(i)) \\ &= \frac{1}{N} \sum_i \chi_E(i) + \frac{1}{N} \sum_i \chi_F(i) \\ &= P_\mu(E) + P_\mu(F). \end{aligned}$$

The finite-family version follows by induction on the length of the family. This is the finite form of Kolmogorov additivity; countable additivity has not been assumed and is not derived here. ■

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Corollary B.4.7 (Probability monotonicity). *Let $\mu = (N, h, f) : \text{Prob}_{\mathbb{Q}}^{\text{urn}}(X)$, and let $E, F : X \rightarrow \text{Bool}$ be decidable events with $E \subseteq F$, that is, $E(x) = \text{true}$ implies $F(x) = \text{true}$ for every $x : X$. Then*

$$P_{\mu}(E) \leq P_{\mu}(F).$$

Proof. Define the decidable event $G(x) := F(x) \wedge \neg E(x)$, which is decidable because E and F are. By construction, E and G are disjoint, and the union $E \cup G$ coincides with F pointwise: any x with $F(x) = \text{true}$ either satisfies $E(x) = \text{true}$, lying in E , or satisfies $E(x) = \text{false}$, lying in G ; in either case it lies in $E \cup G$. Conversely, any x in $E \cup G$ satisfies $F(x) = \text{true}$, since $E \subseteq F$ and $G \subseteq F$. Disjoint additivity from Theorem B.4.6 therefore gives

$$P_{\mu}(F) = P_{\mu}(E) + P_{\mu}(G).$$

Non-negativity from Theorem B.4.6 gives $P_{\mu}(G) \geq 0$, so $P_{\mu}(F) \geq P_{\mu}(E)$. ■

Corollary B.4.8 (Two-event inclusion-exclusion). *Let $\mu = (N, h, f) : \text{Prob}_{\mathbb{Q}}^{\text{urn}}(X)$, and let $E, F : X \rightarrow \text{Bool}$ be decidable events. Then*

$$P_{\mu}(E \cup F) + P_{\mu}(E \cap F) = P_{\mu}(E) + P_{\mu}(F).$$

Proof. Decompose E and F into pieces relative to $E \cap F$. Define decidable events

$$E^{\text{only}}(x) := E(x) \wedge \neg F(x), \quad F^{\text{only}}(x) := F(x) \wedge \neg E(x).$$

Then E^{only} , F^{only} and $E \cap F$ are pairwise disjoint, and

$$E \equiv E^{\text{only}} \cup (E \cap F), \quad F \equiv F^{\text{only}} \cup (E \cap F), \quad E \cup F \equiv E^{\text{only}} \cup F^{\text{only}} \cup (E \cap F),$$

where $\equiv$ is pointwise equality of decidable predicates. Disjoint additivity from Theorem B.4.6 therefore gives

$$P_{\mu}(E) + P_{\mu}(F) = P_{\mu}(E^{\text{only}}) + P_{\mu}(F^{\text{only}}) + 2 P_{\mu}(E \cap F),$$

and

$$P_{\mu}(E \cup F) + P_{\mu}(E \cap F) = P_{\mu}(E^{\text{only}}) + P_{\mu}(F^{\text{only}}) + 2 P_{\mu}(E \cap F).$$

The two right-hand sides agree, so the two left-hand sides agree. The corresponding Lean equality is `UrnProb.inclusion_exclusion` in `lean/Thesis/Probability.lean`. ■

Corollary B.4.9 (Resolution-status probabilities). *Let S be a finite event type, let $\mu : \text{Prob}_{\mathbb{Q}}^{\text{urn}}(S)$, and let $\iota_A : S \rightarrow \text{Status}_A$ be a finite inspection procedure. Then*

$$P_{\mu}(\text{Prov}_A) + P_{\mu}(\text{Ref}_A) + P_{\mu}(\text{Unr}_A) = 1.$$

Proof. By Proposition B.4.1, the three resolution-status events are pairwise disjoint and jointly exhaustive. Finite additivity from Theorem B.4.6 gives

$$P_{\mu}(\text{Prov}_A \cup \text{Ref}_A \cup \text{Unr}_A) = P_{\mu}(\text{Prov}_A) + P_{\mu}(\text{Ref}_A) + P_{\mu}(\text{Unr}_A).$$

The union is the total event $\top_S$, whose probability is 1 by normalisation. ■

Theorem B.4.10 (Bayes’ theorem for decidable finite events). *Let $\mu : \text{Prob}_{\mathbb{Q}}^{\text{urn}}(X)$, and let $E, F : X \rightarrow \text{Bool}$ be decidable events with $P_{\mu}(E) > 0$ and $P_{\mu}(F) > 0$. With conditional probability defined by*

$$P_{\mu}(E \mid F) := \frac{P_{\mu}(E \cap F)}{P_{\mu}(F)},$$

one has

$$P_{\mu}(E \mid F) = \frac{P_{\mu}(F \mid E)P_{\mu}(E)}{P_{\mu}(F)}.$$

Proof. By the definition of conditional probability,

$$P_{\mu}(E \mid F)P_{\mu}(F) = P_{\mu}(E \cap F).$$

Since intersection is commutative,

$$P_{\mu}(E \cap F) = P_{\mu}(F \cap E).$$

Using the definition of conditional probability again, and using $P_{\mu}(E) > 0$, gives

$$P_{\mu}(F \cap E) = P_{\mu}(F \mid E)P_{\mu}(E).$$

Combining these equalities yields

$$P_{\mu}(E \mid F)P_{\mu}(F) = P_{\mu}(F \mid E)P_{\mu}(E).$$

Since $P_{\mu}(F) > 0$, division by $P_{\mu}(F)$ gives the desired formula. ■

Proposition B.4.11 (Bayesian update is conditional probability). *Let $\mu = (p, h) : \text{Prob}_{\mathbb{Q}}(\Omega)$ and let $E \subseteq \Omega$ be a decidable event with $P_{\mu}(E) > 0$. Let μ_E be the Bayesian-updated probability assignment of Definition A.3.9, with mass function $p_E(\omega) := \mathbb{1}_E(\omega)p(\omega)/P_{\mu}(E)$. Then for every decidable event $F \subseteq \Omega$,*

$$P_{\mu_E}(F) = P_{\mu}(F \mid E).$$

Proof. Unfolding the definition of p_E and using finite linearity of summation,

$$P_{\mu_E}(F) = \sum_{\omega:\Omega} \mathbb{1}_F(\omega)p_E(\omega) = \sum_{\omega:\Omega} \mathbb{1}_F(\omega) \frac{\mathbb{1}_E(\omega)p(\omega)}{P_{\mu}(E)} = \frac{1}{P_{\mu}(E)} \sum_{\omega:\Omega} \mathbb{1}_F(\omega)\mathbb{1}_E(\omega)p(\omega).$$

For each $\omega : \Omega$, $\mathbb{1}_F(\omega)\mathbb{1}_E(\omega) = \mathbb{1}_{F \cap E}(\omega)$ because both indicators are Boolean and their product is the indicator of the conjunction. Hence

$$P_{\mu_E}(F) = \frac{1}{P_{\mu}(E)} \sum_{\omega:\Omega} \mathbb{1}_{F \cap E}(\omega)p(\omega) = \frac{P_{\mu}(F \cap E)}{P_{\mu}(E)} = P_{\mu}(F \mid E),$$

the last equality being the definition of conditional probability from Definition A.3.9. The corresponding Lean equality is `UrnProb.bayesian_update_consistency` in `lean/Thesis/Probability.lean`, which checks the equality at the urn level via cross multiplication of `QProb` records. ■

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Proposition B.4.12 (Bayesian update is a probability assignment). *Let $\mu = (p, h) : \text{Prob}_{\mathbb{Q}}(\Omega)$ and let $E \subseteq \Omega$ be a decidable event with $P_{\mu}(E) > 0$. The updated mass function*

$$p_E(\omega) := \frac{\mathbb{1}_E(\omega)p(\omega)}{P_{\mu}(E)}$$

is non-negative and normalised:

$$\sum_{\omega:\Omega} p_E(\omega) = 1.$$

Hence μ_E is again an inhabitant of $\text{Prob}_{\mathbb{Q}}(\Omega)$.

Proof. Non-negativity follows pointwise: $\mathbb{1}_E(\omega)$ is either 0 or 1, $p(\omega) \geq 0$ by the definition of $\text{Prob}_{\mathbb{Q}}(\Omega)$, and the denominator $P_{\mu}(E)$ is positive by assumption.

For normalisation, compute by finite linearity:

$$\sum_{\omega:\Omega} p_E(\omega) = \sum_{\omega:\Omega} \frac{\mathbb{1}_E(\omega)p(\omega)}{P_{\mu}(E)} = \frac{1}{P_{\mu}(E)} \sum_{\omega:\Omega} \mathbb{1}_E(\omega)p(\omega).$$

The final sum is exactly $P_{\mu}(E)$ by definition of event probability. Since $P_{\mu}(E) > 0$,

$$\frac{1}{P_{\mu}(E)} P_{\mu}(E) = 1.$$

The corresponding urn-level normalisation check is formalised as `UrnProb.bayesian_update_normalization` in `lean/Thesis/Probability.lean`. ■

The theorem applies immediately to the resolution-status events Prov_A , Ref_A and Unr_A , provided the conditioning event has positive probability. It does not turn an arbitrary proposition A into a decidable proposition. In particular, there is no general constructive theorem

$$P(\neg A \mid I) = 1 - P(A \mid I).$$

What is available is the finite resolution-status decomposition:

$$P(\text{Prov}_A \mid B) + P(\text{Ref}_A \mid B) + P(\text{Unr}_A \mid B) = 1$$

whenever B is a decidable conditioning event of positive probability. Thus the probability of proof and refutation may sum to less than 1; the remaining mass is exactly $P(\text{Unr}_A \mid B)$, the probability assigned to unresolved cases under the chosen finite inspection procedure.

B.5 Structural-causal-model equivalence proof

The purpose of this appendix is not to reprove Pearl’s identification theory. It is to prove the bridge needed for importing that theory into the constructive setting used in this thesis. The fragment is deliberately narrow: finite, discrete, recursive structural causal models with finite rational probability distributions. Recursive means acyclic: the endogenous variables can be ordered so that

each structural equation depends only on earlier endogenous variables and on exogenous noise.

For the completeness transfer below, the relevant classical fragment is the standard finite recursive semi-Markovian fragment: observed endogenous variables are finite, the directed part of the graph is acyclic, latent exogenous variables are finite roots with finite rational product distribution, and the observed graph records latent common causes by bidirected edges. The Markovian case is the special case with no bidirected edges. The semi-Markovian representation may be taken in its usual canonical edge-latent form, where bidirected structure records which observed variables share an unobserved finite cause.

I write $\llbracket - \rrbracket$ for the translation from the classical finite structural-causal-model notation to the Martin-Löf-style encoding. The translation is not meant to change the mathematics; it repackages finite sets as finite types, functions as terms, products of value spaces as dependent products, and finite rational distributions as probability types.

Conversely, not every arbitrary type-theoretic object counts as a type-theoretic structural causal model for the correspondence theorem. The type-theoretic side is restricted to the record fragment generated by the same data: finite variable indices, finite value types, finite exogenous value types, finite rational probability records, structural terms typed over the declared parents, a recursive order, and, when latent confounding is represented, a finite incidence relation saying which exogenous components feed which observed variables. This restriction is what makes decoding back to an ordinary finite structural causal model possible.

Definition B.5.1 (Restricted finite type-theoretic causal-model fragment). The fragment $\text{SCM}_{\text{fin}}^{\text{tt}}$, also abbreviated FinSCM when the type-theoretic side is clear from context, consists of records with the following data:

B.5.1.1. finite endogenous and exogenous index types, with chosen finite enumerations and decidable equality;

B.5.1.2. for each index, a finite value type, so that endogenous and exogenous assignments are finite dependent products;

B.5.1.3. a rational probability record on exogenous assignments, consisting of a non-negative rational mass function whose finite sum is 1;

B.5.1.4. a decidable declared parent relation on endogenous variables, a recursive order, and a proof that every declared endogenous parent of a variable occurs earlier in that order;

B.5.1.5. for each endogenous variable, a structural term whose context contains exactly the declared endogenous parents and the declared exogenous inputs of that variable;

B.5.1.6. in the semi-Markovian case, a finite decidable incidence relation recording which exogenous components feed which observed structural mechanisms.

The fourth and fifth clauses are the formal version of the ordered-telescope reading: the recursive order is a topological order, and the type-theoretic context

B. Proofs and arguments

for the structural term of V_i exposes only the permitted earlier endogenous variables, together with the declared exogenous inputs. Cyclic dependencies and dependencies on later endogenous variables are therefore excluded by the typing discipline of this restricted record. Only records of this form are quantified over in the correspondence and completeness transfer. General Martin-Löf-style terms that do not decode to such finite recursive structural-causal-model data are outside the fragment.

Theorem B.5.2 (Finite recursive causal-model correspondence). *Let $\text{SCM}_{\text{fin}}^{\text{cl}}$ be the classical finite recursive structural-causal-model fragment just described, and let $\text{SCM}_{\text{fin}}^{\text{tt}}$ be the corresponding type-theoretic record fragment. There are translations*

$$\llbracket - \rrbracket : \text{SCM}_{\text{fin}}^{\text{cl}} \rightarrow \text{SCM}_{\text{fin}}^{\text{tt}}, \quad \text{cl}(-) : \text{SCM}_{\text{fin}}^{\text{tt}} \rightarrow \text{SCM}_{\text{fin}}^{\text{cl}}$$

such that $\text{cl}(\llbracket \mathcal{M} \rrbracket)$ is the same classical structural causal model as $\mathcal{M}$ up to finite renaming of values and variables, and $\llbracket \text{cl}(\mathcal{N}) \rrbracket$ is the same type-theoretic record as $\mathcal{N}$ up to the corresponding finite equivalences. Under these equivalences, the following data and operations are preserved and reflected:

- B.5.2.1. *finite variables and their value spaces;*
- B.5.2.2. *exogenous assignments and their finite rational probability distribution;*
- B.5.2.3. *structural functions and their recursive evaluation order;*
- B.5.2.4. *observational distributions induced by the model;*
- B.5.2.5. *interventions and the resulting interventional distributions;*
- B.5.2.6. *the induced directed graph and the semi-Markovian bidirected structure.*

Proof. The proof has two parts: construction of the two translations, and verification that the operations commute with them.

Let

$$\mathcal{M} = (\mathcal{U}, \mathcal{V}, \mathcal{F}, P_{\mathcal{U}})$$

be a finite recursive classical structural causal model. Since the model is recursive, fix a topological ordering

$$V_1, \dots, V_n$$

of the endogenous variables. For each exogenous variable U and endogenous variable V_i , the classical finite value set is represented by a finite type together with an enumeration. Thus

$$\llbracket \text{Val}(U) \rrbracket, \quad \llbracket \text{Val}(V_i) \rrbracket.$$

Joint assignments are represented by finite dependent products:

$$\llbracket \text{Val}(\mathcal{U}) \rrbracket \simeq \prod_{U \in \mathcal{U}} \llbracket \text{Val}(U) \rrbracket, \quad \llbracket \text{Val}(\mathcal{V}) \rrbracket \simeq \prod_{i=1}^n \llbracket \text{Val}(V_i) \rrbracket.$$

Because all domains are finite, these products are again finite types with decidable equality.

B.5. Structural-causal-model equivalence proof

The exogenous probability distribution is translated directly:

$$P_{\mathcal{U}} : \text{Val}(\mathcal{U}) \rightarrow \mathbb{Q}_{\geq 0} \quad \mapsto \quad \llbracket P_{\mathcal{U}} \rrbracket : \text{Prob}_{\mathbb{Q}}(\llbracket \text{Val}(\mathcal{U}) \rrbracket).$$

Non-negativity and normalisation are finite equalities transported along the chosen finite equivalence. Hence this is preservation and reflection rather than a new probabilistic construction.

For the structural functions, write the classical equation for V_i as

$$f_i : \text{Val}(\text{Pa}_{\mathcal{G}}(V_i)) \times \text{Val}(\mathcal{U}_i) \rightarrow \text{Val}(V_i).$$

Under the chosen topological order, all endogenous parents of V_i occur among $V_1, \dots, V_{i-1}$. Hence its translation is a term in the context containing the exogenous assignment and the earlier endogenous values:

$$\begin{aligned} u : \llbracket \text{Val}(\mathcal{U}) \rrbracket, v_1 : \llbracket \text{Val}(V_1) \rrbracket, \dots, v_{i-1} : \llbracket \text{Val}(V_{i-1}) \rrbracket \\ \vdash \llbracket f_i \rrbracket(u, v_1, \dots, v_{i-1}) : \llbracket \text{Val}(V_i) \rrbracket. \end{aligned}$$

This is exactly the type-theoretic form of recursive evaluation. Given u , the values $v_1, \dots, v_n$ are constructed by induction over the topological order. The induction step computes v_i from u and the already constructed values $v_1, \dots, v_{i-1}$. Therefore, for every exogenous assignment u and every endogenous variable V_i , there is a propositional equality between the transported values,

$$\text{sol}_{\llbracket \mathcal{M} \rrbracket}(\llbracket u \rrbracket)_i = \llbracket \text{sol}_{\mathcal{M}}(u)_i \rrbracket.$$

This proves preservation and reflection of pointwise solutions.

The decoding map $\text{cl}(-)$ is the same construction read in the other direction. Given a type-theoretic structural-causal-model record $\mathcal{N}$, its finite types are decoded to ordinary finite value sets by their finite enumerations, its probability record is decoded to a rational mass function, and its structural terms are decoded to total functions on the corresponding finite products. The declared parent relation, recursive order and exogenous-incidence relation are finite decidable relations, so they decode to the usual graph data. Applying cl after $\llbracket - \rrbracket$ returns the original classical model up to the chosen finite names. Applying $\llbracket - \rrbracket$ after cl returns the original type-theoretic record up to the corresponding finite type equivalences. These are the expected isomorphisms rather than judgemental equalities, because finite sets and finite types may use different names and enumerations for the same values.

Observational distributions are pushforwards of the exogenous distribution along the solution map. For any event $E \subseteq \text{Val}(\mathcal{V})$,

$$P_{\mathcal{M}}(E) = \sum_{u : \text{Val}(\mathcal{U})} \mathbb{1}_E(\text{sol}_{\mathcal{M}}(u)) P_{\mathcal{U}}(u).$$

The translated model computes

$$P_{\llbracket \mathcal{M} \rrbracket}(\llbracket E \rrbracket) = \sum_{\llbracket u \rrbracket : \llbracket \text{Val}(\mathcal{U}) \rrbracket} \mathbb{1}_{\llbracket E \rrbracket}(\text{sol}_{\llbracket \mathcal{M} \rrbracket}(\llbracket u \rrbracket)) \llbracket P_{\mathcal{U}} \rrbracket(\llbracket u \rrbracket).$$

The index type, mass function and solution map are the translations of the classical ones. The equality of the two displayed sums is therefore an instance of Lemma B.5.3. Thus observational distributions are preserved and reflected.

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Now let $\text{do}(X = x)$ be an intervention. Classically, $\mathcal{M}_x$ is obtained by replacing the structural function for each intervened variable X by the constant function x , while leaving all other structural functions unchanged. The type-theoretic intervention performs the same substitution on the corresponding terms. Hence

$$\llbracket \mathcal{M}_x \rrbracket \simeq \llbracket \mathcal{M} \rrbracket_x.$$

Applying the previous pushforward argument to the intervened model gives, for every finite event E ,

$$P_{\mathcal{M}_x}(E) = P_{\llbracket \mathcal{M} \rrbracket_x}(\llbracket E \rrbracket).$$

Therefore interventional distributions are also preserved and reflected.

Finally, consider the graph structure. Directed edges are part of the declared endogenous parent relation in the model record, and the structural terms are typed over those declared parents. Since the translation changes finite sets into finite types but does not change the declared parent arguments of each structural function, directed edges are preserved and reflected. In the semi-Markovian case, the bidirected edge $V_i \leftrightarrow V_j$ is present exactly when the finite exogenous-incidence relation says that some latent exogenous component feeds both structural mechanisms. This is again a finite decidable relation and is unchanged by the translation. Thus the directed part and the bidirected latent projection commute with encoding and decoding.

The remaining graph operations used by do-calculus and identification are finite combinatorial predicates built from the same directed and bidirected edge relations. Their predicate-by-predicate preservation is derived separately in Proposition B.5.5. This completes the two-sided correspondence proof for the stated finite graph and structural-causal-model data. $\blacksquare$

Lemma B.5.3 (Finite reindexing for equivalent finite types). *Let A and B be finite types with decidable equality, let $e : A \simeq B$ be a finite equivalence, and let*

$$f : A \rightarrow \mathbb{Q}, \quad g : B \rightarrow \mathbb{Q}$$

be rational summands such that $g(e(a)) = f(a)$ for every $a : A$. Then

$$\sum_{a:A} f(a) = \sum_{b:B} g(b).$$

Proof. Choose a finite enumeration $a_1, \dots, a_n$ of A without repetition. Since e is an equivalence, $e(a_1), \dots, e(a_n)$ is a finite enumeration of B without repetition. Replacing each summand $g(e(a_i))$ by $f(a_i)$ gives equality of the two finite sums. This is a propositional equality proved by induction over the finite enumeration, not a judgemental equality; different enumerations give the same result because finite sums over decidable finite types are invariant under permutation. $\blacksquare$

Proposition B.5.4 (Finite probability-expression transport). *Let E be an expression built from finite observational or interventional probabilities in the fragment using finite sums, finite products, marginalisation, substitution of interventions, and quotients whose denominators are positive. Then E is defined in a classical finite structural causal model if and only if $\llbracket E \rrbracket$ is defined in its type-theoretic encoding, and the two expressions have the same rational value after transport along the finite equivalence of assignments.*

Proof. The proof is by induction on the construction of E . The base cases are observational and interventional probabilities, already covered by Theorem B.5.2. Finite sums and products are preserved by Lemma B.5.3. Marginalisation is a finite sum over a finite fibre, so it is another instance of the same lemma. Substitution of interventions is preserved because $\llbracket \mathcal{M}_x \rrbracket \simeq \llbracket \mathcal{M} \rrbracket_x$. For quotients, the denominator is a rational number preserved by the induction hypothesis; positivity is therefore preserved and reflected, so the quotient is defined on one side exactly when it is defined on the other. ■

Proposition B.5.5 (Finite graph-predicate transport). *Every graph predicate used by identification, conditional identification, hedges and do-calculus in the finite semi-Markovian fragment is preserved and reflected by the encoding. This includes membership, inclusion, parents, ancestors, descendants, graph mutilation, d-separation, c-components, c-forests, root sets and hedge conditions.*

Proof. The encoding preserves and reflects the directed parent relation, the bidirected latent-incidence relation, and the finite node equivalence. The verification then proceeds predicate by predicate.

For membership and inclusion, a finite subset is represented by a decidable predicate or finite list of nodes. Transporting along the node equivalence sends each node to its encoded node and preserves decidable equality. Hence $X \in S$ and $S \subseteq T$ hold classically if and only if their encoded versions hold type-theoretically.

For parents, the predicate $X \in \text{Pa}_{\mathcal{G}}(Y)$ is exactly the directed edge predicate $X \rightarrow Y$. Directed edges are declared graph data in the model record and are unchanged by encoding and decoding, so parenthood is preserved and reflected.

For ancestors and descendants, $X \in \text{An}_{\mathcal{G}}(Y)$ and $X \in \text{De}_{\mathcal{G}}(Y)$ mean that a finite directed path exists. Encoding maps a directed path

$$X = X_0 \rightarrow X_1 \rightarrow \cdots \rightarrow X_k = Y$$

to the corresponding encoded directed path, and decoding maps it back. Since the graphs are finite and acyclic, the relevant path search is finite. Thus directed reachability is preserved and reflected.

For graph mutilation, $\mathcal{G}_{\overline{X}}$ deletes directed arrows into nodes in X , and $\mathcal{G}_{\underline{X}}$ deletes directed arrows out of nodes in X ; in the semi-Markovian case this is read in the canonical finite latent-variable directed graph of Definition A.3.25. Deletion is defined by a finite edge predicate together with finite membership in X . Both ingredients have already been shown invariant, so mutilation commutes with encoding:

$$\llbracket \mathcal{G}_{\overline{X}} \rrbracket = \llbracket \mathcal{G} \rrbracket_{\llbracket \overline{X} \rrbracket}, \quad \llbracket \mathcal{G}_{\underline{X}} \rrbracket = \llbracket \mathcal{G} \rrbracket_{\llbracket \underline{X} \rrbracket}.$$

For d-separation, first replace a semi-Markovian graph by its canonical finite latent-variable directed acyclic graph as in Definition A.3.23; in the Markovian case this step is trivial. The canonical replacement is finite and is determined by the same bidirected edge relation, so it is preserved by encoding. A path in the canonical directed acyclic graph is mapped to a corresponding encoded path; collider status, non-collider status, membership in the conditioning set and descendant membership are all preserved by the previous clauses. Therefore a path is blocked by Z if and only if its encoded path is blocked by $\llbracket Z \rrbracket$. To turn this into a finite universal check, it suffices to quantify only over *simple*

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paths — paths in which no node repeats — rather than over all walks. The standard reduction lemma for d-separation [Pea09, §1.2.3] states that, in a finite directed graph, two nodes are d-connected by Z along a walk if and only if they are d-connected by Z along a simple path; the proof proceeds by induction on walk length, repeatedly contracting a repeated-vertex segment and using a standard case analysis on the collider status of the join node to argue that each contraction preserves d-connection. Since the directed-and-bidirected graph is finite, the set of simple paths between any two nodes is finite, and the universal statement “every simple path between X and Y is blocked by Z ” is a finite decidable check. Encoding preserves and reflects this finite check, so d-separation is preserved and reflected.

For c-components, two observed nodes are in the same c-component exactly when they are connected by a finite bidirected path. Bidirected paths are preserved by the latent-incidence relation. Maximality of a component is a finite inclusion condition, already handled above. Thus c-components are preserved and reflected.

For c-forests, the defining predicates are: the observed nodes form one c-component; each observed node has at most one directed child inside the subgraph; and the roots are precisely the nodes with no directed child inside the subgraph. The first clause is c-component preservation, the second is a finite bounded child-count predicate over preserved directed edges, and the third is a finite negated-existence predicate over the same child relation. Hence c-forests and their root sets are preserved and reflected.

Finally, a hedge for $P_x(y)$ is the conjunction of finite predicates: F and F' are c-forests, they have the same roots, $F' \subset F$, F meets the intervention set, F' does not meet it, and the common roots are ancestors of Y in the graph with arrows into X deleted. Each conjunct has already been shown invariant: c-forests, roots, inclusion, finite intersection, graph mutilation and ancestry. Therefore hedge witnesses are preserved and reflected. ■

Proposition B.5.6 (Compatible-model class equivalence). *For every finite recursive Markovian or semi-Markovian graph G , the translations restrict to an equivalence between classical finite structural-causal models compatible with G and restricted type-theoretic structural-causal-model records compatible with $\llbracket G \rrbracket$. This equivalence preserves and reflects observational distributions and all ordinary finite interventional queries in the fragment.*

Proof. Compatibility with G in the sense of Definition A.3.31 is preserved and reflected by Theorem B.5.2: the declared directed and bidirected graph data and the finite recursive structure are exactly the data the correspondence preserves. The two composites $\text{cl}(\llbracket - \rrbracket)$ and $\llbracket \text{cl}(-) \rrbracket$ return the original compatible model up to finite renaming and finite type equivalence. Observational and interventional distributions are preserved by the same theorem. ■

Proposition B.5.7 (Modal locality). *If an epistemic causal mode m contains a restricted finite type-theoretic structural-causal-model record, then the correspondence above holds internally at that mode. Modal transitions between epistemic causal modes do not change the transferred Pearl-style completeness result unless they are equipped with additional structure specifying how model records are transformed.*

Proof. The structural-causal-model correspondence is local to the record available at a mode: all variables, value types, structural terms, probability records, graph relations and finite proofs are judged in the context of that mode. The identification, conditional identification and do-calculus operations used here act on that local finite record. A modality can move information between modes only according to the modal structure supplied for that transition. Therefore the ordinary completeness transfer applies at each mode containing a restricted finite model record, while claims about changing from one causal model to another require a separate theorem about the chosen modality. ■

Definition B.5.8 (Finite epistemic transition layer). The finite transition layer used in the mechanisation consists of the following data. An *epistemic record* is a tuple

$$R = (\mathcal{M}, p, e, \rho, b)$$

where $\mathcal{M}$ is a finite recursive structural-causal-model record, p proves that its exogenous finite distribution is a probability distribution, and e is a finite bound for exogenous references. A variable reference is either an endogenous reference $\text{endo}(i)$ with $i < |\mathcal{M}|$, or an exogenous reference $\text{exo}(j)$ with $j < e$. The list ρ is a finite list of relation edges between such references, and b proves that every edge in ρ is bounded by the corresponding endogenous and exogenous counts. An *epistemic mode* is a name paired with such a record. A *finite modal transition* is a label

learn, forget, condition, relate, unrelate, set, unset

together with a source mode and a target mode.

The labels are fixed transition names; they do not inspect the source context. The actual work is in the source–target record transformation. Learning appends a fresh last endogenous variable, with all old structural equations unchanged. Forgetting removes the last endogenous variable and requires a proof that all remaining relation edges are still bounded after the removal. Conditioning replaces the exogenous distribution by a posterior finite probability distribution and leaves the structural equations unchanged. The posterior may be supplied explicitly with a proof of normalisation, or it may be constructed from a decidable evidence event by finite Bayesian update. Setting replaces one endogenous structural equation by a constant value. Unsetting restores a stored old structural equation. Relating inserts a bounded relation edge between any two variable references, endogenous or exogenous. Unrelating erases the inserted edge.

Proposition B.5.9 (Finite modal transition preservation). *Each generating transition of Definition B.5.8 sends a well-formed epistemic record to a well-formed epistemic record, provided the explicit side conditions of the transition are supplied: a learned variable is appended at the end, a forgotten variable is unused by the remaining relation data, a conditioning step is given either a posterior finite probability distribution or a positive-probability evidence event from which such a posterior is constructed, a related edge is bounded by the current endogenous and exogenous reference counts, and an unset operation is given the structural equation it restores.*

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Proof. Learning, forgetting, relating, unrelating, setting and unsetting preserve the exogenous distribution, so the old probability proof p is reused for those cases. Conditioning is the one exception: it replaces the exogenous distribution by the supplied posterior and uses the supplied proof that this posterior is normalised. Learning increases the endogenous variable count from n to $n + 1$ and leaves the exogenous reference count e unchanged, hence every previously bounded relation edge remains bounded. Forgetting has the boundedness proof as an explicit premise: all edges must already be bounded by $n - 1$ and e , which says that the removed last endogenous variable is not mentioned in the remaining relation data. Setting, unsetting and conditioning leave the endogenous count, exogenous reference count and relation list unchanged. Relating requires a proof that the inserted edge is bounded by both counts; all old edges retain their existing proof. Unrelating removes an edge from a finite list, so any edge still present was already present before removal and keeps its previous boundedness proof. These are exactly the preservation lemmas in `lean/Thesis/Modalities.lean`:

```
learnFresh_preserves_wellFormed,
forgetLast_preserves_wellFormed,
conditionNoise_preserves_wellFormed,
setVariable_preserves_wellFormed,
unsetVariable_preserves_wellFormed,
relate_preserves_wellFormed,
unrelate_preserves_wellFormed.
```

■

Proposition B.5.10 (Bayesian conditioning as a modal transition). *Let $R = (\mathcal{M}, p, e, \rho, b)$ be a well-formed epistemic record, and let E be a decidable event on exogenous assignments with $P_{\mathcal{M}}(E) > 0$. Define the posterior exogenous distribution by*

$$p_E(u) := \begin{cases} p(u)/P_{\mathcal{M}}(E), & E(u), \\ 0, & \text{otherwise.} \end{cases}$$

Then p_E is a probability distribution, and the conditioning transition that replaces p by p_E is well formed. Moreover, for every decidable event F on exogenous assignments,

$$P_{p_E}(F) = \frac{P_{\mathcal{M}}(E \cap F)}{P_{\mathcal{M}}(E)}.$$

If O is a decidable event on evaluated endogenous assignments, the same construction applies to the lifted exogenous event $E_O(u) := O(\mathcal{M}(u))$ whenever $P_{\mathcal{M}}(E_O) > 0$.

Proof. Normalisation is the finite Bayesian-update calculation of Proposition B.4.12, applied to the finite exogenous distribution. Non-negativity is preserved because the original masses are non-negative and the denominator $P_{\mathcal{M}}(E)$ is positive. The displayed equation is finite conditional probability: summing p_E over F sums exactly the original mass over $E \cap F$, divided by

$P_{\mathcal{M}}(E)$. In Lean, the checked posterior facts are

```
FinDist.bayesPosterior,
FinDist.bayesPosterior_isProbability,
FinDist.bayesPosterior_probOf
```

in `lean/Thesis/SCM.lean`. The corresponding modal record operations and facts in `lean/Thesis/Modalities.lean` are

```
conditionOnExogenousEvidence,
conditionOnObservation,
conditionOnExogenousEvidence_preserves_wellFormed,
conditionOnObservation_preserves_wellFormed,
conditionOnExogenousEvidence_probOf.
```

■

Proposition B.5.11 (Finite modal cancellation checks). *The basic inverse-looking transition pairs cancel at the finite record level:*

$$\text{forget}(\text{learn}(R)) = R, \quad \text{unset}(\text{set}(R)) = R, \quad \text{unrelate}(\text{relate}(R)) = R,$$

with the intended side conditions: learning appends a fresh last variable, forgetting removes that same last variable, unsetting is supplied with the stored old structural equation, and unrelating erases the inserted relation edge.

Proof. For learning followed by forgetting, the learned variable is appended as the last variable. In a recursive finite SCM, earlier structural equations cannot depend on a later variable; the Lean lemma `learnFreshModel_old_fn` records that the old equations are unchanged when viewed inside the extended model. Removing the last variable therefore recovers exactly the original model, and the relation boundedness proof reduces to the original boundedness proof. This is formalised as `forgetLast_learnFresh_cancel`.

For setting followed by unsetting, setting changes only the selected structural equation. If the old equation has been stored as finite data, unsetting puts that equation back and leaves all other equations untouched. This is `unset_set_cancel`. For relating followed by unrelating, the relation list is extended by one edge and then that inserted occurrence is erased, giving the previous list back. This is `unrelate_relate_cancel`. Conditioning has no such cancellation law in general: replacing a prior by a posterior loses information unless the prior and update evidence are stored separately. These cancellation facts are not consequences of Gratzer’s modal rules alone; they are the record-level equations that justify treating the corresponding finite transitions as inverse-looking sanity checks. ■

Remark B.5.12 (Twin networks as finite composites). The transition layer therefore does not contain a primitive modality for “adding a twin network”. A twin network is too specific: it is a particular construction using ordinary record changes. In the finite recursive fragment it can be represented as a composite of existing transitions. First, when there is evidence about the actual unit, a `condition` transition replaces the exogenous distribution by the

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posterior distribution required by the abduction step. Second, repeated `learn` transitions append the copied endogenous variables and copied structural equations in topological order. Third, `relate` records the intended sharing: copied and actual variables may be related to the same exogenous references, and actual/counterfactual endogenous counterparts may also be related. Fourth, a `set` transition imposes the counterfactual intervention in the copied sector. Prediction is then ordinary evaluation of the resulting finite record.

This is enough to express the construction at the record level, but it is not yet a full counterfactual completeness theorem. Such a theorem would additionally have to prove that the composite just described is equivalent to Pearl’s twin-network or multi-world semantics for the class of counterfactual queries under discussion. That correspondence is deliberately left outside the transfer theorem of Theorem B.6.13.

A representative Lean 4 formalisation of the computational core of this correspondence is included in the modules `Thesis.SCM` and `Thesis.Modalities`, in `lean/Thesis/SCM.lean` and `lean/Thesis/Modalities.lean`. Their role is deliberately narrow. They check the shared finite record, the encode/decode skeleton, recursive evaluation, intervention, observational and interventional probability, the model-level counterfactual probability helper, the finite modal transition layer above, and an abstract completeness-transport lemma. The modal module also checks the same tenure-track counterfactual through the explicit record composite `conditionOnObservationThenSet`, and proves that this composite agrees with the model-level counterfactual helper for a single set action. The SCM module also checks the local binary conditional-probability-table row functionalisation used by the finite causal-Bayesian-network argument: a finite uniform tagged seed type with prescribed true and false cell counts realises the corresponding rational row.

It does not mechanise Pearl’s graph theory. D-separation, c-components, c-forests, hedges, hedge countermodels, identification, conditional identification, arbitrary finite causal-Bayesian-network functionalisation and semi-Markovian projection remain ordinary mathematical arguments in this appendix. Put plainly, this is where the Lean check stops and the appendix proof takes over. In particular, the encode and decode maps in `Thesis.SCM` are the identity on the shared finite record, so the mechanised correspondence lemmas check that the classical and type-theoretic presentations carry the same finite data by construction; they are not an independent mechanical verification of Pearl’s identification theory, which remains the cited external result.

Corollary B.5.13 (Interventional-query adequacy). *For every finite ordinary interventional query $Q = P_x(y)$ in the recursive structural-causal-model fragment, where $P_x(y)$ abbreviates $P(Y = y \mid \text{do}(X = x))$ or the corresponding finite joint event,*

$$P_{\mathcal{M}_x}(Y = y) = q \iff P_{\llbracket \mathcal{M} \rrbracket_x}(\llbracket Y = y \rrbracket) = q.$$

Moreover, two classical models in the fragment agree on their finite observational distribution if and only if their type-theoretic encodings agree on the encoded observational distribution.

Proof. The first claim is the interventional-distribution part of Theorem B.5.2 applied to the singleton event or finite event that defines y . The second

claim follows from the observational-distribution part of the same theorem, since observational distributions are finite pushforwards of the same exogenous distribution through corresponding solution maps. ■

Proposition B.5.14 (Identifiability is transported). *Let G be a finite recursive causal graph in the Markovian or semi-Markovian representation used above, and let $Q = P_x(y)$ be an ordinary finite interventional query. Then Q is identifiable in the classical finite structural-causal-model fragment over G if and only if $\llbracket Q \rrbracket$ is identifiable in the type-theoretic record fragment over $\llbracket G \rrbracket$.*

Proof. Identifiability says that any two compatible models with the same observational distribution give the same value to the target interventional query. By Proposition B.5.6, the classical compatible-model class over G and the restricted type-theoretic compatible-record class over $\llbracket G \rrbracket$ are equivalent. Observational equality and interventional-query values are preserved and reflected by Theorem B.5.2. Thus a classical counterexample to identifiability translates into a type-theoretic counterexample. Conversely, because the type-theoretic quantification is restricted to the record fragment decoded by $\text{cl}(-)$, a type-theoretic counterexample decodes into a classical counterexample. Equivalently, absence of such counterexamples is preserved in both directions. ■

Proposition B.5.15 (Do-calculus derivations are transported). *Any finite classical do-calculus derivation whose expressions, graph operations and probability manipulations lie in the fragment covered by Theorem B.5.2 and Proposition B.5.5 translates into a type-theoretic derivation of the encoded equality of expressions.*

Proof. By Definition A.3.27, a derivation is a finite sequence of reflexivity, model-evaluation, probability and do-calculus steps. Reflexivity steps are preserved by definitional equality of encoded expressions. Each model-evaluation step rewrites $P_{\mathcal{M}_x}(y)$ to a finite rational expression in the exogenous distribution and structural functions of $\mathcal{M}_x$; this is licensed by Theorem B.5.2 and so is preserved by encoding. Each probability step is an equality of finite sums, finite products, marginalisations or conditionings with non-zero denominator; these are transported by Proposition B.5.4. Each do-calculus step is licensed by a d-separation claim in a mutilated graph; these graph conditions are transported by Proposition B.5.5. Therefore each step in the classical derivation has an encoded type-theoretic step. Induction on the length of the derivation gives the translated derivation. ■

B.6 Causality-completeness proof

The completeness theorem used here is an external mathematical result. In the finite recursive semi-Markovian setting, the Shpitser–Pearl identification algorithm is sound and complete for ordinary joint interventional distributions, and the rules of do-calculus together with ordinary probability manipulations are complete for identifying such effects [HV06; Pea12; SP06; TS09]. This thesis does not rederive that algorithmic proof. It proves that the result applies to the restricted type-theoretic structural-causal-model record fragment by transport along the two-sided correspondence of Theorem B.5.2.

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The theorem-by-theorem scope is as follows. From Shpitser–Pearl, the transferred fragment includes termination and soundness of the finite identification procedure, the hedge failure criterion, completeness of identification for effects of the form $P_x(y)$, and completeness of do-calculus with standard probability manipulations for those unconditional effects [SP06]. From Huang–Valtorta, it includes the finite soundness and completeness of the identification algorithm for ordinary causal effects $P_T(S)$ and the resulting completeness of Pearl’s three do-calculus rules for such effects, after restricting to finite models and applying the finite functionalisation and semi-Markovian correspondence below [HV06]. From Tian–Shpitser, it includes the reviewed completeness result for the unconditional identification of effects and, with the positivity side condition made explicit, the completeness result for the conditional identification of effects $P_x(y \mid z)$ [TS09]. The counterfactual sections of Tian–Shpitser are not covered here; they require a separate correspondence theorem for counterfactual graphs and multi-world assignments.

Lemma B.6.1 (Disjoint exogenous parents give independence). *Let $\mathcal{M} = (\mathcal{U}, \mathcal{V}, \mathcal{F}, P_{\mathcal{U}})$ be a finite recursive structural causal model in the sense of Definition A.3.29 whose exogenous distribution factorises as a finite product*

$$P_{\mathcal{U}}(u_1, \dots, u_k) = \prod_{j=1}^k P_{U_j}(u_j),$$

with each $P_{U_j} : \text{Prob}_{\mathbb{Q}}(\text{Val}(U_j))$. Suppose that $X, Y \in \mathcal{V}$ are endogenous variables and that the finite sets of exogenous variables that functionally feed into the recursive evaluation of X and Y are disjoint:

$$\mathcal{U}_{\text{anc}}(X) \cap \mathcal{U}_{\text{anc}}(Y) = \emptyset.$$

Here $\mathcal{U}_{\text{anc}}(V)$ denotes the finite set of exogenous variables that occur in the recursive expansion of the structural equations along the ancestry of V in the induced graph. Then X and Y are independent under the observational distribution induced by $\mathcal{M}$, in the sense of Definition A.3.10: for every pair of decidable events A on $\text{Val}(X)$ and B on $\text{Val}(Y)$,

$$P_{\mathcal{M}}(A(X) \cap B(Y)) = P_{\mathcal{M}}(A(X)) \cdot P_{\mathcal{M}}(B(Y)).$$

Proof. Because the model is recursive, the values of X and Y under exogenous assignment $u : \text{Val}(\mathcal{U})$ depend only on the components of u in $\mathcal{U}_{\text{anc}}(X)$ and $\mathcal{U}_{\text{anc}}(Y)$ respectively. Write $u = (u_X, u_Y, u_R)$ for the partition into the X -ancestral block, the Y -ancestral block, and the remainder. The disjointness assumption guarantees that this partition is well defined. Since $P_{\mathcal{U}}$ is a finite product distribution, finite Fubini for finite rational sums gives

$$P_{\mathcal{U}}(u_X, u_Y, u_R) = P_{\mathcal{U}_{\text{anc}}(X)}(u_X) \cdot P_{\mathcal{U}_{\text{anc}}(Y)}(u_Y) \cdot P_{\mathcal{U}_R}(u_R),$$

with each factor a finite rational marginal of $P_{\mathcal{U}}$.

Let $\text{val}_X(u_X)$ and $\text{val}_Y(u_Y)$ denote the values of X and Y obtained by recursive evaluation; by the recursive structure these depend only on their

respective blocks. Then

$$\begin{aligned}
 P_{\mathcal{M}}(A(X) \cap B(Y)) &= \sum_u \mathbb{1}_A(\text{val}_X(u_X)) \mathbb{1}_B(\text{val}_Y(u_Y)) P_{\mathcal{U}}(u) \\
 &= \left(\sum_{u_X} \mathbb{1}_A(\text{val}_X(u_X)) P_{\mathcal{U}_{\text{anc}}(X)}(u_X) \right) \\
 &\quad \cdot \left(\sum_{u_Y} \mathbb{1}_B(\text{val}_Y(u_Y)) P_{\mathcal{U}_{\text{anc}}(Y)}(u_Y) \right) \cdot 1 \\
 &= P_{\mathcal{M}}(A(X)) \cdot P_{\mathcal{M}}(B(Y)),
 \end{aligned}$$

where the second equality is finite distributivity of multiplication over the three independent finite sums and the third equality marginalises the u_R block (which sums to 1) and recognises the remaining two finite sums as the marginal probabilities of the events $A(X)$ and $B(Y)$. The factorisation extends from pairs to arbitrary finite families of variables with pairwise disjoint exogenous-ancestor sets by an analogous finite Fubini step. ■

Lemma B.6.2 (Finite rational hedge countermodels). *Let a finite hedge witness non-identifiability in the Shpitser–Pearl semi-Markovian setting. Then the separating countermodels used for the hedge can be chosen inside the finite rational recursive structural-causal-model fragment: the variables on the hedge are binary, the exogenous variables are independent fair coins, and the structural functions are deterministic parity functions. If the ambient value types are larger than binary, the construction applies after choosing two distinct values for each variable used in the hedge and making the structural functions land only in those chosen values. Thus this support is a finite-value claim for binary value types, or for value types with two distinguished values; singleton value types do not carry the usual hedge countermodel.*

Proof. This lemma extracts the finite-rational content of the Shpitser–Pearl hedge countermodel construction [SP06]. Let $F' \subset F$ be the hedge, with common root set R . The published proof first reduces the separating query to a root query on an R -rooted c-forest. On that c-forest, take every observed variable to have value type $\{0, 1\}$. For each bidirected latent connection used in the forest, use a binary exogenous root $U_e : \{0, 1\}$ with the fair-coin distribution, and take all such roots to be independent.

The two countermodels have the same finite exogenous variables and the same exogenous distribution. In the first model, every observed node in F is the bit parity, i.e. the sum modulo 2, of its observed parents in F and its incident binary latent parents. In the second model, the same parity equations are used except that the nodes in F' disregard parent values contributed by the grown part $F \setminus F'$. Outside the hedge, the models are extended in the same way on both sides, so any separation produced on the hedge is preserved in the ambient graph.

The parity argument is the finite counting step in the Shpitser–Pearl construction. In an R -rooted c-forest, each latent bit in the relevant forest has exactly two downward paths to the root set R . Hence, before intervention, its contribution to the root parity is counted twice and cancels modulo 2. Thus in the first model the xor of the root values counts each latent bit from F twice, while in the second model the xor of the root values counts each latent bit

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from F' twice. In both cases the possible observed assignments are therefore constrained to have even root parity.

The multiplicities are also the same. If two latent assignments differ, let U^* be the finite set of latent bits on which they differ. In the c-forest construction there is a topmost observed node W with an odd number of parents in U^* ; flipping the bits in U^* flips the value of W . Hence the map from latent assignments to observed assignments is injective for the relevant c-forest in each of the two models. Since the binary c-forest construction has $|U| + 1 = |V|$, the number of latent assignments equals the number of observed assignments with even root parity. Therefore every even-root-parity observed assignment occurs with the same fair-coin mass in the corresponding forest, and every odd-root-parity assignment has probability zero. The two models agree on the functions and distributions outside the modified part of the hedge, so summing over the same finite fibres gives the same observational distribution on the observed variables.

The hedge condition gives $F \cap X \neq \emptyset$ and $F' \cap X = \emptyset$. After the intervention $\text{do}(x)$, the incoming arrows into the intervened variables remove part of the grown structure $F \setminus F'$. The double-counting parity cancellation is then broken in one model but not in the other. Consequently there is a root assignment r such that

$$P_{\mathcal{M}_1, x}(R = r) \neq P_{\mathcal{M}_2, x}(R = r).$$

Since the roots lie among the ancestors of Y in the graph with arrows into X deleted, the standard extension step makes downstream variables copy or parity-code the root assignment, giving a difference in the target effect while preserving equality of the observational distribution.

The construction is finite throughout. A product of k fair coins assigns each exogenous assignment the rational mass 2^{-k} . Thus both separating models lie in the finite rational recursive structural-causal-model fragment. For a larger finite value type, choose an injection of the two bit values into that type and define all structural functions to land in its image. The induced probabilities of assignments outside the image are then zero. This preserves the finite graph witness and the separating binary event. The lemma does not assert a positive-mass countermodel for every ambient assignment; positivity assumptions, when needed for conditional effects, are carried separately in Proposition B.6.6. ■

The restriction to finite rational probabilities is part of the transfer. Successful identification and conditional-identification expressions are built from finite sums, products, marginalisations and quotients of the given observational probabilities, so they remain inside the finite rational probability fragment when the input distribution is rational. On the failure side, Lemma B.6.2 gives finite binary countermodels with rational masses for the hedge failures used in the completeness proof. Thus the rational finite subfragment preserves the positive and negative cases needed for the transferred theorem; the claim is not a claim about arbitrary real-valued probability records inside Martin-Löf-style type theory.

This transfer is constructive in the following precise sense. It does not require excluded middle for arbitrary propositions. The case distinctions used by identification, conditional identification and hedge checking are decisions about finite data: membership in a finite set, equality in a finite value type, existence of

a finite path, d-separation in a finite graph, containment of finite subgraphs, and positivity or non-zerosness of a rational finite probability. These are decidable predicates in the finite type-theoretic fragment. Thus the algorithmic content imported from the classical proofs can be re-expressed using finite decidable case analysis, not unrestricted classical reasoning.

Proposition B.6.3 (Constructive finite identification). *In the finite recursive semi-Markovian fragment, the identification procedure can be read as a constructive finite algorithm in the type-theoretic record fragment. It terminates, and each branch condition is decidable.*

Proof. The inputs to identification are finite lists or finite indexed families of variables, finite rational probability records, and a finite directed-and-bidirected graph. The branch tests ask whether certain finite sets are empty, whether one finite set is contained in another, whether a finite graph has one or several c-components, whether a variable lies among finite ancestors, and whether a displayed finite product or sum expression is defined. Each of these questions is decidable by finite search.

Termination is by well-founded recursion on an explicit lexicographic measure. For an identification call with input data

$$(G, V, X, Y),$$

where G is the current finite directed-and-bidirected graph, V is the finite set of active observed variables, and $X, Y \subseteq V$ are the finite intervention and outcome sets, define

$$m(G, V, X, Y) := (|V|, |V \setminus X|) \in \mathbb{N}^2,$$

with $\mathbb{N}^2$ carrying the lexicographic order. The recursive cases of the Shpitser–Pearl identification algorithm [SP06], applied in order so that the first matching case is taken, have the following decrement behaviour:

- if $X = \emptyset$, the call returns a finite marginalisation and does not recurse;
- if $V \setminus \text{An}_G(Y) \neq \emptyset$, the call recurses on $V' = \text{An}_G(Y) \subsetneq V$, strictly decreasing the first component of m ;
- if $W := (V \setminus X) \setminus \text{An}_{G_{\bar{X}}}(Y) \neq \emptyset$, the call recurses with $X' = X \cup W \supsetneq X$, leaving V unchanged and strictly decreasing $|V \setminus X|$;
- if $C(G[V \setminus X])$ has at least two c-components $S_1, \dots, S_k$, the call recurses, for each component, as $\text{ID}(s_i, v \setminus s_i, P, G)$ with the graph G and hence V unchanged; since $k \geq 2$ and the components partition $V \setminus X$, each $|s_i| < |V \setminus X|$, so the first component of m is unchanged and the second component $|V \setminus X|$ strictly decreases;
- if $C(G[V \setminus X]) = \{C\}$ with $C \notin C(G)$ and $C \subsetneq C'$ for some c-component $C' \in C(G)$ with $C' \subsetneq V$, the call recurses on $V' = C'$, strictly decreasing the first component of m ;
- if $C(G[V \setminus X]) = \{C\}$ with $C \in C(G)$, the call returns a finite product formula and does not recurse;

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- if $C(G[V \setminus X]) = \{C\}$ with $C(G) = \{V\}$, the call returns failure with a hedge witness and does not recurse.

Every recursive call therefore has m -value strictly less than its parent's. The lexicographic order on $\mathbb{N}^2$ is well founded, so finite induction on m shows that identification terminates on every input in the fragment. Conditional-identification calls reduce to a finite d-separation search and a finite number of unconditional identification calls on strictly smaller subproblems, so the same lexicographic measure handles them after a single decrement.

No appeal is made to excluded middle for arbitrary propositions: every branch test is a finite decidable predicate. Since Theorem B.5.2 preserves and reflects the finite graph and probability data used by the branch tests, the same finite computation and the same lexicographic measure are available in the type-theoretic record fragment. ■

Proposition B.6.4 (Identification and conditional-identification trace transport). *Run identification or conditional identification on a finite classical input in the covered fragment, and run the encoded procedure on the corresponding type-theoretic input. The two runs take corresponding branches at every recursive call. If the classical run returns an expression, the type-theoretic run returns the encoded expression with the same rational interpretation. If the classical run returns failure with a hedge witness, the type-theoretic run returns the encoded hedge witness, and conversely for decoded type-theoretic runs.*

Proof. A run of identification or conditional identification is a finite well-founded recursion tree whose branch labels are finite graph predicates and finite probability-expression side conditions. Predicate by predicate, the identification branch tests ask whether the intervention set is empty; whether the active node set equals the ancestor set of Y ; whether a displayed finite set W is empty; whether the c-component decomposition of a mutilated graph has one component or several; whether the active graph itself is one c-component; whether a component belongs to $C(G)$; and whether there exists a strictly larger c-component containing a given component. These are finite combinations of membership, inclusion, ancestry, mutilation and c-component predicates, all transported by Proposition B.5.5. The conditional-identification branch test asks whether some $W \in Z$ satisfies the relevant d-separation condition in the appropriate mutilated graph; this is again a finite search over the transported d-separation predicate. The returned probability expressions and their positivity conditions are transported by Proposition B.5.4. Therefore corresponding inputs choose corresponding branches. Induction on the well-founded recursion tree gives correspondence of the returned expression or failure witness. The converse direction follows by applying the decoding map $\text{cl}(-)$ and using the same preservation-and-reflection facts. ■

Proposition B.6.5 (Constructive hedge witnesses). *In the same fragment, failure of identification transports to an explicit type-theoretic hedge witness, and a type-theoretic hedge witness decodes to an ordinary classical hedge witness.*

Proof. A hedge is finite data: two c-forests F and F' with the same roots, a finite inclusion $F' \subset F$, a finite proof that F meets the intervention set, a finite proof that F' does not, and a finite ancestry condition in the graph with arrows

into the intervention deleted. By Theorem B.5.2, c-components, c-forests, roots, graph containment, ancestry and graph mutilation are preserved and reflected by the translations. Hence the hedge produced by a classical identification failure encodes as the same finite witness type-theoretically. Conversely, because the type-theoretic side is restricted to records decoded by $\text{cl}(-)$, any encoded hedge decodes to the corresponding classical finite graph witness. ■

Proposition B.6.6 (Conditional-effect correspondence). *Let $P_x(y \mid z)$ be a finite conditional interventional query in the recursive semi-Markovian fragment. If the denominator $P_x(z)$ is positive, then*

$$P_{\mathcal{M}_x}(Y = y \mid Z = z) = q \iff P_{\llbracket \mathcal{M} \rrbracket_x}(\llbracket Y = y \rrbracket \mid \llbracket Z = z \rrbracket) = q.$$

Positivity of the denominator is also preserved and reflected.

Proof. Conditional probability in this thesis is defined only when the conditioning event has positive probability. In the finite rational setting,

$$P_{\mathcal{M}_x}(Y = y \mid Z = z) = \frac{P_{\mathcal{M}_x}(Y = y, Z = z)}{P_{\mathcal{M}_x}(Z = z)}.$$

The numerator and denominator are ordinary finite interventional probabilities, so they are preserved and reflected by Corollary B.5.13. Positivity of the denominator is a decidable order statement about a rational number, and the rational number itself is preserved by the correspondence. Therefore the quotient is defined on one side exactly when it is defined on the other, and the two values agree. ■

Corollary B.6.7 (Transferred conditional-identification completeness). *The finite conditional-identification completeness result for identifiable conditional effects $P_x(y \mid z)$ transfers to the type-theoretic record fragment whenever the required conditioning denominators are positive.*

Proof. Conditional identification reduces a conditional effect to unconditional identification calls after finite d-separation tests and applications of the action-observation exchange rule. The graph tests are finite decidable predicates preserved by Proposition B.5.5, and the conditional probabilities are preserved by Proposition B.6.6. The unconditional calls are covered by Proposition B.6.3, and the branch-by-branch algorithmic correspondence is Proposition B.6.4. Thus the finite classical conditional-identification derivation transports step by step to the type-theoretic record fragment, with every conditioning step carrying its explicit positive-denominator assumption. ■

Lemma B.6.8 (Finite rational causal-Bayesian-network functionalisation). *Every finite rational causal Bayesian network on a finite acyclic graph can be represented by a finite recursive structural causal model with independent finite exogenous seeds, deterministic structural functions, and the same induced joint distribution. The same construction applies to the full finite graph before some variables are hidden.*

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Proof. Fix a topological order $V_1, \dots, V_n$ of the finite directed acyclic graph. For each variable V_i and each finite parent assignment pa_i , the conditional table gives rational numbers

$$P(v_i \mid pa_i)$$

over the finite value type $\text{Val}(V_i)$. Since there are only finitely many entries in all rows of this table, choose a common positive denominator N_i such that every number $N_i P(v_i \mid pa_i)$ is a natural number. Let the exogenous seed for V_i be the finite type $\text{Fin}(N_i)$ with the uniform rational distribution.

For each parent assignment pa_i , partition $\text{Fin}(N_i)$ into blocks indexed by $v_i : \text{Val}(V_i)$, with the block for v_i having size $N_i P(v_i \mid pa_i)$. Define the structural function $f_i(pa_i, u_i)$ to be the value whose block contains u_i . This is a finite deterministic function. The exogenous distribution of the model is the finite product of the uniform seed distributions, so the seeds are independent and rational.

By construction,

$$\Pr(f_i(pa_i, U_i) = v_i) = P(v_i \mid pa_i)$$

for every row of every conditional table. Evaluating the recursive model in the chosen topological order and using independence of the seeds therefore gives

$$\Pr(V_1 = v_1, \dots, V_n = v_n) = \prod_i P(v_i \mid pa_i),$$

which is exactly the Bayesian-network factorisation. All choices made here are finite choices over listed rational tables and finite types, so the construction is available in the finite type-theoretic record fragment. ■

Definition B.6.9 (Finite latent projection). Let G be a finite acyclic directed graph whose nodes are partitioned into observed nodes V and latent nodes L , in the sense of Definition A.3.28: observed nodes are the endogenous and observed exogenous variables of the model, latent nodes are the unobserved exogenous variables. The finite latent projection of G onto V is the directed-and-bidirected graph over V defined as follows. There is a directed edge $A \rightarrow B$ between observed nodes when G contains a directed path from A to B whose intermediate nodes, if any, are latent. There is a bidirected edge $A \leftrightarrow B$ when G contains a path between A and B whose intermediate nodes are latent, whose endpoint arrowheads point into A and B , and whose intermediate latent nodes are non-colliders on that path. Equivalently, in the acyclic latent-variable setting used here, the bidirected edge records a finite latent common-cause path between the two observed endpoints.

The canonical edge-latent representative of the projected graph is obtained by replacing each bidirected edge $A \leftrightarrow B$ by a fresh latent root U_{AB} with arrows $U_{AB} \rightarrow A$ and $U_{AB} \rightarrow B$. This representative is used for the semi-Markovian graph predicates in the completeness transfer. It is not asserted to be judgmentally identical to the original latent causal Bayesian network; the relevant claim is the classical identifiability-preservation result for this finite projection. Since G , V and L are finite, the path searches and edge tests in the projection are decidable finite predicates and can be represented in the restricted type-theoretic record fragment.

Proposition B.6.10 (Finite semi-Markovian projection). *The finite latent projection of Definition B.6.9, followed by its canonical edge-latent semi-Markovian representative, is representable in the type-theoretic fragment, and its classical identifiability-preservation theorem transports across the correspondence.*

Proof. By Lemma B.6.8, the finite rational causal Bayesian network can first be represented as a finite recursive structural causal model on the full graph, including latent variables. The graph step is then exactly the finite latent projection of Definition B.6.9: retain observed variables, add directed edges for directed latent-mediated paths, and add bidirected edges for latent common-cause paths. Its canonical edge-latent representative adds one fresh finite latent root for each bidirected edge. In the type-theoretic setting, all of these are finite list or finite indexed-family operations with decidable edge and path predicates.

The claim needed here is not that the original latent model and the canonical edge-latent representative are judgmentally identical, nor that the thesis has reproved the Huang–Valtorta reduction. The claim is that the finite projection and the target query are represented on both sides of the correspondence, so the classical identifiability-preservation theorem used by Huang–Valtorta can be transported [HV06]. Since the functionalised structural causal model, the finite projected graph, its canonical edge-latent representative and the relevant interventional expressions are finite data preserved by Theorem B.5.2, the preservation result applies to the restricted type-theoretic record fragment. ■

Theorem B.6.11 (Transferred do-calculus completeness). *In the finite, discrete, recursive structural-causal-model fragment covered by Theorem B.5.2, every identifiable encoded ordinary interventional distribution $\llbracket P_x(y) \rrbracket$, with $P_x(y)$ read as $P(Y = y \mid \text{do}(X = x))$, has a type-theoretic do-calculus derivation reducing it to an encoded observational expression.*

Proof. Suppose $\llbracket P_x(y) \rrbracket$ is identifiable in the type-theoretic record fragment. By Proposition B.5.14, the corresponding classical query $P_x(y)$ is identifiable in the classical finite structural-causal-model fragment. By the published Shpitser–Pearl and Huang–Valtorta completeness results, there exists a finite classical derivation, using do-calculus and standard probability manipulations, that reduces $P_x(y)$ to an expression containing only observational quantities.

By Proposition B.5.15, this finite derivation translates step by step into the type-theoretic setting. The final expression is observational classically, and observability is preserved by the encoding because the expression contains no remaining intervention operator. Hence the translated derivation reduces $\llbracket P_x(y) \rrbracket$ to an encoded observational expression. That is exactly completeness of do-calculus for the encoded finite fragment. ■

Corollary B.6.12 (Transferred failure witnesses). *If the classical identification algorithm fails because a hedge witnesses non-identifiability, then the encoded hedge witnesses non-identifiability in the type-theoretic fragment.*

Proof. Hedges are finite graph-theoretic objects built from c-forests, ancestry, containment and the intervention-mutilated graph. These operations are preserved and reflected by Proposition B.5.5. Therefore a classical hedge translates into an encoded hedge with the same root, containment and ancestry

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properties. Since identifiability itself is transported by Proposition B.5.14, and since identification failure traces are transported by Proposition B.6.4, the encoded hedge witnesses the corresponding encoded failure of identifiability. ■

Theorem B.6.13 (Finite causal-completeness transfer). *In the restricted finite rational recursive type-theoretic structural-causal-model fragment of Definition B.5.1, the ordinary Pearl–Tian–Shpitser completeness results covered in this section carry over in the stated transfer sense: completeness of identification for finite ordinary interventional distributions, completeness of do-calculus for those effects, hedge witnesses for identification failure, completeness of conditional identification for finite conditional effects with positive denominators, and the finite functionalisation and semi-Markovian projection steps used to reduce the Huang–Valtorta setting.*

Proof. The underlying model classes are equivalent by Proposition B.5.6. Observational and interventional probabilities are preserved by Theorem B.5.2, composite probability expressions by Proposition B.5.4, graph predicates by Proposition B.5.5, and the identification and conditional-identification branch traces by Proposition B.6.4. Identifiability itself is transported by Proposition B.5.14. The finite rational negative cases are supported by Lemma B.6.2, and the causal-Bayesian-network reduction is supported by Lemma B.6.8 and Proposition B.6.10. The published finite semi-Markovian completeness theorems can therefore be applied on the classical side and then transported back to the restricted type-theoretic side. Conditional effects use Proposition B.6.6, so their positive-denominator assumptions are part of the transported statement. ■

The scope of this transfer should be kept explicit. It covers the finite ordinary interventional-distribution fragment used above, quantified over the restricted type-theoretic structural-causal-model records that decode to ordinary finite recursive structural causal models. It also covers finite conditional effects once their positive-denominator side conditions are made explicit, and it covers the finite causal-Bayesian-network functionalisation and semi-Markovian projection steps used by the Huang–Valtorta reduction. Finite rational probability is the intended target of the theorem, not an approximation to a hidden real-valued result. The transfer does not by itself cover cyclic or non-recursive structural causal models, arbitrary type-theoretic causal records outside the decoded fragment, general counterfactual identification, transportability, meta-synthesis or path-specific effects. Those results may be treated similarly only after their additional objects and operations receive their own correspondence theorems.

Bibliography/sources

“Contrarily, it is characteristic of a truly wise man that, being just a mortal, he doesn’t want to transgress any boundaries with his wisdom and together with humanity either turns a blind eye, or, out of courtesy, wanders along”

Folly in *The Praise of Folly*

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- [Ait13] Aitkenhead, D. *Peter Higgs: I Wouldn't Be Productive Enough for Today's Academic System*. <https://www.theguardian.com/science/2013/dec/06/peter-higgs-boson-academic-system>. The Guardian. 6th Dec. 2013. URL: <https://www.theguardian.com/science/2013/dec/06/peter-higgs-boson-academic-system> (visited on 12/05/2026).
- [AK91] Allis, V. and Koetsier, T. 'On Some Paradoxes of the Infinite'. In: *The British Journal for the Philosophy of Science* vol. 42, no. 2 (1st June 1991). <https://www.jstor.org/stable/688052>, pp. 187–194.
- [AK95] Allis, V. and Koetsier, T. 'On Some Paradoxes of the Infinite II'. In: *The British Journal for the Philosophy of Science* vol. 46, no. 2 (1st June 1995). <https://www.jstor.org/stable/687814>, pp. 235–247.
- [AR11] Awodey, S. and Rabe, F. 'Kripke Semantics for Martin-Löf's Extensional Type Theory'. In: *Logical Methods in Computer Science* vol. 7, no. 3, 18 (2011), pp. 1–25.
- [AS00] Arnborg, S. and Sjödin, G. 'Bayes Rules in Finite Models'. In: *Proceedings of the 14th European Conference on Artificial Intelligence*. <https://frontiersinai.com/ecai/ecai2000/pdf/p0571.pdf>. Berlin, 2000, pp. 571–575.
- [AS01] Arnborg, S. and Sjödin, G. 'On the Foundations of Bayesianism'. In: *AIP Conference Proceedings*. Vol. 568. AIP Publishing, 2001, pp. 61–71.
- [AS14] Alvesson, M. and Sandberg, J. 'Habitat and Habitus: Boxed-in versus Box-Breaking Research'. In: *Organization Studies* vol. 35, no. 7 (2014), pp. 967–987.
- [AS99] Arnborg, S. and Sjödin, G. *A Note on Foundations of Bayesianism*. Tech. rep. Royal Institute of Technology, 1999.
- [Bar26] Barber, G. *What Can We Gain by Losing Infinity?* <https://www.quantamagazine.org/what-can-we-gain-by-losing-infinity-20260429/>. Quanta Magazine. 29th Apr. 2026. URL: <https://www.quantamagazine.org/what-can-we-gain-by-losing-infinity-20260429/> (visited on 12/05/2026).
- [BDG25] Bentzen, B., Del Santo, F. and Gisin, N. *Naturalistic Intuitionism for Physics*. Preprint, PhilSci-Archive. <https://philsci-archive.pitt.edu/26743/>. 2025.
- [BDK02] Beasley, M. R., Datta, S. and Kinyoun, H. H. *Report of the Investigation Committee on the Possibility of Scientific Misconduct in the Work of Hendrik Schön and Coauthors*. Tech. rep. https://media-bell-labs-com.s3.amazonaws.com/pages/20170403_1709/misconduct-review-report-lucent.pdf. Bell Labs, Lucent Technologies, 2002.
- [Bec+01] Beck, K. et al. *Manifesto for Agile Software Development*. <https://agilemanifesto.org/>. 2001. URL: <https://agilemanifesto.org/> (visited on 26/05/2026).

- [BHM21] Bornmann, L., Haunschild, R. and Mutz, R. ‘Growth Rates of Modern Science: A Latent Piecewise Growth Curve Approach to Model Publication Numbers from Established and New Literature Databases’. In: *Humanities and Social Sciences Communications* vol. 8 (2021), p. 224.
- [Bot+20] Botvinik-Nezer, R. et al. ‘Variability in the Analysis of a Single Neuroimaging Dataset by Many Teams’. In: *Nature* vol. 582 (2020), pp. 84–88.
- [Bre+22] Breznau, N. et al. ‘Observing Many Researchers Using the Same Data and Hypothesis Reveals a Hidden Universe of Uncertainty’. In: *Proceedings of the National Academy of Sciences of the United States of America* vol. 119, no. 44 (2022), e2203150119.
- [BT24] Banach, S. and Tarski, A. ‘Sur la décomposition des ensembles de points en parties respectivement congruentes’. In: *Fundamenta Mathematicae* vol. 6, no. 1 (1924). <https://eudml.org/doc/214280>, pp. 244–277.
- [But+13] Button, K. S. et al. ‘Power Failure: Why Small Sample Size Undermines the Reliability of Neuroscience’. In: *Nature Reviews Neuroscience* vol. 14 (2013), pp. 365–376.
- [Cam+16] Camerer, C. F. et al. ‘Evaluating Replicability of Laboratory Experiments in Economics’. In: *Science* vol. 351, no. 6280 (2016), pp. 1433–1436.
- [Cam98] Campanario, J. M. ‘Peer Review for Journals as It Stands Today—Part 2’. In: *Science Communication* vol. 19, no. 4 (1998), pp. 277–306.
- [Cas26] Castelvechi, D. ‘It is incredible’: How AI is transforming mathematics’. In: *Nature* vol. 653, no. 8115 (19th May 2026). <https://www.nature.com/articles/d41586-026-01553-1>, pp. 664–665.
- [Cen] Center for Open Science. *Registered Reports*. Center for Open Science. <https://www.cos.io/initiatives/registered-reports>.
- [Col04] Colyvan, M. ‘The Philosophical Significance of Cox’s Theorem’. In: *International Journal of Approximate Reasoning* vol. 37, no. 1 (2004), pp. 71–85.
- [Cou23] Council of the European Union. *Council Calls for Transparent, Equitable, and Open Access to Scholarly Publications*. <https://www.consilium.europa.eu/en/press/press-releases/2023/05/23/council-calls-for-transparent-equitable-and-open-access-to-scholarly-publications/>. Council of the European Union. 23rd May 2023. URL: <https://www.consilium.europa.eu/en/press/press-releases/2023/05/23/council-calls-for-transparent-equitable-and-open-access-to-scholarly-publications/> (visited on 12/05/2026).
- [Cox46] Cox, R. T. ‘Probability, Frequency and Reasonable Expectation’. In: *American Journal of Physics* vol. 14, no. 1 (1946), pp. 1–13.
- [Cox61] Cox, R. T. *The Algebra of Probable Inference*. <https://www.press.jhu.edu/books/title/1089/algebra-probable-inference>. Baltimore: Johns Hopkins Press, 1961.

B. Proofs and arguments

- [CW17] Clayton, A. and Waddington, T. ‘Bridging the Intuition Gap in Cox’s Theorem: A Jaynesian Argument for Universality’. In: *International Journal of Approximate Reasoning* vol. 80 (2017), pp. 36–51.
- [Cyr06] Cyranoski, D. ‘Verdict: Hwang’s Human Stem Cells Were All Fakes’. In: *Nature* vol. 439 (2006), pp. 122–123.
- [DG19] Del Santo, F. and Gisin, N. ‘Physics Without Determinism: Alternative Interpretations of Classical Physics’. In: *Physical Review A* vol. 100 (2019), p. 062107.
- [DOR12] DORA. *San Francisco Declaration on Research Assessment*. <https://sfdora.org/read/>. DORA. 16th Dec. 2012. URL: <https://sfdora.org/read/> (visited on 12/05/2026).
- [DP24] Dybjer, P. and Palmgren, E. ‘Intuitionistic Type Theory’. In: *The Stanford Encyclopedia of Philosophy*. Ed. by Zalta, E. N. and Nodelman, U. Winter 2024. <https://plato.stanford.edu/archives/win2024/entries/type-theory-intuitionistic/>. Metaphysics Research Lab, Stanford University, 2024.
- [Eis11] Eisen, M. *Research Bought, Then Paid For*. <https://www.michaelseisen.org/blog/?p=890>. It Is NOT Junk. 5th Dec. 2011. URL: <https://www.michaelseisen.org/blog/?p=890> (visited on 12/05/2026).
- [Era71] Erasmus, D. *Praise of Folly and Letter to Martin Dorp*. Ed. by Levi, A. H. T. Trans. by Radice, B. Penguin Classics. Latin original *Moriae Encomium*, Paris: Gilles de Gourmont, 1511. Harmondsworth: Penguin, 1971.
- [Fan09] Fanelli, D. ‘How Many Scientists Fabricate and Falsify Research? A Systematic Review and Meta-Analysis of Survey Data’. In: *PLOS ONE* vol. 4, no. 5 (2009), e5738.
- [Fey75] Feyerabend, P. *Against Method: Outline of an Anarchistic Theory of Knowledge*. London: New Left Books, 1975.
- [FK23] Fritz, T. and Klingler, A. ‘The d-Separation Criterion in Categorical Probability’. In: *Journal of Machine Learning Research* vol. 24, no. 46 (2023). <https://jmlr.org/papers/v24/22-0993.html>, pp. 1–49.
- [Fle79] Fleck, L. *Genesis and Development of a Scientific Fact*. Ed. by Trenn, T. J. and Merton, R. K. Trans. by Bradley, F. and Trenn, T. J. Foreword by Thomas S. Kuhn. German original *Entstehung und Entwicklung einer wissenschaftlichen Tatsache*, Schwabe, Basel, 1935. Chicago: University of Chicago Press, 1979.
- [FSC12] Fang, F. C., Steen, R. G. and Casadevall, A. ‘Misconduct Accounts for the Majority of Retracted Scientific Publications’. In: *Proceedings of the National Academy of Sciences of the United States of America* vol. 109, no. 42 (2012), pp. 17028–17033.
- [GB19] Gross, K. and Bergstrom, C. T. ‘Rationalizing Risk Aversion in Science: Why Incentives to Work Hard Clash with Incentives to Take Risks’. In: *PLOS Biology* vol. 17, no. 6 (2019), e3000302.

- [GHR17] Gyenis, Z., Hofer-Szabó, G. and Rédei, M. ‘Conditioning Using Conditional Expectations: The Borel–Kolmogorov Paradox’. In: *Synthese* vol. 194, no. 7 (2017), pp. 2595–2630.
- [Gis20] Gisin, N. ‘Mathematical Languages Shape Our Understanding of Time in Physics’. In: *Nature Physics* vol. 16 (2020), pp. 114–116.
- [GL13] Gelman, A. and Loken, E. *The Garden of Forking Paths: Why Multiple Comparisons Can Be a Problem, Even When There Is No “Fishing Expedition” or “p-Hacking” and the Research Hypothesis Was Posited Ahead of Time*. Tech. rep. https://www.stat.columbia.edu/~gelman/research/unpublished/p_hacking.pdf. Department of Statistics, Columbia University, 2013.
- [GL14] Gelman, A. and Loken, E. ‘The Statistical Crisis in Science’. In: *American Scientist* vol. 102, no. 6 (2014), pp. 460–465.
- [Gra+21] Gratzer, D. et al. *Multimodal Dependent Type Theory*. arXiv preprint. <https://arxiv.org/abs/2011.15021>. 2021. arXiv: 2011.15021.
- [Gra23] Gratzer, D. ‘Syntax and Semantics of Modal Type Theory’. <https://iris-project.org/pdfs/2023-phd-gratzer.pdf>. PhD thesis. Aarhus University, 2023.
- [GSB19] Gratzer, D., Sterling, J. and Birkedal, L. ‘Implementing a Modal Dependent Type Theory’. In: *Proceedings of the ACM on Programming Languages* vol. 3, no. ICFP (2019), 107:1–107:29.
- [GSM11] Godlee, F., Smith, J. and Marcovitch, H. ‘Wakefield’s Article Linking MMR Vaccine and Autism Was Fraudulent’. In: *BMJ* vol. 342 (2011). <https://www.bmj.com/content/342/bmj.c7452>, p. c7452.
- [Hac83] Hacking, I. *Representing and Intervening: Introductory Topics in the Philosophy of Natural Science*. Cambridge: Cambridge University Press, 1983.
- [Hal99a] Halpern, J. Y. ‘A Counterexample to Theorems of Cox and Fine’. In: *Journal of Artificial Intelligence Research* vol. 10 (1999). <https://www.jair.org/index.php/jair/article/view/10223>, pp. 67–85.
- [Hal99b] Halpern, J. Y. ‘Cox’s Theorem Revisited’. In: *Journal of Artificial Intelligence Research* vol. 11 (1999). <https://arxiv.org/abs/cs/9911012>, pp. 429–435.
- [Hof23] Hofweber, T. ‘Logic and Ontology’. In: *The Stanford Encyclopedia of Philosophy*. Ed. by Zalta, E. N. and Nodelman, U. Summer 2023. <https://plato.stanford.edu/archives/sum2023/entries/logic-ontology/>. Metaphysics Research Lab, Stanford University, 2023.
- [HP07] Halpern, J. Y. and Pucella, R. ‘Dealing with Logical Omniscience’. In: *Proceedings of the Tenth Conference on Theoretical Aspects of Rationality and Knowledge*. http://www.tark.org/proceedings/tark_jun25_07/p169-halpern.pdf. Singapore, 2007, pp. 169–181.
- [HS10] Harrigan, N. and Spekkens, R. W. ‘Einstein, Incompleteness, and the Epistemic View of Quantum States’. In: *Foundations of Physics* vol. 40, no. 2 (2010), pp. 125–157.

B. Proofs and arguments

- [Hui49] Huizinga, J. *Homo Ludens: A Study of the Play-Element in Culture*. London: Routledge & Kegan Paul, 1949.
- [Hum75] Hume, D. *An Enquiry concerning Human Understanding*. Ed. by Selby-Bigge, L. A. and Nidditch, P. H. 3rd ed. Originally published as *Philosophical Essays concerning Human Understanding*, London: A. Millar, 1748. Oxford: Clarendon Press, 1975.
- [Hum78] Hume, D. *A Treatise of Human Nature: Being an Attempt to Introduce the Experimental Method of Reasoning into Moral Subjects*. Ed. by Selby-Bigge, L. A. and Nidditch, P. H. 2nd ed. Originally published London: John Noon, 1739–1740. Oxford: Clarendon Press, 1978.
- [HV06] Huang, Y. and Valtorta, M. ‘Pearl’s Calculus of Intervention Is Complete’. In: *Proceedings of the Twenty-Second Conference on Uncertainty in Artificial Intelligence*. <https://arxiv.org/abs/1206.6831>. AUA Press, 2006, pp. 217–224.
- [Ibá99] Ibáñez, C. F. ‘Researchers Face ‘Catch-22’ Grants Trap’. In: *Nature* vol. 398 (1999), p. 280.
- [Ioa05] Ioannidis, J. P. A. ‘Why Most Published Research Findings Are False’. In: *PLOS Medicine* vol. 2, no. 8 (2005), e124.
- [Jay03] Jaynes, E. T. *Probability Theory: The Logic of Science*. Cambridge: Cambridge University Press, 2003.
- [Jef61] Jeffreys, H. *Theory of Probability*. 3rd ed. Oxford: Clarendon Press, 1961.
- [JLP12] John, L. K., Loewenstein, G. and Prelec, D. ‘Measuring the Prevalence of Questionable Research Practices with Incentives for Truth Telling’. In: *Psychological Science* vol. 23, no. 5 (2012), pp. 524–532.
- [Kak26] Kakaes, K. *The AI Revolution in Math Has Arrived*. <https://www.quantamagazine.org/the-ai-revolution-in-math-has-arrived-20260413/>. Quanta Magazine. 13th Apr. 2026. URL: <https://www.quantamagazine.org/the-ai-revolution-in-math-has-arrived-20260413/> (visited on 12/05/2026).
- [Kan98] Kant, I. *Critique of Pure Reason*. Ed. and trans. by Guyer, P. and Wood, A. W. The Cambridge Edition of the Works of Immanuel Kant. German original *Kritik der reinen Vernunft*, Riga: Hartknoch, 1st ed. 1781 (A) / 2nd ed. 1787 (B). Cambridge: Cambridge University Press, 1998.
- [Key21] Keynes, J. M. *A Treatise on Probability*. <https://archive.org/details/treatiseonprobab007528mbp>. London: Macmillan, 1921.
- [Kie80] Kierkegaard, S. *The Concept of Anxiety: A Simple Psychologically Orienting Deliberation on the Dogmatic Issue of Hereditary Sin*. Ed. and trans. by Thomte, R. and Anderson, A. B. Vol. VIII. Kierkegaard’s Writings. Danish original *Begrebet Angest*, Reitzel, Copenhagen, 1844. Princeton, NJ: Princeton University Press, 1980.

- [Kie83] Kierkegaard, S. *Fear and Trembling / Repetition*. Ed. and trans. by Hong, H. V. and Hong, E. H. Vol. VI. Kierkegaard's Writings. Danish original *Frygt og Bæven*, Reitzel, Copenhagen, 1843. Princeton, NJ: Princeton University Press, 1983.
- [Kie89] Kierkegaard, S. *The Concept of Irony, with Continual Reference to Socrates*. Ed. and trans. by Hong, H. V. and Hong, E. H. Vol. II. Kierkegaard's Writings. Danish original *Om Begrebet Ironi med stadigt Hensyn til Socrates*, Reitzel, Copenhagen, 1841. Princeton, NJ: Princeton University Press, 1989.
- [Kle+14] Klein, R. A. et al. 'Investigating Variation in Replicability: A "Many Labs" Replication Project'. In: *Social Psychology* vol. 45, no. 3 (2014), pp. 142–152.
- [Lan+22] Lane, J. N. et al. 'Conservatism Gets Funded? A Field Experiment on the Role of Negative Information in Novel Project Evaluation'. In: *Management Science* vol. 68, no. 6 (2022), pp. 4478–4495.
- [Lan16] Landhuis, E. 'Information Overload'. In: *Nature* vol. 535 (2016), pp. 457–458.
- [Los11] Los Angeles Times. *Dutch Psychologist Admits Faking Data in Dozens of Studies*. <https://www.latimes.com/science/la-sci-science-fraud-20111106-story.html>. Los Angeles Times. 6th Nov. 2011. URL: <https://www.latimes.com/science/la-sci-science-fraud-20111106-story.html> (visited on 12/05/2026).
- [Mar84] Martin-Löf, P. *Intuitionistic Type Theory*. Naples: Bibliopolis, 1984.
- [McE20] McElreath, R. *Statistical Rethinking: A Bayesian Course with Examples in R and Stan*. 2nd ed. <https://xcelab.net/rm/>. Boca Raton: CRC Press, 2020.
- [Mel14] Mellor, D. *Higgs Boson Scientist: Publish or Perish Culture Is Bad for Science*. <https://www.theguardian.com/commentisfree/2014/feb/14/higgs-boson-publish-or-perish-science-culture>. The Guardian. 14th Feb. 2014. URL: <https://www.theguardian.com/commentisfree/2014/feb/14/higgs-boson-publish-or-perish-science-culture> (visited on 12/05/2026).
- [MO15] Marcus, A. and Oransky, I. *How the Biggest Fabricator in Science Got Caught*. <https://nautil.us/how-the-biggest-fabricator-in-science-got-caught-235421/>. Nautilus. 18th June 2015. URL: <https://nautil.us/how-the-biggest-fabricator-in-science-got-caught-235421/> (visited on 12/05/2026).
- [MU21] Moura, L. de and Ullrich, S. 'The Lean 4 Theorem Prover and Programming Language'. In: *Automated Deduction — CADE 28*. Ed. by Platzer, A. and Sutcliffe, G. Vol. 12699. Lecture Notes in Computer Science. https://doi.org/10.1007/978-3-030-79876-5_37. Cham: Springer, 2021, pp. 625–635.
- [Mun+17] Munafò, M. R. et al. 'A Manifesto for Reproducible Science'. In: *Nature Human Behaviour* vol. 1 (10th Jan. 2017), p. 0021.
- [NI12] Nicholson, J. M. and Ioannidis, J. P. A. 'Conform and Be Funded'. In: *Nature* vol. 492 (2012), pp. 34–36.

B. Proofs and arguments

- [Nos+15] Nosek, B. A. et al. ‘Promoting an Open Research Culture’. In: *Science* vol. 348, no. 6242 (2015), pp. 1422–1425.
- [NPS90] Nordström, B., Petersson, K. and Smith, J. M. *Programming in Martin-Löf’s Type Theory: An Introduction*. <https://intuitionistic.wordpress.com/wp-content/uploads/2010/07/nordstrom-et-al-type-theory.pdf>. Oxford: Oxford University Press, 1990.
- [NR19] Nguyen, H. N. and Rakib, A. ‘A Probabilistic Logic for Resource-Bounded Multi-Agent Systems’. In: *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence*. International Joint Conferences on Artificial Intelligence Organization, 2019, pp. 521–527.
- [NSM12] Nosek, B. A., Spies, J. R. and Motyl, M. ‘Scientific Utopia: II. Restructuring Incentives and Practices to Promote Truth over Publishability’. In: *Perspectives on Psychological Science* vol. 7, no. 6 (2012), pp. 615–631.
- [OGr24] O’Grady, C. *Failure at Every Level: How Science Sleuths Exposed Massive Ethics Violations at a Famed French Institute*. <https://www.science.org/content/article/failure-every-level-how-science-sleuths-exposed-massive-ethics-violations-famed-french>. Science. 5th Sept. 2024. URL: <https://www.science.org/content/article/failure-every-level-how-science-sleuths-exposed-massive-ethics-violations-famed-french> (visited on 12/05/2026).
- [Ope15] Open Science Collaboration. ‘Estimating the Reproducibility of Psychological Science’. In: *Science* vol. 349, no. 6251 (2015), aac4716.
- [Orn26] Ornes, S. *Math Long Resisted a Digital Disruption. AI Is Poised to Change That*. <https://www.sciencenews.org/article/math-disrupted-by-ai-verify-proofs>. Science News. 15th Apr. 2026. URL: <https://www.sciencenews.org/article/math-disrupted-by-ai-verify-proofs> (visited on 12/05/2026).
- [PBR12] Pusey, M. F., Barrett, J. and Rudolph, T. ‘On the Reality of the Quantum State’. In: *Nature Physics* vol. 8 (2012), pp. 475–478.
- [PD01] Pfenning, F. and Davies, R. ‘A Judgmental Reconstruction of Modal Logic’. In: *Mathematical Structures in Computer Science* vol. 11, no. 4 (2001), pp. 511–540.
- [Pea09] Pearl, J. *Causality: Models, Reasoning, and Inference*. 2nd ed. Cambridge: Cambridge University Press, 2009.
- [Pea12] Pearl, J. ‘The Do-Calculus Revisited’. In: *Proceedings of the Twenty-Eighth Conference on Uncertainty in Artificial Intelligence*. https://ftp.cs.ucla.edu/pub/stat_ser/r402.pdf. AUAI Press, 2012, pp. 4–11.
- [PM18] Pearl, J. and Mackenzie, D. *The Book of Why: The New Science of Cause and Effect*. <https://www.basicbooks.com/titles/judea-pearl/the-book-of-why/9780465097616/>. New York: Basic Books, 2018.

- [Ram31] Ramsey, F. P. ‘Truth and Probability’. In: *The Foundations of Mathematics and Other Logical Essays*. Ed. by Braithwaite, R. B. Written in 1926. London: Kegan Paul, Trench, Trubner & Co., 1931, pp. 156–198.
- [Ric+25] Richardson, R. A. K. et al. ‘The Entities Enabling Scientific Fraud at Scale Are Large, Resilient, and Growing Rapidly’. In: *Proceedings of the National Academy of Sciences of the United States of America* vol. 122, no. 32 (12th Aug. 2025), e2420092122.
- [Sch+21] Schweinsberg, M. et al. ‘Same Data, Different Conclusions: Radical Dispersion in Empirical Results When Independent Analysts Operationalize and Test the Same Hypothesis’. In: *Organizational Behavior and Human Decision Processes* vol. 165 (2021), pp. 228–249.
- [Sch13] Schekman, R. *How Journals Like Nature, Cell and Science Are Damaging Science*. <https://www.theguardian.com/commentisfree/2013/dec/09/how-journals-nature-science-cell-damage-science>. The Guardian. 9th Dec. 2013. URL: <https://www.theguardian.com/commentisfree/2013/dec/09/how-journals-nature-science-cell-damage-science> (visited on 12/05/2026).
- [Sha04] Shafer, G. ‘Comments on “Constructing a Logic of Plausible Inference: A Guide to Cox’s Theorem”, by Kevin S. Van Horn’. In: *International Journal of Approximate Reasoning* vol. 35 (2004), pp. 97–105.
- [Sil+18] Silberzahn, R. et al. ‘Many Analysts, One Dataset: Making Transparent How Variations in Analytical Choices Affect Results’. In: *Advances in Methods and Practices in Psychological Science* vol. 1, no. 3 (2018), pp. 337–356.
- [Sim55] Simon, H. A. ‘A Behavioral Model of Rational Choice’. In: *The Quarterly Journal of Economics* vol. 69, no. 1 (1955), pp. 99–118.
- [Slo26] Sloman, L. *In Math, Rigor Is Vital. But Are Digitized Proofs Taking It Too Far?* <https://www.quantamagazine.org/in-math-rigor-is-vital-but-are-digitized-proofs-taking-it-too-far-20260325/>. Quanta Magazine. 25th Mar. 2026. URL: <https://www.quantamagazine.org/in-math-rigor-is-vital-but-are-digitized-proofs-taking-it-too-far-20260325/> (visited on 12/05/2026).
- [SN25] Steup, M. and Neta, R. ‘Epistemology’. In: *The Stanford Encyclopedia of Philosophy*. Ed. by Zalta, E. N. and Nodelman, U. Spring 2025. <https://plato.stanford.edu/archives/spr2025/entries/epistemology/>. Metaphysics Research Lab, Stanford University, 2025.
- [Sno98] Snow, P. ‘On the Correctness and Reasonableness of Cox’s Theorem for Finite Domains’. In: *Computational Intelligence* vol. 14, no. 3 (1998), pp. 452–459.
- [SNS11] Simmons, J. P., Nelson, L. D. and Simonsohn, U. ‘False-Positive Psychology: Undisclosed Flexibility in Data Collection and Analysis Allows Presenting Anything as Significant’. In: *Psychological Science* vol. 22, no. 11 (2011), pp. 1359–1366.

B. Proofs and arguments

- [SP06] Shpitser, I. and Pearl, J. ‘Identification of Joint Interventional Distributions in Recursive Semi-Markovian Causal Models’. In: *Proceedings of the Twenty-First National Conference on Artificial Intelligence*. AAAI Press, 2006, pp. 1219–1226.
- [Spe07] Spekkens, R. W. ‘Evidence for the Epistemic View of Quantum States: A Toy Theory’. In: *Physical Review A* vol. 75, no. 3 (2007), p. 032110. arXiv: [quant-ph/0401052](https://arxiv.org/abs/quant-ph/0401052).
- [SS20] Schwaber, K. and Sutherland, J. *The Scrum Guide: The Definitive Guide to Scrum: The Rules of the Game*. <https://scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf>. 2020. URL: <https://scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf> (visited on 26/05/2026).
- [The16] The New Yorker. *The Stem-Cell Scandal*. <https://www.newyorker.com/magazine/2016/02/29/the-stem-cell-scandal>. The New Yorker. 29th Feb. 2016. URL: <https://www.newyorker.com/magazine/2016/02/29/the-stem-cell-scandal> (visited on 12/05/2026).
- [The24] ThePrint. *India’s Research Crime Is Getting Worse. Scientists Are Gaming Peer Review System*. <https://theprint.in/ground-reports/indias-research-crime-is-getting-worse-scientists-are-gaming-peer-review-system/2261884/>. ThePrint. 14th Sept. 2024. URL: <https://theprint.in/ground-reports/indias-research-crime-is-getting-worse-scientists-are-gaming-peer-review-system/2261884/> (visited on 12/05/2026).
- [Tho54] Thomson, J. F. ‘Tasks and Super-Tasks’. In: *Analysis* vol. 15, no. 1 (1st Oct. 1954). <https://academic.oup.com/analysis/article-abstract/15/1/1/170170>, pp. 1–13.
- [TP02] Tian, J. and Pearl, J. ‘A General Identification Condition for Causal Effects’. In: *Proceedings of the Eighteenth National Conference on Artificial Intelligence*. AAAI Press, 2002, pp. 567–573.
- [TS09] Tian, J. and Shpitser, I. ‘On Identifying Causal Effects’. In: *Proceedings of the Twenty-Fifth Conference on Uncertainty in Artificial Intelligence*. AUAI Press, 2009, pp. 722–729.
- [Van03] Van Horn, K. S. ‘Constructing a Logic of Plausible Inference: A Guide to Cox’s Theorem’. In: *International Journal of Approximate Reasoning* vol. 34, no. 1 (2003), pp. 3–24.
- [Vit05] Vitali, G. *Sul problema della misura dei gruppi di punti di una retta: Nota*. <https://web.unica.it/unica/protected/417040/0/def/ref/MAT246835/>. Bologna: Tip. Gamberini e Parmeggiani, 1905, p. 5.

